Open KPZ equation from a class of nonlinear SPDEs

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ABSTRACT. We derive the open KPZ equation as a continuum limit for a general class of Ginzburg-Landau SPDEs under weak nonlinearity scaling. To our knowledge, this is a first universality result for the open KPZ equation beyond integrable models and their perturbations.

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1. INTRODUCTION {section:intro}

The *open Kardar-Parisi-Zhang (*KPZ*) equation* is a stochastic PDE (SPDE) that describes the fluctuations of a random dynamical interface, which grows on the substrate $[0,1]$ subject to fixed slopes near two "walls" located at the boundary $\{0,1\} \subseteq [0,1]$. See Figure 1 for a sample interface. To be precise, it is the following SPDE:

$$\partial_{\mathrm{t}} \mathbf{h}_{\mathrm{t},\mathrm{X}} = \alpha \partial_{\mathrm{X}}^2 \mathbf{h}_{\mathrm{t},\mathrm{X}} + \beta (\partial_{\mathrm{X}} \mathbf{h}_{\mathrm{t},\mathrm{X}})^2 + \xi_{\mathrm{t},\mathrm{X}}, \quad \text{where } (\mathrm{t},\mathrm{X}) \in [0,\infty) \times [0,1]. \tag{1.1}$$ {eq:kpzI}

Here, the constants $\alpha > 0$ and $\beta \neq 0$ are fixed, and ξ is the Gaussian space-time white noise. The aforementioned boundary is given by the following inhomogeneous Neumann data, in which $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$ are also fixed:

$$\partial_{\mathrm{X}} \mathbf{h}_{\mathrm{t},\mathrm{X}}|_{\mathrm{X}=0} = \widetilde{\mathbf{A}}, \tag{1.2}$$ {eq:kpzII}

$$\partial_{\mathrm{X}} \mathbf{h}_{\mathrm{t},\mathrm{X}}|_{\mathrm{X}=1} = \widetilde{\mathbf{B}}. \tag{1.3}$$ {eq:kpzIII}

Strictly speaking, the SPDE (1.1)-(1.3) is *singular* (and thus analytically ill-posed) [6, 10, 12]. To address this, we will adopt the *Cole-Hopf solution* herein [6]. That is, we define $\mathbf{h} := \beta^{-1}\alpha \log \mathbf{Z}$, where $\mathbf{Z}$ satisfies the following *open stochastic heat equation* (SHE) with *renormalized boundary parameters* $\mathbf{A}, \mathbf{B} \in \mathbb{R}$ [10]:

$$\partial_{\mathrm{t}} \mathbf{Z}_{\mathrm{t},\mathrm{X}} = \alpha \partial_{\mathrm{X}}^2 \mathbf{Z}_{\mathrm{t},\mathrm{X}} + \tfrac{\beta}{\alpha} \mathbf{Z}_{\mathrm{t},\mathrm{X}} \xi_{\mathrm{t},\mathrm{X}}, \tag{1.4}$$ {eq:sheI}

$$\partial_{\mathrm{X}} \mathbf{Z}_{\mathrm{t},\mathrm{X}}|_{\mathrm{X}=0} = \mathbf{A} \mathbf{Z}_{\mathrm{t},0}, \tag{1.5}$$ {eq:sheII}

$$\partial_{\mathrm{X}} \mathbf{Z}_{\mathrm{t},\mathrm{X}}|_{\mathrm{X}=1} = \mathbf{B} \mathbf{Z}_{\mathrm{t},1}. \tag{1.6}$$ {eq:sheIII}

Physically, the open KPZ equation is a conjecturally universal model for the fluctuations in many models of interface growth, as well as a toy model for turbulence in $1+1$-dimensional systems, both of which are in contact with a "wall" or a "reservoir" [4]. This is similar to the original proposal of the full-space KPZ equation in [21] as a conjecturally universal approximation for the large-scale fluctuations of random interfaces.



FIGURE 1. Sample of the open KPZ interface at a fixed time, with positive slopes at the two walls. {figure:kpzoper

The focus of this work is the *universality* of the open KPZ equation for classes of random dynamical interfaces. Our motivation comes from the feature that the open KPZ equation has recently attracted a great deal of attention, which has led to a thorough investigation of its long-time (non-Gaussian!) behaviors (including the construction of invariant measures [1, 2, 3, 5, 8, 19].) But progress on universality, which would imply the same probabilistic behavior for a larger class of interfaces, is very minimal, despite the fact that universality of full-space and half-space versions of the KPZ equation has witnessed considerable progress. The reason is that general models on a compact interval lack explicit or simple invariant measures. Indeed, even for the special case of open KPZ itself, the identification of its invariant measures is a highly difficult challenge, resolved in [1, 2, 3, 5, 8, 19].

Perhaps the most *fundamental* universality problem for open KPZ, which is motivated by the original heuristic in [21] for universality of the full-space equation, concerns the following family of *Ginzburg-Landau* SPDEs:

{eq:gl}
$$\partial_{\mathrm{t}}\mathbf{j}_{\mathrm{t},\mathrm{X}} = \partial_{\mathrm{X}}\mathscr{U}'(\partial_{\mathrm{X}}\mathbf{j}_{\mathrm{t},\mathrm{X}}) + \mathbf{F}[\mathscr{U}'(\partial_{\mathrm{X}}\mathbf{j}_{\mathrm{t},\mathrm{X}})] + \xi_{\mathrm{t},\mathrm{X}}. \tag{1.7}$$

We equip (1.7) with inhomogeneous Neumann boundary data as well. Above, $\mathscr{U} : \mathbb{R} \to \mathbb{R}$ is a smooth *potential*, and $\mathbf{F} : \mathbb{R} \to \mathbb{R}$ is a smooth *nonlinearity*. The Ginzburg-Landau SPDEs, which were systematized in physics [20], are popular models in statistical mechanics due to the flexible functional parameters $\mathscr{U}, \mathbf{F}$, which let us model a variety of physical systems. They have also played a key role in the mathematics of hydrodynamic limits [26]. In any case, we note that $\mathscr{U}, \mathbf{F}$ both being quadratic recovers (1.1).

The universality conjecture of interest for (1.7) is the following:

(1) Discretize (1.7) in space according to some scaling parameter, say N, and apply a "weak nonlinearity scaling" depending on N to the $\mathbf{F}$-nonlinearity. (This turns the singular SPDE (1.7) into a well-posed SDE.)
(2) Take the continuum limit of solutions to the discretized equations. Conjecturally, this limit is the solution to (1.1)-(1.3) for some effective constants $\alpha, \beta, \widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$.

The main result of this work is a resolution of this conjecture for a large set of potentials $\mathscr{U}$ and non-equilibrium initial data. To our knowledge, this yields a first universality result for open KPZ *beyond integrable models and their perturbations*. In this paper, we focus on quadratic $\mathbf{F}$ (as opposed to higher-degree polynomials) for mostly technical reasons; since the potential $\mathscr{U}$ is allowed to be very general (described in Assumption 2.2), the models (1.7) at hand are still quite general and far from (1.1). Moreover, we choose a space-discretization that preserves invariant measures of (the bulk dynamics of) (1.7), which is natural if we view (1.7) as a dynamical field theory. We refer to Theorem 2.4 for a precise statement of our main result.

ubsection:tech}

1.1. **Technical aspects.** The desired convergence to (1.1)-(1.3) goes through the convergence of exponentials to (1.4)-(1.6). The primary source of the technical challenges herein is that the evolution equation of the exponential of (a space-discretization of) (1.7) does *not* satisfy a simple discretization of (1.4)-(1.6) in general. In particular, said space-discretized exponential satisfies a lattice (1.4)-(1.6) with additional (and complicated) "fluctuations". (Roughly put, "fluctuations" are spatially local functions of $\partial_{\mathrm{X}}\mathbf{j}$ whose cancellations from space-time integration are encoded via projections to non-Gaussian versions of the Wiener chaos. There are two types of fluctuations, more or less, namely "bulk fluctuations" arising from the dynamics away from the edge, and "edge fluctuations" arising from the boundary data. See Definitions 3.1 and 3.2 for precise definitions.) In fact,

because the evolution (1.7) is spatially local, we will be able to derive the desired evolution equation by patching together the evolution equation in the half-space.

Precise estimates for a variety of space-time integrals of the aforementioned fluctuations is the technical key to the analysis in this paper. To this end, the strategy which we develop herein uses crucially that we only require integrals of fluctuations *on local space-time scales*. This allows us to construct the relevant dynamics by gluing half-space dynamics. Indeed, at local scales, dynamics away from the left edge look like dynamics in the positive half-space, and dynamics away from the right edge at local scales look like dynamics in the negative half-space (which are just dynamics in the positive half-space after a reflection). Then, we compare space-time integrals of these fluctuations to their half-space version and thus inherit the relevant estimates from the half-space problem in [33]. We emphasize that this gluing idea was implemented for the half-space problem in [33], in which dynamics away from the edge are compared to the full-space, while the edge was addressed separately.

However, the aforementioned local gluing mechanism requires a global-scale input that ultimately amounts to a priori (sub-optimal) regularity estimates for (a space-discretization of) $\mathbf{j}^{\mathsf{N}}$. (This is for technical reasons, and the situation is the same in both the full-space [32] and half-space [33] settings.) *The key challenge is that, unlike the full-space or half-space, there is no explicit invariant measure of (a space-discretized)* (1.7) *in general.* In this spirit, we note that even in the special case of open KPZ (1.1)-(1.3), invariant measures (with fairly sophisticated descriptions) have only recently been constructed. We also clarify that key to a priori regularity estimates in the full-space and half-space settings of [32, 33] is the existence of *at least one* explicit and simple invariant measure (which is unavailable here).

The key technical innovation in this paper is the construction of some perturbation of (a space-discretization of) (1.7) *which both has an explicit invariant measure and is mutually absolutely continuous with respect to the path-space law of (the space-discretization of)* (1.7). In particular, regularity estimates for the perturbed process transfer immediately to (a space-discretization of) (1.7). We clarify that this construction is a *dynamical* version of the statement that the invariant measure of open KPZ is mutually absolutely continuous with respect to that of open KPZ with $\beta = 0$ in (1.1). We also note that, to our knowledge, this is the first such dynamical construction, even for open KPZ (let alone the much more general family of SPDEs in (1.7)). Finally, we note that the method of constructing global perturbations to the dynamics seems to be new in the context of hydrodynamic limits and related problems (in which the lack of explicit invariant measures often produces serious challenges). Moreover, the construction of the perturbation only uses a calculus of infinitesimal generators, so we anticipate the method to be widely applicable (e.g. to universality of open KPZ) for a much larger family of models.

We emphasize that the perturbation requires a fairly sophisticated construction. For instance, one cannot just remove the nonlinearity $\mathbf{F}$ from (1.7), as the resulting process would not be mutually absolutely continuous with respect to (1.7) itself (essentially since this nonlinearity is what makes (1.7) singular).

Ultimately, as a result of this analysis, we are able to establish the necessary fluctuation estimates, which are often dubbed *Boltzmann-Gibbs principles*. In particular, we are able to establish these principles despite the lack of an explicit and simple invariant measure, which is one of the main technical contributions here.

{subsection:ba

1.2. **Background.** The derivation of open KPZ from microscopic systems, more generally, starts with the works [6, 12, 25, 28, 31], which obtain the open KPZ equation as a large-scale limit from the *open* ASEP and its perturbations. In these works, it is important that the exponentiated "height function" of interest solves an exact discretization of the limiting SHE (1.4)-(1.6). (Exceptions to this include [28, 31], where hydrodynamic-limit-type arguments are needed to address additional error terms, though the nature of the particle systems therein being perturbations of open ASEP is used crucially. The other exception is [12], which instead uses the theory of energy solutions [11, 13] to show convergence to open KPZ via martingale problems instead of going through open SHE. However, this is currently limited to the setting where $\widetilde{\mathbf{A}} = -\widetilde{\mathbf{B}}$, in which case the open KPZ equation, as well as the open ASEP of interest in [12], has explicit and simple invariant measures.) Finally, of these papers, [6, 25, 31] also extend to the half-space in addition to the compact interval.

As noted earlier, the work [33] establishes the half-space analogue of the results herein. Moreover, the analysis and methods in [33] play a key role in this paper. (Indeed, as explained earlier, a key step in this paper is reducing the analysis of fluctuations to the half-space setting in [33].)

Next, let us cite [7, 14, 16, 17, 18, 23, 24, 29, 30, 32], which establish a similar Hairer-Quastel universality for the KPZ equation but on domains without boundaries (namely, the torus and the full-space).

Finally, we comment that the open KPZ equation (1.1)-(1.3) with inhomogeneous Neumann (and even Dirichlet) boundary data was solved using the theory of regularity structures in [10]. Although natural, it does not seem that these regularity structures methods apply to the equation (1.7) in general, in large part since the linearization of (1.7) is a parabolic PDE with singular coefficients if $\mathscr{U}$ is not quadratic.

1.3. **AI disclosure.** An LLM[1] was used in the following mathematical capacity. The analysis in Section 5 uses a generalization of Gaussian integration-by-parts for general potentials $\mathscr{U}$. The LLM reminded the author of this identity (which was already shown and used crucially in prior work [32, 33] of the author). However, computations given by the LLM to illustrate the utility of this integration-by-parts formula were so complicated that we did not bother to look at them (beyond noting the importance of this formula) and continued with our own computations. Regardless, the author is highly grateful for this pointer. LLMs were not used in any other capacity; in particular, the analysis and writing are all human-performed.

2. Main results

2.1. **Model.** We now present the dynamics of interest precisely. Before we do so, we will require the following notation throughout this paper (including what follows in the introduction).

- We set $[\![\mathrm{a},\mathrm{b}]\!] := [\mathrm{a},\mathrm{b}] \cap \mathbb{Z}$ for any $\mathrm{a},\mathrm{b} \in \mathbb{R}$ (with $\mathrm{a} \leqslant \mathrm{b}$).
- We let $\mathsf{N} > 0$ be a large scaling parameter (to be taken to ∞; this is the "large-N limit").
- We denote elements in $\mathbb{R}^{\mathbb{Z}}$ by $\boldsymbol{\phi} = (\boldsymbol{\phi}_{\mathrm{x}})_{\mathrm{x}\in\mathbb{Z}}$. Sometimes, we will only need projections of $\boldsymbol{\phi} \in \mathbb{R}^{\mathbb{Z}}$ onto $\mathbb{R}^{\mathbb{I}}$ with various subsets $\mathbb{I} \subseteq \mathbb{Z}$, including the discrete interval $\mathbb{I} = [\![0, \mathrm{K}\mathsf{N}]\!]$ for some fixed $\mathrm{K} \geqslant 0$. We will still use the $\boldsymbol{\phi}$ notation for said projections.
- For any $\mathrm{x} \in \mathbb{Z}$, we let $\tau_{\mathrm{x}} : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}^{\mathbb{Z}}$ be the shift map $\tau_{\mathrm{x}}\boldsymbol{\phi} = (\tau_{\mathrm{x}}\boldsymbol{\phi}_{\mathrm{w}})_{\mathrm{w}\in\mathbb{Z}} := (\boldsymbol{\phi}_{\mathrm{x}+\mathrm{w}})_{\mathrm{w}\in\mathbb{Z}}$.

Let us introduce the SDE of interest in three pieces, consisting of the bulk and edge SDEs. Similar to [33], the bulk equation will mirror that in the full-space of [32], and the edge ones correspond to fluxes of independent Langevin dynamics (which mirror the "heat bath" dynamics in open ASEP, for instance). These dynamics are given below, in which $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times [\![1, \mathsf{N}-1]\!]$, and in which we use notation to be explained afterwards:

{eq:currI}
$$\mathrm{d}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{1}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\tau_{\mathrm{x}}\boldsymbol{\phi}_{\mathrm{t}}] + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}}, \tag{2.1}$$

{eq:currII}
$$\mathrm{d}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},0} = \mathsf{N}^{\frac{3}{2}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},1}]\mathrm{d}\mathrm{t} + \mathsf{N}\widetilde{\mathbf{A}}\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},0}, \tag{2.2}$$

{eq:currIII}
$$\mathrm{d}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathsf{N}} = -\mathsf{N}^{\frac{3}{2}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathrm{d}\mathrm{t} + \mathsf{N}\widetilde{\mathbf{B}}\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathsf{N}}. \tag{2.3}$$

- The $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}} \in \mathbb{R}$ are fixed constants.
- The smooth function $\mathscr{U} : \mathbb{R} \to \mathbb{R}$ is fixed and will be referred to as the "potential".
- We use the field-theoretic notation $\boldsymbol{\phi}_{\mathrm{t}} := (\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}})_{\mathrm{x}\in[\![1,\mathsf{N}]\!]}$ with $\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{1/2}(\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}-1})$. It will be convenient to adopt the convention that $\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}$ extends to $\mathrm{x} \in \mathbb{Z}$ (from $\mathrm{x} \in [\![1,\mathsf{N}]\!]$) via the rule $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}] = 0$ if $\mathrm{x} \in \mathbb{Z} \setminus [\![1,\mathsf{N}]\!]$; this will be well-posed by Assumption 2.2 below (which puts assumptions on the potential $\mathscr{U}$).
- For any $\mathfrak{l} \in \mathbb{Z}$, let $\nabla^{\mathbf{X}}_{\mathfrak{l}}$ be the length-$\mathfrak{l}$ discrete-gradient on $\mathbb{Z}$, so for any $\mathsf{f} : \mathbb{Z} \to \mathbb{Z}$, we have $\nabla^{\mathbf{X}}_{\mathfrak{l}}\mathsf{f}_{\mathrm{x}} = \mathsf{f}_{\mathrm{x}+\mathfrak{l}} - \mathsf{f}_{\mathrm{x}}$.
- The processes $\mathrm{t} \mapsto \mathbf{b}_{\mathrm{t},\mathrm{x}}$ are independent standard Brownian motions.
- The nonlinearity $\mathbf{F}$ is defined as follows. For any $\boldsymbol{\phi} \in \mathbb{R}^{\mathbb{Z}}$ (recall our extension of $\boldsymbol{\phi}_{\mathrm{t}}$ to $\mathbb{R}^{\mathbb{Z}}$ above), we set

{eq:nl}
$$\mathbf{F}[\boldsymbol{\phi}] := \tfrac{1}{3}\beta(\mathscr{U}'[\boldsymbol{\phi}_2]\mathscr{U}'[\boldsymbol{\phi}_1] + \mathscr{U}'[\boldsymbol{\phi}_1]\mathscr{U}'[\boldsymbol{\phi}_0] + \mathscr{U}'[\boldsymbol{\phi}_0]\mathscr{U}'[\boldsymbol{\phi}_{-1}]). \tag{2.4}$$

Above, the constant $\beta \neq 0$ is a fixed real number.

The RHS of (2.1)-(2.3) depends only on $\boldsymbol{\phi}_{\mathrm{t}}$, so it will be helpful to derive an equation for this process as well by applying (2.1)-(2.3) to $\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{1/2}(\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}-1}) = -\mathsf{N}^{1/2}\nabla^{\mathbf{X}}_{-1}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}$. We get, with notation explained afterwards, the system below, in which $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times [\![2,\mathsf{N}]\!]$:

{eq:phiI}
$$\mathrm{d}\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^2\Delta\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]\mathrm{d}\mathrm{t} + \mathsf{N}^{\frac{3}{2}}\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} - \sqrt{2}\mathsf{N}\nabla^{\mathbf{X}}_{-1}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}}, \tag{2.5}$$

{eq:phiII}
$$\mathrm{d}\boldsymbol{\phi}_{\mathrm{t},1} = -\mathsf{N}^2\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},1}]\mathrm{d}\mathrm{t} - \mathsf{N}^{\frac{3}{2}}\widetilde{\mathbf{A}}\mathrm{d}\mathrm{t} - \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},0} + \mathsf{N}^2\nabla^{\mathbf{X}}_{1}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},1}]\mathrm{d}\mathrm{t} + \widetilde{\mathbf{F}}[\tau_1\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}\mathrm{d}\mathbf{b}_{\mathrm{t},1},, \tag{2.6}$$

[1]Claude Opus 5.5, with access due to the Claude-for-scientists program.

$$\mathrm{d}\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}} = -\mathsf{N}^2\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathrm{d}\mathrm{t} + \mathsf{N}^{\frac{3}{2}}\widetilde{\mathbf{B}}\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathsf{N}} + \mathsf{N}^2\nabla_{-1}^{\mathbf{X}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathrm{d}\mathrm{t} + \widetilde{\mathbf{F}}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathrm{t} - \sqrt{2}\mathsf{N}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathsf{N}-1}. \tag{2.7}$$

{eq:phiIII}

The notation $\Delta = -\nabla_{-1}^{\mathbf{X}}\nabla_{1}^{\mathbf{X}}$ is the discrete Laplacian on $\mathbb{Z}$, and $\widetilde{\mathbf{F}}[\boldsymbol{\phi}] := \mathbf{F}[\boldsymbol{\phi}] - \mathbf{F}[\boldsymbol{\phi}_{-1}]$ has the form

$$\widetilde{\mathbf{F}}[\boldsymbol{\phi}] = \tfrac{1}{3}\beta(\mathscr{U}'[\boldsymbol{\phi}_2]\mathscr{U}'[\boldsymbol{\phi}_1] - \mathscr{U}'[\boldsymbol{\phi}_{-1}]\mathscr{U}'[\boldsymbol{\phi}_{-2}]). \tag{2.8}$$

{eq:wtnl}

Let us clarify the structure of (2.5)-(2.7). The first line (2.5) is the bulk dynamic, and it agrees with the full-space $\boldsymbol{\phi}$-dynamic in [32] as the "swap" (or "Kawasaki") dynamics; in particular, it is a stochastic conservation law. The first three terms on the RHS of (2.6) correspond to Langevin (or "Glauber") dynamics associated to the potential $\mathscr{U}$ at the left-edge (with linear tilt $\mathsf{N}^{-1/2}\widetilde{\mathbf{A}}$). The other terms in (2.6) correspond to the aforementioned stochastic conservation law dynamics "at the points $1, 2$". The SDE (2.7) has the same structure as (2.6) after taking a mirror image (reversing orientation on $\mathbb{Z}$) and at the right-edge.

As stated in the introduction, convergence to the open KPZ equation will go through convergence to the open SHE for an exponentiated version of $\mathbf{j}^{\mathsf{N}}$. To define this exponentiation (and the relevant constants), we will need to introduce the following family of probability measures.

{definition:gc}

Definition 2.1. Take any $\sigma \in \mathbb{R}$. We let the associated *grand-canonical ensemble* (or *grand-canonical measure*) of density σ be the following product measure on $\mathbb{R}^{\mathbb{Z}}$:

$$\mathbb{P}^{\sigma}[\mathrm{d}\boldsymbol{\phi}] := \prod_{\mathrm{x}\in\mathbb{Z}} \boldsymbol{\mathcal{Z}}_{\sigma,\mathscr{U}}^{-1}\mathbf{Exp}(-\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}] + \upsilon_{\sigma}\boldsymbol{\phi}_{\mathrm{x}})\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}}. \tag{2.9}$$

{eq:gcI}

Above, $\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}}$ denotes the Lebesgue measure on $\boldsymbol{\phi}_{\mathrm{x}} \in \mathbb{R}$. The constant $\upsilon_{\sigma} \in \mathbb{R}$ is chosen so that $\mathbb{E}^{\sigma}\boldsymbol{\phi}_{\mathrm{x}} = \sigma$ for any $\mathrm{x} \in \mathbb{Z}$. We will sometimes consider projections (or pushforwards) of $\mathbb{P}^{\sigma}$ onto $\mathbb{R}^{\mathbb{I}}$ for various subsets $\mathbb{I} \subseteq \mathbb{Z}$, but we may not include this dependence in the notation (without lack of clarity, since $\mathbb{P}^{\sigma}$ is a product measure).

We clarify that the grand-canonical ensembles are generally *not* invariant measures for $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$ unless $\widetilde{\mathbf{A}} = -\widetilde{\mathbf{B}}$ (so that the densities at both edges are the same). However, they are still "invariant for the bulk dynamics" (that is, the full-space version of (2.5)-(2.7)). It is ultimately for this reason that these grand-canonical measures are relevant for the limiting constants in the limit open KPZ equation.

As promised earlier, we now define the following *microscopic Cole-Hopf transform*:

$$\mathbf{Z}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}} := \mathbf{Exp}(\lambda\mathbf{j}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}} - \lambda\mathscr{R}_{\lambda}\mathrm{t}). \tag{2.10}$$

{eq:ch}

This is defined for all $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times [\![0,\mathsf{N}]\!]$. The constants $\lambda, \mathscr{R}_{\lambda}$ are defined by the formulas

$$\lambda := \tfrac{\beta}{\alpha} \quad \text{and} \quad \mathscr{R}_{\lambda} := \tfrac{1}{12}\lambda^3\mathbb{E}^0(\mathscr{U}'[\boldsymbol{\phi}_1]\boldsymbol{\phi}_1^3) - \tfrac{1}{6}\beta\lambda^2. \tag{2.11}$$

{eq:rg}

Above, β is from (2.4) (and is assumed to be non-zero), whereas α is instead defined by

$$\alpha := \partial_{\sigma}\mathbb{E}^{\sigma}\mathscr{U}'[\boldsymbol{\phi}_1]|_{\sigma=0}. \tag{2.12}$$

{eq:alpha}

We clarify that the point 1 is not important in (2.11)-(2.12), because $\mathbb{P}^{\sigma}$ is a homogeneous product measure. It is shown in Appendix A.2 of [32] that $\alpha > 0$.

2.2. **The results.** Before we record the main results of this paper, we introduce the two key assumptions, the first of which focuses on general (and standard) properties of the potential and the second of which concerns the allowed initial data.

{assump:potent

Assumption 2.2. *There exists a constant $\Lambda \geqslant 1$ such that $\Lambda^{-1} \leqslant \mathscr{U}''[\mathrm{a}] \leqslant \Lambda$ for all $\mathrm{a} \in \mathbb{R}$. Moreover, $\upsilon_0 = 0$.*

The last piece $\upsilon_0 = 0$ is for convenience and is without loss of generality via shifting $\mathscr{U}$ by a linear function.

{assump:data}

Assumption 2.3. *Let $\mathfrak{P}$ denote the density for the law of the initial data of $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$ with respect to (the projection onto $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$ of) $\mathbb{P}^0$; see Definition 2.1. There exists a sequence $\gamma_{\mathsf{N}} \to 0$ such that*

$$\mathbb{E}^0\mathfrak{P}\log\mathfrak{P} \leqslant \gamma_{\mathsf{N}}\log\mathsf{N}. \tag{2.13}$$

{eq:dataI}

Let us clarify Assumption 2.3. The LHS of (2.13) is the relative entropy of $\mathfrak{P}\mathrm{d}\mathbb{P}^0$ with respect to $\mathrm{d}\mathbb{P}^0$. The fact that the RHS of (2.13) can diverge in the large-N limit is the statement of allowing non-equilibrium initial data; in particular, this is what allows us to ultimately derive the open KPZ equation with essentially initial data which it can make sense of; this includes the invariant measure initial data discussed in Section 1.

We emphasize the peculiar nature of Assumption 2.3. The relative entropy on the LHS of (2.13) is with respect to a *non-invariant* measure for the $\mathrm{t} \mapsto \phi_{\mathrm{t}}$ dynamics. Ultimately, however, we will show that *these dynamics are a perturbation of ones whose invariant measure is itself a type of perturbation of* $\mathbb{P}^0$, as discussed in Section 1.

We now present the main result. Besides Assumptions 2.2 and 2.3, the result only requires standard Bertini-Giacomin-style estimates on the initial data. Before we write the result, we first introduce important notation for this paper. We write $\mathrm{a} = \mathrm{O}(\mathrm{b})$ (or equivalently, $\mathrm{a} \lesssim \mathrm{b}$) if $|\mathrm{a}| \leqslant \mathrm{C}|\mathrm{b}|$ for some $\mathrm{C} \geqslant 0$. The parameters on which C depends will be listed as subscripts for O or $\lesssim$.

{theorem:main}

Theorem 2.4. *Suppose that in addition to Assumptions 2.2 and 2.3, we have:*

(1) Fix any $p \geqslant 1$ and $\zeta > 0$. We have

{eq:mainI}
$$\max_{\mathrm{x} \in [\![0, \mathsf{N}]\!]} \mathbb{E} |\mathbf{Z}^{\mathsf{N}}_{0,\mathrm{x}}|^{2p} \lesssim_p 1 \quad \text{and} \quad \max_{\substack{\mathrm{x}, \mathrm{y} \in [\![0, \mathsf{N}]\!] \\ \mathfrak{l} := \mathrm{y} - \mathrm{x} \neq 0}} (\mathsf{N}^{-p+p\zeta} |\mathfrak{l}|^{p-p\zeta})^{-1} \mathbb{E} |\nabla^{\mathbf{X}}_{\mathfrak{l}} \mathbf{Z}^{\mathsf{N}}_{0,\mathrm{x}}|^{2p} \lesssim_{p,\zeta} 1. \tag{2.14}$$

(2) The initial data $\mathbf{Z}^{\mathsf{N}}_{0,\cdot}$ is independent of the Brownian motions in (2.1)-(2.3). In addition, the re-scaled initial data $\mathrm{X} \mapsto \mathbf{Z}^{\mathsf{N}}_{0,\mathsf{N}\mathrm{X}}$, which is defined on $\mathrm{X} \in [0,1]$ upon extension-via-linear-interpolation from $\mathrm{X} \in \mathsf{N}^{-1}[0,1]$, converges uniformly in $[0,1]$ to a continuous non-zero function $\mathrm{X} \mapsto \mathbf{Z}_{0,\mathrm{X}}$, in probability.

There is a coupling between $\mathbf{Z}^{\mathsf{N}}$ and the solution $(\mathrm{t}, \mathrm{X}) \mapsto \mathbf{Z}_{\mathrm{t},\mathrm{X}}$ to (1.4)-(1.6) with initial data $\mathbf{Z}_{0,\cdot}$ and parameters $\mathbf{A}, \mathbf{B}$ given below such that $\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathsf{N}\mathrm{X}} - \mathbf{Z}_{\mathrm{t},\mathrm{X}} \to 0$ locally uniformly on $[0,\infty) \times [0,1]$, in probability:

{eq:mainII}
$$\mathbf{A} := -\tfrac{1}{2} \lambda \alpha^{-1} - \alpha^{-1} \widetilde{\mathbf{A}}, \tag{2.15}$$

{eq:mainIII}
$$\mathbf{B} := \tfrac{1}{2} \lambda \alpha^{-1} + \alpha^{-1} \widetilde{\mathbf{B}}. \tag{2.16}$$

We now illustrate the generic nature of the assumptions in Theorem 2.4. Consider any initial data of interest for the limiting open KPZ equation; we can assume that it is deterministic by conditioning on it. Now, sample $\mathbf{j}^{\mathsf{N}}$ first by $\mathbb{P}^0$, and then conditioning on it to be within $\varepsilon(\mathsf{N})$ of the data of interest. Here, $\varepsilon(\mathsf{N}) \to 0$ slowly. It follows by standard support properties for Wiener measure, more or less, that if $\varepsilon(\mathsf{N}) \to 0$ slowly, then the probability of the event we condition on vanishes slowly as well. Thus, the slightly randomized initial data, which converges to the data of interest since $\varepsilon(\mathsf{N}) \to 0$, satisfies (2.13). We illustrate the importance that the RHS of (2.13) is allowed to diverge in the large-N limit. Otherwise, we would not be able to take $\varepsilon(\mathsf{N}) \to 0$ at all, never-mind slowly.

2.3. **Plan for the paper.** In Section 3, we reduce the proof for Theorem 2.4 to Proposition 3.7 (which presents a form of the SDE for the evolution of (2.10) that is amenable to analysis) and Proposition 3.9 (which compares (2.10) to a lattice SHE). In Section 4, we prove Proposition 3.7. In Section 5, we develop analysis via relative entropy of the law of the ϕ_{t}-process, which includes the construction of globally perturbed dynamics we described briefly in Section 1, as well as consequences of our construction and estimates. In Section 6, we provide the main fluctuation estimates (also alluded to in Section 1), which usually go by *Boltzmann-Gibbs principles*. In Section 7, we use the Boltzmann-Gibbs principles and other stochastic estimates from Section 5 to prove Proposition 3.9, which completes the proof of Theorem 2.4. Finally, the appendix collects a number of technical discussions.

ction:outline}

3. Outline of the proof

In this section, we list the key ingredients needed to deduce Theorem 2.4. These ingredients consist of the evolution equation for the microscopic Cole-Hopf transform (2.10) (see Proposition 3.7) and a comparison of (2.10) to a lattice version of (1.4)-(1.6) (Proposition 3.9). After presenting these, we will assume and use them to prove Theorem 2.4. The rest of this paper is dedicated to proving Propositions 3.7 and 3.9.

3.1. **Evolution equation.** We start by borrowing notions from the half-space setting in [33] that will be used to package our SDE for the microscopic Cole-Hopf transform. Indeed, as noted in Section 1, this SDE of interest is obtained by gluing together two copies of the half-space SDE, since the evolution of the microscopic Cole-Hopf transform is local in space. We start with the following class of *bulk admissible* functions, which are essentially fluctuations with respect to the bulk dynamics.

ulkadmissible}

Definition 3.1. First, for any $\mathfrak{k} = 0, 1, 2$, let the orthogonal jet space $\mathrm{Jet}_{\mathfrak{k}}^{\perp}$ consist of all $\mathsf{F} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ such that

{eq:jet}
$$\partial_{\sigma}^{\ell} \mathbb{E}^{\sigma} \mathsf{F}[\boldsymbol{\phi}]|_{\sigma=0} = 0 \quad \text{for all } \ell = 0, \dots, \mathfrak{k}. \tag{3.1}$$

Let us now say that a function $\mathfrak{q} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ is *bulk admissible with respect to a point* $\mathrm{x} \in [\![0, \mathsf{N}]\!]$ if the following conditions are satisfied:

(1) We have $\mathfrak{q} = \mathsf{N}^{-1+\mathfrak{k}/2}\mathfrak{f}$, where $\mathfrak{f} \in \mathrm{Jet}_{\mathfrak{k}}^{\perp}$, for some $\mathfrak{k} = 0, 1, 2$.
(2) We have exactly one of the following two scenarios:
   - For any $\boldsymbol{\phi} \in \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, the value $\mathfrak{f}[\boldsymbol{\phi}]$ depends only on $\boldsymbol{\phi}_{\mathrm{y}}$ for $\mathrm{y} \in [\![\mathrm{x}+1, \mathrm{x}+\mathfrak{l}]\!] \cap [\![1, \mathsf{N}]\!]$ for some $0 < \mathfrak{l} \lesssim 1$.
   - For any $\boldsymbol{\phi} \in \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, the value $\mathfrak{f}[\boldsymbol{\phi}]$ depends only on $\boldsymbol{\phi}_{\mathrm{y}}$ for $\mathrm{y} \in [\![\mathrm{x}-\mathfrak{l}, \mathrm{x}]\!] \cap [\![1, \mathsf{N}]\!]$ for some $0 < \mathfrak{l} \lesssim 1$.

   In words, the "support" of $\mathfrak{f}$ has $\mathrm{O}(1)$ size, and it is either to the right or to the left of x, roughly speaking. In the former case, we say that $\mathfrak{q}$ is *forwards-oriented*; in the latter, we say that $\mathfrak{q}$ is *backwards-oriented*.
(3) The function $\mathfrak{f}$ satisfies the following estimate uniformly in $\boldsymbol{\phi} \in \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, where $0 \leqslant \mathrm{C} = \mathrm{O}(1)$:

{eq:bulkadmissi
$$|\mathfrak{f}[\boldsymbol{\phi}]| + \sup_{\mathrm{y} \in [\![1,\mathsf{N}]\!]} |\partial_{\boldsymbol{\phi}_{\mathrm{y}}} \mathfrak{f}[\boldsymbol{\phi}]| \lesssim 1 + \sum_{|\mathrm{w}-\mathrm{x}| \lesssim 1} |\boldsymbol{\phi}_{\mathrm{w}}|^{\mathrm{C}}. \tag{3.2}$$

We now briefly clarify Definition 3.1. First, the jet spaces can be seen as generalizations of the Wiener chaos for non-Gaussian potentials $\mathscr{U}$. As in [16], for instance, having a vanishing projection onto higher degree Wiener chaos suggests smallness. Indeed, local equilibrium suggests the replacement $\mathsf{F} \mapsto \mathbb{E}^{\sigma}\mathsf{F}$, where σ is the $\boldsymbol{\phi}$-density in a small macroscopic interval of size $\mathrm{o}(\mathsf{N})$. Taylor expansion then yields $\mathbb{E}^{\sigma}\mathsf{F} = \mathrm{O}(\sigma^{\mathfrak{k}+1})$ if $\mathsf{F} \in \mathrm{Jet}_{\mathfrak{k}}^{\perp}$, while the typical CLT-cancellation heuristic suggests that $\sigma \lesssim \varepsilon \mathsf{N}^{-1/2}$ for some $\varepsilon > 0$ small but fixed. In fact, this suggests that bulk admissible functions are $\lesssim \mathsf{N}^{-3/2}$ in size. We emphasize the bulk nature of this paragraph; indeed, the $\mathbb{P}^{\sigma}$ measures are all invariant for the bulk dynamics.

Now, we clarify that the support assumptions for bulk admissible functions are important as we will eventually need to understand certain "dynamical correlations" between said functions and j^{N} itself. However, the dynamics of j^{N} at point x depend only on the spin-exchange dynamics along the bond $\{\mathrm{x}, \mathrm{x}+1\}$; said bond dynamics do not play a role in the dynamics of a function that is bulk admissible with respect to x by construction. Lastly, the estimate (3.2) is a technical a priori bound.

We now introduce the following class of *edge admissible* functions, which are similarly fluctuations in some sense, but with respect to the edge dynamics. We clarify this particular point more precisely below.

{definition:edg
Definition 3.2. We say that $\mathfrak{b} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ is *edge-admissible* if the following conditions are all met:

(1) We have $\mathbb{E}^{0}\mathfrak{b}[\boldsymbol{\phi}] = 0$.
(2) We have exactly one of the following two scenarios:
   - For any $\boldsymbol{\phi} \in \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, the value $\mathfrak{b}[\phi]$ depends only on $\boldsymbol{\phi}_{\mathrm{y}}$ for $\mathrm{y} \in [\![1, \mathrm{O}(1)]\!]$.
   - For any $\boldsymbol{\phi} \in \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, the value $\mathfrak{b}[\phi]$ depends only on $\boldsymbol{\phi}_{\mathrm{y}}$ for $\mathrm{y} \in [\![\mathsf{N} - \mathrm{O}(1), \mathsf{N}]\!]$.
(3) If the first case in (2) holds, then the function $\mathfrak{b}$ satisfies the estimate (3.2) with $\mathrm{x} = 0$ therein. If the second case in (2) holds, then the function $\mathfrak{b}$ satisfies the estimate (3.2) with $\mathrm{x} = \mathsf{N}$ therein.

Let us clarify Definition 3.2; in a nutshell, these are fluctuations with respect to the *edge dynamics*. We note, however, that it is only $\mathbb{P}^{\sigma}$ for $\sigma = \mathrm{O}(\mathsf{N}^{-1/2})$ (depending on $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$ in (2.2)-(2.3)) that is invariant for the individual edge dynamics (2.6) and (2.7). Indeed, each edge dynamic fixes a typical density depending on $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$. However, we will only require power-saving estimates for (time-averages of) $\mathfrak{b}$ which have the form $\lesssim \mathsf{N}^{-\rho}$ for some $\rho > 0$ fixed. Thus, it will not matter if we impose the condition $\mathbb{E}^{0}\mathfrak{b}[\boldsymbol{\phi}] = 0$ or $\mathbb{E}^{\sigma}\mathfrak{b}[\boldsymbol{\phi}] = 0$ for $\sigma \ll 1$.

Let us now introduce notation for a family of deterministic heat kernels that will be important throughout the paper for stating and analyzing the SDE for the microscopic Cole-Hopf transform, as well as introducing spatial averages for bulk admissible functions.

{definition:hea
Definition 3.3. Take any $\mathbf{a}, \mathbf{b} \in \mathbb{R}$. We let $\mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}}$ be a function of $(\mathrm{s}, \mathrm{t}, \mathrm{x}, \mathrm{y}) \in [0,\infty) \times [0,\infty) \times [\![0, \mathsf{N}]\!] \times [\![0, \mathsf{N}]\!]$, restricted to the time-slice $\mathrm{s} \leqslant \mathrm{t}$, which satisfies the following PDE for all $\mathrm{t} \geqslant \mathrm{s}$, in which all discrete differential operators act on x (so that (s, y) are both treated as fixed variables):

{eq:heatI}
$$\partial_{\mathrm{t}} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} = \mathscr{T}_{\mathsf{N},\mathbf{a},\mathbf{b}} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} := (\alpha \mathsf{N}^2 + \tfrac{1}{4}\mathsf{N}\lambda) \Delta_{\mathbf{a},\mathbf{b}} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} \quad \text{and} \quad \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{s},\mathrm{x},\mathrm{y}} = \mathbf{1}_{\mathrm{x}=\mathrm{y}}. \tag{3.3}$$

Above, $\Delta_{\mathbf{a},\mathbf{b}}$ denotes the Robin Laplacian on $[\![0,\mathsf{N}]\!]$, so $\Delta_{\mathbf{a},\mathbf{b}}\mathsf{f}$ restricted to $[\![0,\mathsf{N}]\!]$ is the usual discrete Laplacian (as in (2.5)) after we extend $\mathsf{f}:[\![0,\mathsf{N}]\!]\to\mathbb{R}$ to $[\![-1,\mathsf{N}+1]\!]$ via the Robin rules $\mathsf{f}_0-\mathsf{f}_{-1}=\mathsf{N}^{-1}\mathbf{a}\lambda\mathsf{f}_0$ and $\mathsf{f}_{\mathsf{N}+1}-\mathsf{f}_{\mathsf{N}}=\mathsf{N}^{-1}\mathbf{b}\lambda\mathsf{f}_{\mathsf{N}}$. See (2.12) and (2.11) for the α,λ constants here. Next, we let $\mathrm{t}\mapsto\mathbf{e}^{\mathrm{t}\mathscr{T}_{\mathsf{N},\mathbf{a},\mathbf{b}}}$ be the associated semigroup, so that for any $\mathsf{f}:[\![0,\mathsf{N}]\!]\to\mathbb{R}$, we have

{eq:heatII}
$$[\mathbf{e}^{\mathrm{t}\mathscr{T}_{\mathsf{N},\mathbf{a},\mathbf{b}}}(\mathsf{f})]_{\mathrm{x}}=[\mathbf{e}^{\mathrm{t}\mathscr{T}_{\mathsf{N},\mathbf{a},\mathbf{b}}}(\mathsf{f}_\cdot)]_{\mathrm{x}}:=\sum_{\mathrm{y}\in[\![0,\mathsf{N}]\!]}\mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{0,\mathrm{t},\mathrm{x},\mathrm{y}}\cdot\mathsf{f}_{\mathrm{y}}. \tag{3.4}$$

Finally, we introduce a smoothed version of the microscopic Cole-Hopf transform $\mathbf{Z}^{\mathsf{N}}$ and the associated ratio:

{eq:heatIII}
$$\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}:=[\mathbf{e}^{\mathsf{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N},\mathbf{A},\mathbf{B}}}(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})]_{\mathrm{x}}\quad\text{and}\quad\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}:=\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\cdot(\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}})^{-1}. \tag{3.5}$$

Above, we used the time-scale $\mathsf{t}_{\mathsf{N}}:=\mathsf{N}^{-\delta_{\mathbf{S}}}$, where $\delta_{\mathbf{S}}>0$ is a small parameter that does *not* vanish in the large-N limit. The boundary parameters $\mathbf{A},\mathbf{B}$ are from (2.15)-(2.16).

We briefly clarify the smoothing in (3.5). The reason we use $\mathscr{T}_{\mathsf{N},\mathbf{A},\mathbf{B}}$ in particular is because it will be important for our choice of smoothing to be "compatible" with the evolution of $\mathbf{Z}^{\mathsf{N}}$ itself, which involves $\mathscr{T}_{\mathsf{N},\mathbf{A},\mathbf{B}}$ specifically in view of the emergent parameters $\mathbf{A},\mathbf{B}$ from Theorem 2.4. Let us clarify that $\mathsf{N}\mathsf{t}_{\mathsf{N}}^{1/2}$, namely the averaging scale, is just below the macroscopic scale, so that the smoothing allows us to now (discrete) differentiate $\mathbf{S}^{\mathsf{N}}$, with each derivative costing only a small power of $\mathsf{N}^{\delta_{\mathbf{S}}/2}$. At the same time, since $\mathsf{N}\mathsf{t}_{\mathsf{N}}^{1/2}=\mathsf{N}^{1-\delta_{\mathbf{S}}/2}$ is small, we will be able to later show that $\mathbf{S}^{\mathsf{N}}\approx\mathbf{Z}^{\mathsf{N}}$ (i.e. $\mathbf{R}^{\mathsf{N}}\approx 1$) in a sufficiently strong sense due to regularity estimates for $\mathbf{Z}^{\mathsf{N}}$ (which mirror regularity for (1.4)-(1.6)).

We now introduce the averaging mechanisms for analyzing bulk and edge admissible functions. We start with the space-time averaging for bulk admissible ones, which we explain afterwards.

bulkaveraging}

Definition 3.4. First, let us introduce the time-scale $\mathsf{t}_{\mathbf{Av}}:=\mathsf{N}^{-2/3-\mathrm{C}_0\delta_{\mathbf{S}}}$, in which $\mathrm{C}_0>0$ is large but fixed, and $\delta_{\mathbf{S}}>0$ denotes the small constant from Definition 3.3. Let us also introduce the length-scale $\mathsf{l}_{\mathbf{Av}}=\mathsf{N}^{1-3\delta_{\mathbf{S}}/2}$, as well as $\tau_{\mathbf{Av}}:=\mathsf{N}^{-2}\mathsf{l}_{\mathbf{Av}}^2=\mathsf{N}^{-3\delta_{\mathbf{S}}}$ as the associated diffusive time-scale.

Take a collection $\mathscr{Q}=(\mathsf{q}_{\mathrm{x}})_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}$ of functions $\mathbb{R}^{[\![0,\mathsf{N}]\!]}\to\mathbb{R}$. Define the space-time average

bulkaveragingI}
$$\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}}_{\mathrm{t},\mathrm{x}}:=\mathsf{t}_{\mathbf{Av}}^{-1}\int_0^{\mathsf{t}_{\mathbf{Av}}}[\mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}(\mathbf{1}_{|\mathrm{x}-\cdot|\lesssim(\log\mathsf{N})^2\mathsf{l}_{\mathbf{Av}}}\mathsf{q}_\cdot[\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{s},\cdot})]_{\mathrm{x}}\mathrm{d}\mathrm{s}. \tag{3.6}$$

That is, we average (backwards) in time at scale $\mathsf{t}_{\mathbf{Av}}$ and in space at scale $\approx\mathsf{l}_{\mathbf{Av}}$; the cutoff on the RHS is technical and harmless, as the main benefit behind it is to localize the support of what is inside the semigroup operator as a function of $\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}$. (We emphasize the use of $\mathscr{T}_{\mathsf{N},0,0}$; it is for these parameters that the corresponding heat kernel is a probability measure in either spatial-variable on $[\![0,\mathsf{N}]\!]$, which will be important for us.) In (3.6), we drop all s for which $\mathrm{t}-\mathrm{s}\leqslant 0$ (i.e. we avoid times for which the processes at hand are not defined).

We now consider the following two cases:

(1) Suppose that q_{x} is forwards-oriented with respect to x for all x (see Definition 3.1). We define

kaveragingIIa}
$$\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}}_{\mathrm{t},\mathrm{x}}:=\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}}_{\mathrm{t},\mathrm{x}}\cdot(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}_-})^{-1},\quad\text{where }\mathrm{x}_-:=\max\{0,\mathrm{x}-(\log\mathsf{N})^3\mathsf{l}_{\mathbf{Av}}\}. \tag{3.7}$$

That is, we factor from $\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}}$ a copy of $\mathbf{Z}^{\mathsf{N}}$ to the left, i.e. the direction opposite of orientation of q_{x}.

(2) Suppose that q_{x} is backwards-oriented with respect to x for all x (see Definition 3.1). We define

kaveragingIIb}
$$\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}}_{\mathrm{t},\mathrm{x}}:=\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}}_{\mathrm{t},\mathrm{x}}\cdot(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}_+})^{-1},\quad\text{where }\mathrm{x}_+:=\min\{\mathsf{N},\mathrm{x}+(\log\mathsf{N})^3\mathsf{l}_{\mathbf{Av}}\}. \tag{3.8}$$

Finally, let $\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}}$ denote the quantity (3.6) upon setting $\mathsf{t}_{\mathbf{Av}}=0$, i.e. remove the time-averaging and set $\mathrm{s}=0$. We define $\mathbf{Av}^{\mathbf{X},\mathscr{Q}}$ by (3.7)-(3.8) (depending on the orientation of q_{x}) upon replacing $\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}}$ therein by $\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}}$.

We will now briefly clarify the use of $\mathrm{x}_\pm$ in (3.7)-(3.8). Suppose that q_{x} is forwards-oriented. Given any point w, the factor $\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{s},\mathrm{w}}\cdot(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}_-})^{-1}$ involves $\boldsymbol{\phi}_{\mathrm{r},\mathrm{y}}$-variables for $\mathrm{r}\in\{\mathrm{t}-\mathrm{s},\mathrm{t}\}$ and $\mathrm{y}\leqslant\mathrm{w}$. This follows from (2.10) and the rule $\boldsymbol{\phi}_{\mathrm{t},\mathrm{y}}=\mathsf{N}^{1/2}(\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{y}}-\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{y}-1})$. It is important for us that the function q_{w} does not depend on any of these spins (which follows from the assumption that it is forwards-oriented). Otherwise, we will have estimate dynamical correlations between $\mathsf{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]$ and $\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}$, as noted after Definition 3.1.

We now introduce the averaging scheme that we will need for edge admissible functions. As these functions are supported near the edge, there will not be any spatial averaging like in Definition 3.4. However, this will be

unimportant, because we only need *some* power-saving for edge admissible functions. (For ultimately the same reason, there will also be no need to use $\mathrm{x}_\pm$-notation as in Definition 3.4.)

{definition:edg

Definition 3.5. Suppose that $\mathfrak{b}$ is edge admissible. Set $\mathfrak{t}_\partial = \mathsf{N}^{-3/2}$. Define

$$\boldsymbol{\Upsilon}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}} := \mathfrak{t}_\partial^{-1} \int_0^{\mathfrak{t}_\partial} \mathfrak{b}[\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}] \mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{s},\mathrm{x}} \mathrm{d}\mathrm{s} \quad \text{and} \quad \mathbf{Av}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}} := \boldsymbol{\Upsilon}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}} \cdot (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}})^{-1}. \tag{3.9}$$

Finally, we introduce notation for a time-averaged gradient, which will encode the errors behind implementing a time-average (in both Definitions 3.4 and 3.5).

{definition:tim

Definition 3.6. Fix any time-scale $\mathfrak{t} \geqslant 0$ and $\mathrm{g} : [0,\infty) \to \mathbb{R}$. Define

$$\nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}} \mathrm{g}_{\mathrm{t}} := \mathbf{1}_{\mathfrak{t}>0} \mathfrak{t}^{-1} \int_0^{\mathfrak{t}} (\mathrm{g}_{\mathrm{t}} - \mathrm{g}_{\mathrm{t}-\mathrm{s}}) \mathrm{d}\mathrm{s}.$$

Again, we drop all s in the integral above for which $\mathrm{t}+\mathrm{s} \leqslant 0$.

We are ready to present the final form of the SDE of interest for the microscopic Cole-Hopf transform. Before we do so, we clarify its structure. This SDE is a lattice (1.4)-(1.6) (with mesoscopic-scale smoothing in space as in (3.5)), with the following additional corrections:

(1) The first is a set of space-time averaged bulk fluctuations (as in Definition 3.4) that are integrated against a nice kernel (ultimately coming the smoothing kernel in (3.5)).
(2) The second is a set of time-averaged edge fluctuations (as in Definition 3.5), again integrated in space against another nice kernel.
(3) The third is a set of space-averaged bulk fluctuations, once again integrated against a nice kernel, that we take a time-averaged gradient of. This encodes the error in replacing a space-average by a space-time average.
(4) The fourth is time-averaged gradients acting on the space-integral of a nice kernel against un-averaged edge fluctuations, which captures the error in replacing an edge fluctuation by its time-average.
(5) The fifth is an error term which is small for (essentially) deterministic reasons.

Finally, let us clarify that the errors in introducing spatial-averages for un-averaged bulk fluctuations, which are not described in the list above, are encoded by spatial gradients that we then move onto the nice kernels. We bake these gradients into the kernels, which is why they do not explicitly appear in the equation below. Moreover, we clarify the relevance of terms $\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}$ appearing below since this product is equal to $\mathbf{Z}^{\mathsf{N}}$ itself by (3.5).

{prop:sde}

Proposition 3.7. *Recall the notation from Definitions 3.1, 3.2, 3.3, 3.4, 3.5, and 3.6. We have the following* SDE *for all* $(\mathrm{t},\mathrm{x}) \in [0,\infty) \times [\![0,\mathsf{N}]\!]$*, in which we use notation to be explained afterwards:*

$$\mathrm{d}\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathscr{T}_{\mathsf{N}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}} (\sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\cdot} \mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} \tag{3.10}$$ {eq:sdeI1}

$$+ \sum_{\ell \in [\![1,\mathrm{L}_1]\!]} \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N} \cdot \mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathrm{d}\mathrm{t} \tag{3.11}$$ {eq:sdeI2}

$$+ \sum_{\ell \in [\![1,\mathrm{L}_1]\!]} \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N} \cdot \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_{\mathbf{Av}}} (\mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}) \mathrm{d}\mathrm{t} \tag{3.12}$$ {eq:sdeI3}

$$+ \sum_{\ell \in [\![1,\mathrm{L}_2]\!]} \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N} \mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathrm{d}\mathrm{t} + \sum_{\ell \in [\![1,\mathrm{L}_2]\!]} \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N} \cdot \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial} (\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}] \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}) \mathrm{d}\mathrm{t}. \tag{3.13}$$ {eq:sdeI4}

(1) The term Err *is the following (possibly random) linear integral operator:*

$$\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}} := \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \mathbf{E}^{\mathsf{N}}_{\mathrm{t},\mathrm{x},\mathrm{w}} \cdot \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}. \tag{3.14}$$ {eq:sdeII1}

Above, the kernel $\mathbf{E}^{\mathsf{N}}$ *satisfies the following deterministic estimate for some* $0 \leqslant \mathrm{C} \lesssim 1$*:*

$$\mathbf{E}^{\mathsf{N}}_{\mathrm{t},\mathrm{x},\mathrm{w}} \lesssim (\mathsf{N}^{-\frac{6}{5}} + \mathsf{N}^{-\frac{1}{3}} \mathbf{1}_{\mathrm{dist}(\mathrm{w},\{0,\mathsf{N}\}) \lesssim 1}) \cdot \Big\{ 1 + \sum_{|\mathrm{z}-\mathrm{w}| \lesssim 1} |\boldsymbol{\phi}_{\mathrm{t},\mathrm{z}}|^{\mathrm{C}} \Big\}. \tag{3.15}$$ {eq:sdeII2}

(The exponent $\frac{6}{5}$ is unimportant, just that it is strictly larger than 1. Similarly, we only need $\frac{1}{3} > 0$.)

(2) *The kernels $\mathcal{K}^{(\ell)}$ are deterministic. Moreover, we have the following pointwise estimate, in which δ_{S} is the small constant from Definition 3.3:*

$$|\mathcal{K}^{(\ell)}_{\mathrm{x},\mathrm{w}}| \lesssim \mathsf{N}^{-1+\mathrm{O}(\delta_{\mathrm{S}})}. \tag{3.16}$$ {eq:sdeIII1}

The kernels $\mathcal{K}^{\partial,\ell}$ are deterministic as well, and they satisfy the following analogous but localized estimate:

$$|\mathcal{K}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}| \lesssim \mathsf{N}^{-1+\mathrm{O}(\delta_{\mathrm{S}})} \cdot \mathbf{1}_{\mathrm{dist}(\mathrm{w},\{0,\mathsf{N}\})\lesssim 1}. \tag{3.17}$$ {eq:sdeIII2}

(3) *The constants $\mathrm{L}_1, \mathrm{L}_2 = \mathrm{O}(1)$ are positive integers.*

(4) *For any $\ell \in [\![1, \mathrm{L}_1]\!]$, the collection $\mathcal{Q}_\ell := (\mathfrak{q}_{\ell,\mathrm{x}})_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}$ are each bulk-admissible with respect to x, respectively. Moreover, either every function in $\mathcal{Q}_\ell$ is forwards-oriented, or every function therein is backwards-oriented. In the former case, we set $\pm = -$ in (3.11)-(3.12); in the latter case, we set $\pm = +$ in (3.11)-(3.12).*

(5) *For any $\ell \in [\![1, \mathrm{L}_2]\!]$, the function $\mathfrak{b}_\ell$ is edge admissible.*

3.2. **Comparison to lattice** SHE**.** Given Proposition 3.7, we anticipate $\mathbf{S}^{\mathsf{N}}$ to be close to the solution $\mathbf{Q}^{\mathsf{N}}$ of

$$\mathrm{d}\mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathscr{T}_{\mathsf{N}}\mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}}. \tag{3.18}$$ {eq:qsde}

We will equip (3.18) with the same initial data as $\mathbf{Z}^{\mathsf{N}}_{0,\cdot}$.

Indeed, all but the first two terms on the RHS of (3.10)-(3.13) are anticipated to have vanishing contribution in the large-N limit, and the mesoscopic smoothing hitting the noise is also anticipated to wash away in the large-N limit since all $\mathbf{S}^{\mathsf{N}}, \mathbf{Z}^{\mathsf{N}}, \mathbf{Q}^{\mathsf{N}}$ should all have macroscopic-scale regularity. Ultimately, we have the following result, which is preluded by a notion of high probability to be used throughout the paper.

{definition:hp}

Definition 3.8. We say an event $\mathscr{E}$ holds with *high probability* (or is high probability) if $\mathbb{P}(\mathscr{E}) = 1 - \mathrm{o}(1)$.

{prop:compare}

Proposition 3.9. *Recall notation from (3.5), and fix any $\mathrm{T} > 0$. With high probability, we have*

$$\sup_{\mathrm{t}\in[0,\mathrm{T}]}\max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - 1| \lesssim \mathsf{N}^{-\frac{1}{6}\delta_{\mathrm{S}}}. \tag{3.19}$$ {eq:compareI}

Moreover, with high probability, we also have the following for some $\gamma > 0$:

$$\sup_{\mathrm{t}\in[0,\mathrm{T}]}\max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\gamma}. \tag{3.20}$$ {eq:compareII}

In principle, Proposition 3.9 is all we need for the proof of Theorem 2.4. However, Proposition 3.7 is crucial for the proof of Proposition 3.9 (and gives us the basis for why Proposition 3.9 should even be true), hence our presentation on it. We are now in position to show Theorem 2.4.

Proof of Theorem 2.4. Standard stability of (1.4)-(1.6) under space-discretization shows that Theorem 2.4 holds true if we replace $\mathbf{Z}^{\mathsf{N}}$ therein by $\mathbf{Q}^{\mathsf{N}}$. (For said stability, see [15], which treats the special case $\mathbf{A}, \mathbf{B} = 0$ but whose argument applies more generally.) By (3.20), we also deduce the same for $\mathbf{S}^{\mathsf{N}}$ in place of $\mathbf{Q}^{\mathsf{N}}$. In particular, this gives $\|\mathbf{S}^{\mathsf{N}}\|_{\mathrm{L}^\infty([0,1]),\ell^\infty([\![0,\mathsf{N}]\!])} \lesssim \mathsf{N}^{\varepsilon}$ with high probability for small but fixed $\varepsilon > 0$. This last estimate combined with the formula $\mathbf{Z}^{\mathsf{N}} = \mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}$ from (3.5) and (3.19) implies that $\|\mathbf{Z}^{\mathsf{N}} - \mathbf{S}^{\mathsf{N}}\|_{\mathrm{L}^\infty([0,1]),\ell^\infty([\![0,\mathsf{N}]\!])} \to 0$. Therefore, we deduce Theorem 2.4 as written, and the proof is complete. □

{section:sde}

4. Proof of Proposition 3.7

The proof of Proposition 3.7 follows by gluing bulk and edge SDEs. Both SDEs were derived in the half-space setting of [33], with exception of the right-edge (which is just a mirror-image of the left-edge dynamics). So, the goal of this section is to record it precisely. We start with bulk dynamics, then edge dynamics. (To be clear, this gives an SDE for $\mathbf{Z}^{\mathsf{N}}$, after which we introduce smoothing as in (3.5).) This gives us with an SDE for $\mathbf{S}^{\mathsf{N}}$ without all the space- and time-averaging in (3.10)-(3.13). In particular, we conclude by introducing this averaging.

4.1. **Bulk dynamics.** We borrow this step from [33] (which borrows the following from the full-space setting of [32]). We start with notation that captures all the bulk and edge fluctuations.

{notation:mshe}

Notation 4.1. Define the "error-term" operator

$$\mathfrak{e}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} := \mathfrak{e}_2[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} + \mathfrak{e}_1[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{N}_{\mathrm{t},\mathrm{x}} + \mathfrak{e}_0[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} + \mathfrak{f}_{\mathrm{total}}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} + \mathsf{N}\Delta(\mathfrak{f}_0[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}). \tag{4.1}$$ {eq:msheII}

Let us now introduce the terms on the RHS of (4.1) as follows. (The reader is invited to think of the ℓ-subscript as indicating the power of $\mathsf{N}^{1/2}$; for example, terms in $\mathfrak{e}_2$ will have an "effective scaling" of $(\mathsf{N}^{1/2})^2 = \mathsf{N}$.)

(1) Assume that $\mathrm{x} \in [\![1, \mathsf{N}-1]\!]$. Indeed, the evolution of $\mathbf{j}^{\mathsf{N}}$ does not see edge dynamics at these interior points; see (2.1)-(2.3). (The same is thus true for $\mathbf{Z}^{\mathsf{N}}$.)

(2) We start with $\mathfrak{e}_2$. First, define the "renormalized" quadratic

$$\mathcal{Q}[\phi] := \tfrac{1}{2}\beta_2 \mathcal{U}'[\phi_0]\mathcal{U}'[\phi_1] - \tfrac{1}{2}\lambda(\mathcal{U}'[\phi_0]\phi_0 - 1). \tag{4.2}$$ {eq:quadratic_r

The first term $\mathfrak{e}_2$ is then defined as

$$\mathfrak{e}_2[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} := \lambda\mathsf{N}\mathcal{Q}[\phi_{\mathrm{t},\mathrm{x}+1}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} + \lambda\mathsf{N}\mathcal{Q}[\phi_{\mathrm{t},\mathrm{x}-1}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \tag{4.3}$$ {eq:msheII(2)a}

$$- \tfrac{1}{6}\lambda^2\mathsf{N}(\Delta\{\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+2}]\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+3}]\}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}). \tag{4.4}$$ {eq:msheII(2)b}

(3) We move to $\mathfrak{e}_1$. Set $\mathcal{V}[\phi_{\mathrm{x}}] := \mathcal{U}[\phi_{\mathrm{x}}] - \frac{1}{2}\alpha\phi_{\mathrm{x}}^2$, so that $\mathcal{V}'[\phi_{\mathrm{x}}] = \mathcal{U}'[\phi_{\mathrm{x}}] - \alpha\phi_{\mathrm{x}}$. We also define

$$\mathcal{G}[\phi] := \phi_2\mathcal{U}'[\phi_2]\cdot(\mathcal{U}'[\phi_1] - \mathcal{U}'[\phi_3]) + \phi_2\cdot(\mathcal{U}'[\phi_4]\mathcal{U}'[\phi_5] - \mathcal{U}'[\phi_3]\mathcal{U}'[\phi_4]). \tag{4.5}$$ {eq:funnygradie

We now define

$$\mathfrak{e}_1[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} := \tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_1(\mathcal{V}'[\phi_{\mathrm{t},\mathrm{x}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}) - \tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{-1}(\mathcal{V}'[\phi_{\mathrm{t},\mathrm{x}+1}]\mathbf{Z}^{N}_{\mathrm{t},\mathrm{x}}) \tag{4.6}$$ {eq:msheII(1)a}

$$- \tfrac{1}{6}\lambda^3\mathsf{N}^{\frac{1}{2}}\mathcal{G}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}. \tag{4.7}$$ {eq:msheII(1)c}

(4) For $\mathfrak{e}_0$, we have

$$\mathfrak{e}_0[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} := \tfrac{1}{12}\lambda^4\cdot(\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+1}]\phi^3_{\mathrm{t},\mathrm{x}+1} - \mathbb{E}^0\{\mathcal{U}'[\phi_0]\phi_0^3\})\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \tag{4.8}$$ {eq:msheII(0)a}

$$- \tfrac{1}{6}\lambda^4(\mathcal{G}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\phi_{\mathrm{t},\mathrm{x}+1} - 1)\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \tag{4.9}$$ {eq:msheII(0)a1

$$+ \tfrac{1}{2}\lambda^2\mathsf{N}\nabla^{\mathbf{X}}_1([\phi^2_{\mathrm{t},\mathrm{x}} - 1]\cdot\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}) \tag{4.10}$$ {eq:msheII(0)b}

$$+ \tfrac{1}{2}\lambda^2\mathsf{N}\nabla^{\mathbf{X}}_{-1}([\phi^2_{\mathrm{t},\mathrm{x}+1} - 1]\cdot\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}) \tag{4.11}$$ {eq:msheII(0)c}

$$+ \tfrac{1}{3}\lambda^2\mathsf{N}\nabla^{\mathbf{X}}_1(\Delta\{\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+2}]\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+3}]\}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}) \tag{4.12}$$ {eq:msheII(0)d}

$$- \tfrac{1}{12}\lambda^4(\Delta\{\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+2}]\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+3}]\})\phi^2_{\mathrm{t},\mathrm{x}+1}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}. \tag{4.13}$$ {eq:msheII(0)e}

$$- \tfrac{1}{12}\lambda^4(\Delta\{\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+2}]\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+3}]\})\phi^2_{\mathrm{t},\mathrm{x}+2}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}. \tag{4.14}$$ {eq:msheII(0)e1

$$+ \tfrac{1}{12}\lambda^4(\nabla^{\mathbf{X}}_1\{\mathcal{U}'[\phi_{\mathrm{t},\mathrm{x}+1}]\phi^3_{\mathrm{t},\mathrm{x}+1}\})\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}. \tag{4.15}$$ {eq:msheII(0)f}

(5) The first $\mathfrak{f}$-error term is

$$\mathfrak{f}_{\mathrm{total}}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{\frac{1}{2}}\nabla^{\mathbf{X}}_1\left(\mathfrak{f}_1[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\right) \tag{4.16}$$ {eq:msheIIfa}

$$+ \mathsf{N}^{\frac{1}{2}}\nabla^{\mathbf{X}}_{-1}\left(\mathfrak{f}_2[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\right) \tag{4.17}$$ {eq:msheIIfb}

$$+ \mathsf{N}^{-\frac{1}{2}}\mathfrak{f}_3[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}. \tag{4.18}$$ {eq:msheIIfc}

Above, the functions $\mathfrak{f}_i : \mathbb{R}^{\mathbb{Z}} \to \mathbb{R}$ satisfy the following a priori estimate for some $0 \leqslant \mathrm{c}, \mathrm{C} \lesssim 1$:

$$|\mathfrak{f}_i[\phi]| \lesssim 1 + \sum_{|\mathrm{w}|\leqslant \mathrm{c}} |\phi_{\mathrm{w}}|^{\mathrm{C}}. \tag{4.19}$$ {eq:msheIIff}

The exact form of these functions is unimportant; each is scaled too weakly to contribute for large N.

(6) The last error $\mathfrak{f}_0$ is a function, whose exact form is also not important, which satisfies the estimate (4.19).

We are now in position to present the SDE for the $\mathbf{Z}^{\mathsf{N}}$ on the set $[\![2, \mathsf{N}-1]\!]$. {lemma:msheinte

Lemma 4.2. *Fix $\mathrm{x} \in [\![1, \mathsf{N}-1]\!]$. With notation explained after, we have*

$$\mathrm{d}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathcal{T}_{\mathsf{N}}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}} + \mathfrak{e}[\tau_{\mathrm{x}}\phi_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t}. \tag{4.20}$$ {eq:msheinteric

The independent standard Brownian motions $\mathbf{b}$ are from (2.1), and $\mathcal{T}_{\mathsf{N}}$ is from Proposition 3.7.

Proof. This follows by Lemmas 4.14 and 4.15 in [32]. Indeed, the evolution for $\mathbf{j}^{\mathsf{N}}$ matches that in the half-space (and even the full-space) when we restrict to $\mathrm{x} \in [\![1, \mathsf{N}-1]\!]$; see (2.1)-(2.3). (We clarify that Lemma 4.15 in [32]

pays extra attention to $\mathrm{x} = 1$; it also packages all the error terms in Notation 4.1, besides those in $\mathfrak{e}_2$ and on the RHS of (4.6), into a single $\mathfrak{f}$-function, again at the single point $\mathrm{x} = 1$.) □

4.2. **Edge dynamics.** We now move to edge dynamics, starting with those at $\mathrm{x} = 0$. In this case, we can directly inherit the computation from the half-space setting in [33], because said dynamics appear therein as well (and are local around $\mathrm{x} = 0$, so there is no influence of the right-edge-dynamics).

emma:msheedge1}

Lemma 4.3. *We have, for some function $\mathfrak{f} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ which satisfies (4.19), that*

eq:msheedge1I}

$$\begin{aligned} \mathrm{d}\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}} &= \mathscr{T}_{\mathsf{N}}\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathbf{b}_{\mathrm{t},0} + \lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},1}]\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathsf{N}(\lambda\mathbf{F}[\boldsymbol{\phi}_{\mathrm{t}}] + \lambda^2 - \tfrac{1}{2}\alpha\lambda^2\boldsymbol{\phi}_{\mathrm{t},1}^2 + \alpha\lambda\mathbf{A} + \lambda\widetilde{\mathbf{A}})\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \mathsf{N}^{\frac{1}{2}}\mathfrak{f}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t}. \end{aligned} \tag{4.21}$$

Proof. The proof is exactly the same as that of Lemma 4.16 in [33], with one adjustment. In [33], it was assumed (for convenience) that $\widetilde{\mathbf{A}} = 0$. However, the contribution of $\widetilde{\mathbf{A}}$ in the SDE above is the term $\lambda\mathsf{N}\widetilde{\mathbf{A}}\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t}$, which can be seen from the fact that the previous display is shown via the Itô formula and $\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}} = \mathbf{Exp}(\lambda\mathbf{j}_{\mathrm{t},0}^{\mathsf{N}} - \lambda\mathscr{R}_{\lambda}\mathrm{t})$. □

We now take $\mathrm{x} = \mathsf{N}$. Ultimately, by the same argument, which we give details for since the computation does not actually appear in, say, the half-space problem of [33], we obtain a "mirror-image copy" of Lemma 4.3 above.

emma:msheedge2}

Lemma 4.4. *We have, for some $\widetilde{\mathfrak{f}} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ that satisfies (4.19) (but with $|\mathrm{w}| \leqslant \mathrm{c}$ instead of $|\mathrm{w} - \mathsf{N}| \leqslant \mathrm{c}$), that*

eq:msheedge2I}

$$\begin{aligned} \mathrm{d}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} &= \mathscr{T}_{\mathsf{N}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathsf{N}} - \lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathsf{N}(\lambda\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathrm{t}}] + \lambda^2 - \tfrac{1}{2}\lambda^2\alpha\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}^2 - \alpha\lambda\mathbf{B} + \lambda\widetilde{\mathbf{B}})\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \mathsf{N}^{\frac{1}{2}}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t}. \end{aligned} \tag{4.22}$$

Proof. We use (2.10), the Itô formula, and (2.2) to obtain

$$\begin{aligned} \mathrm{d}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} &= \lambda\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathbf{j}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} + \tfrac{1}{2}\lambda^2\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}[\mathbf{j}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}, \mathbf{j}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}] - \lambda\mathscr{R}_{\lambda}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\approx -\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \lambda\mathsf{N}\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \lambda\mathsf{N}\widetilde{\mathbf{B}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathsf{N}} + \lambda^2\mathsf{N}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t}. \end{aligned}$$

Here the notation $\approx$ denotes equality up to terms of the form $\mathsf{N}^{1/2}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t}$. Comparing (4.22) and the previous display, it now suffices to show that

eq:msheedge2I1}

$$\mathscr{T}_{\mathsf{N}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} - \lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} - \tfrac{1}{2}\lambda^2\alpha\mathsf{N}\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}^2 \approx -\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}. \tag{4.23}$$

To compute the first term in the previous display, we first remark that the discrete Laplacian $\Delta = -\nabla_{-1}^{\mathbf{X}}\nabla_{1}^{\mathbf{X}}$ on $\mathbb{Z}$ can also be written as $\Delta = \nabla_{1}^{\mathbf{X}} + \nabla_{-1}^{\mathbf{X}}$. Using this, we recall from Definition 3.3 that

$$\begin{aligned} \mathscr{T}_{\mathsf{N}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} &= (\alpha\mathsf{N}^2 + \tfrac{1}{4}\mathsf{N}\lambda)\Delta_{\mathbf{A},\mathbf{B}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} = (\alpha\mathsf{N}^2 + \tfrac{1}{4}\mathsf{N}\lambda)\cdot(\mathsf{N}^{-1}\mathbf{B}\lambda\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} + \nabla_{-1}^{\mathbf{X}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}) \\ &\approx \alpha\mathbf{B}\lambda\mathsf{N}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} + (\alpha\mathsf{N}^2 + \tfrac{1}{4}\mathsf{N}\lambda)\nabla_{-1}^{\mathbf{X}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} \\ &\approx \alpha\mathbf{B}\lambda\mathsf{N}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} - \lambda\alpha\mathsf{N}^{\frac{3}{2}}\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} + \tfrac{1}{2}\lambda^2\alpha\mathsf{N}\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}^2\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}, \end{aligned}$$

in which the last line uses (2.10) and Taylor expands the exponential therein, while using $\nabla_{-1}^{\mathbf{X}}\mathbf{j}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}} = -\mathsf{N}^{-1/2}\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}$. Now add $-\lambda\mathsf{N}^{3/2}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}$ to the display above and recall via Notation 4.1 that $-\alpha\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}} - \mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}] = -\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]$. This completes the proof of (4.23), so we are done. □

4.3. **Putting it together.** We now combine Lemmas 4.2, 4.3, and 4.4 to get

$$\begin{aligned} \mathrm{d}\mathbf{Z}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}} &= \mathscr{T}_{\mathsf{N}}\mathbf{Z}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{Z}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}}\mathrm{d}\mathbf{b}_{\mathrm{t},\mathrm{x}} + \mathbf{1}_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]}\mathfrak{e}[\tau_{\mathrm{x}}\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathbf{1}_{\mathrm{x}=0}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},1}]\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} - \mathbf{1}_{\mathrm{x}=\mathsf{N}}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathbf{1}_{\mathrm{x}=0}\mathsf{N}(\lambda\mathbf{F}[\boldsymbol{\phi}_{\mathrm{t}}] + \lambda^2 - \tfrac{1}{2}\alpha\lambda^2\boldsymbol{\phi}_{\mathrm{t},1}^2 + \alpha\lambda\mathbf{A} + \lambda\widetilde{\mathbf{A}})\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathbf{1}_{\mathrm{x}=\mathsf{N}}\mathsf{N}(\lambda\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathrm{t}}] + \lambda^2 - \tfrac{1}{2}\lambda^2\alpha\boldsymbol{\phi}_{\mathrm{t},\mathsf{N}}^2 - \alpha\lambda\mathbf{B} + \lambda\widetilde{\mathbf{B}})\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t} \\ &\quad + \mathbf{1}_{\mathrm{x}=0}\mathsf{N}^{\frac{1}{2}}\mathfrak{f}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},0}^{\mathsf{N}}\mathrm{d}\mathrm{t} + \mathbf{1}_{\mathrm{x}=\mathsf{N}}\mathsf{N}^{\frac{1}{2}}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}_{\mathrm{t},\mathsf{N}}^{\mathsf{N}}\mathrm{d}\mathrm{t}. \end{aligned}$$

We now the above to derive an equation for $\mathbf{S}^{\mathsf{N}} = \mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{Z}^{\mathsf{N}})$. We obtain the following, because the $\mathscr{T}_{\mathsf{N}}$-semigroup commutes with itself, and because $\mathbf{Z}^{\mathsf{N}} = \mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}$:

{eq:ssde1}
$$\begin{aligned}
\mathrm{d}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathfrak{x}} &= \mathscr{T}_{\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathfrak{x}}\mathrm{d}\mathfrak{t} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},\cdot}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\cdot}\mathrm{d}\mathbf{b}_{\mathfrak{t},\cdot})]_{\mathfrak{x}} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\mathfrak{e}[\tau_{\cdot}\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} \\
&+ [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=0}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathfrak{t},1}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} - [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=\mathsf{N}}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathfrak{t},\mathsf{N}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} \\
&+ [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=0}\mathsf{N}(\lambda\mathbf{F}[\boldsymbol{\phi}_{\mathfrak{t}}] + \lambda^2 - \tfrac{1}{2}\alpha\lambda^2\boldsymbol{\phi}^2_{\mathfrak{t},1} + \alpha\lambda\mathbf{A} + \lambda\widetilde{\mathbf{A}})\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} \\
&+ [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=\mathsf{N}}\mathsf{N}(\lambda\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}_{\mathfrak{t}}] + \lambda^2 - \tfrac{1}{2}\lambda^2\alpha\boldsymbol{\phi}^2_{\mathfrak{t},\mathsf{N}} - \alpha\lambda\mathbf{B} + \lambda\widetilde{\mathbf{B}})\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} \\
&+ [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=0}\mathsf{N}^{\frac{1}{2}}\mathfrak{f}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},0}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},0})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=\mathsf{N}}\mathsf{N}^{\frac{1}{2}}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}})]_{\mathfrak{x}}\mathrm{d}\mathfrak{t}.
\end{aligned} \tag{4.24}$$

We will now identify each term on the RHS of the previous display, except for the first two therein, with various error terms and admissible functions. We start with the following.

{lemma:errs}
Lemma 4.5. *The following has the form of* $\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathfrak{t},\mathfrak{x}}$ *from Proposition 3.7:*

$$[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\mathsf{N}\Delta\mathfrak{f}_0[\tau_{\cdot}\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathfrak{x}} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=0}\mathsf{N}^{\frac{1}{2}}\mathfrak{f}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},0}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},0})]_{\mathfrak{x}} + [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=\mathsf{N}}\mathsf{N}^{\frac{1}{2}}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}})]_{\mathfrak{x}}.$$

Proof. It suffices to show that each summand here has the form $\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathfrak{t},\mathfrak{x}}$. This holds for the last two terms above; indeed, for the last term, we can rewrite it as

$$[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot=\mathsf{N}}\mathsf{N}^{\frac{1}{2}}\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}})]_{\mathfrak{x}} = \sum_{\mathfrak{w}\in[\![0,\mathsf{N}]\!]}\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},\mathfrak{w}}\mathbf{1}_{\mathfrak{w}=\mathsf{N}}\cdot\mathsf{N}^{\frac{1}{2}}\cdot\widetilde{\mathfrak{f}}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{R}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}. \tag{4.25}$$

We now bake $\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},\mathfrak{w}}\mathbf{1}_{\mathfrak{w}=\mathsf{N}}\mathsf{N}^{1/2}$ into the $\mathbf{E}^{\mathsf{N}}$-kernel in (3.15) (the necessary estimate is satisfied by Proposition A.1, and the fact that $\mathfrak{t}_{\mathsf{N}} = \mathsf{N}^{-\delta_{\mathbf{S}}}$ with $\delta_{\mathbf{S}} > 0$ small by Definition 3.3). The same argument works for the second term in the main display of the lemma. For the first term therein, we will first extend $\mathfrak{f}_0[\tau_{\mathfrak{x}}\boldsymbol{\phi}_{\mathfrak{t}}]$ from $\mathfrak{x}\in[\![1,\mathsf{N}-1]\!]$ to $\mathfrak{x}\in[\![0,\mathsf{N}]\!]$ via $\Delta_{\mathbf{A},\mathbf{B}}\mathfrak{f}_0[\tau_{\mathfrak{x}}\boldsymbol{\phi}_{\mathfrak{t}}] = 0$ for $\mathfrak{x} = 0,\mathsf{N}$. This allows us to write

$$[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\mathsf{N}\Delta\mathfrak{f}_0[\tau_{\cdot}\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathfrak{x}} = [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathsf{N}\Delta_{\mathbf{A},\mathbf{B}}\mathfrak{f}_0[\tau_{\cdot}\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathfrak{x}} = \mathsf{N}\Delta_{\mathbf{A},\mathbf{B}}[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathfrak{f}_0[\tau_{\cdot}\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathfrak{x}},$$

in which the last display follows because $\mathscr{T}_{\mathsf{N}}$ commutes with its semigroup, and $\mathscr{T}_{\mathsf{N}}$ is a scalar multiple of $\Delta_{\mathbf{A},\mathbf{B}}$. Now, it suffices to use that $\mathsf{N}\Delta_{\mathbf{A},\mathbf{B}}\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},\mathfrak{y}} \lesssim \mathsf{N}^{-2}\mathfrak{t}_{\mathsf{N}}^{-3} \lesssim \mathsf{N}^{-6/5}$ by the heat kernel estimates in Proposition A.1. (In words, every gradient introduces a factor of N^{-1}, modulo harmless powers of $\mathfrak{t}_{\mathsf{N}}^{-1}$, because the heat kernel is smooth just below the macroscopic spatial scale.) This completes the proof. □

We now move to the third and fourth lines of (4.24).

{lemma:edgead1}
Lemma 4.6. *The third and fourth lines of* (4.24) *have the form*

$$\sum_{\mathfrak{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathfrak{x},\mathfrak{w}}\cdot\mathsf{N}\mathfrak{b}_{\ell}[\boldsymbol{\phi}_{\mathfrak{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathfrak{w}},$$

where $\mathfrak{b}_{\ell}$ *is edge-admissible, and* $\boldsymbol{\mathcal{K}}^{\partial,\ell}$ *satisfies the estimate* (3.17).

Proof. Note that the $\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}$-kernel has the form $\boldsymbol{\mathcal{K}}^{\partial,\ell}$ by heat kernel estimates in Proposition A.1 and the fact that $\mathfrak{t}_{\mathsf{N}} = \mathsf{N}^{-\delta_{\mathbf{S}}}$ for $\delta_{\mathbf{S}} > 0$ small by Definition 3.3. Thus, it suffices to show that

$$\lambda\mathbf{F}[\boldsymbol{\phi}] + \lambda^2 - \tfrac{1}{2}\alpha\lambda^2\boldsymbol{\phi}_1^2 + \alpha\lambda\mathbf{A} + \lambda\widetilde{\mathbf{A}} \quad\text{and}\quad \lambda\mathbf{F}[\tau_{\mathsf{N}}\boldsymbol{\phi}] + \lambda^2 - \tfrac{1}{2}\lambda^2\alpha\boldsymbol{\phi}_{\mathsf{N}}^2 - \alpha\lambda\mathbf{B} + \lambda\widetilde{\mathbf{B}}$$

are edge-admissible. The a priori polynomial bound on the functions above is satisfied by (2.4) and bounds on the potential in Assumption 2.2. Thus, it suffices to prove that both vanish under $\mathbb{E}^0$. To this end, it is shown in Lemmas 4.20 and 4.21 in [33] that $\mathbb{E}^0\mathbf{F}[\tau_{\mathfrak{x}}\boldsymbol{\phi}] = 0$ and $\mathbb{E}^0\boldsymbol{\phi}_{\mathfrak{x}}^2 = \alpha^{-1}$ for any $\mathfrak{x}\in[\![1,\mathsf{N}]\!]$. Now use (2.15)-(2.16). □

We are left with the last term in the first line and the second line of (4.24). Before we can identify these terms, we will need a summation-by-parts identity.

{lemma:sbp}
Lemma 4.7. *Fix* $\mathfrak{f}:[\![0,\mathsf{N}]\!]\to\mathbb{R}$. *We have*

$$[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\nabla^{\mathbf{X}}_1(\mathfrak{f}_{\cdot}\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot}))]_{\mathfrak{x}} = \sum_{\mathfrak{w}\in[\![2,\mathsf{N}]\!]}\nabla^{\mathbf{X}}_{-1}\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},\mathfrak{w}}\mathfrak{f}_{\mathfrak{w}}\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathfrak{w}} - \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},1}\mathfrak{f}_1\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},1} + \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathfrak{x},\mathsf{N}}\mathfrak{f}_{\mathsf{N}}\mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}},$$

$$[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\cdot\nabla^{\mathbf{X}}_{-1}(\mathsf{f}_\cdot\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}=\sum_{\mathrm{w}\in[\![0,\mathsf{N}-2]\!]}\nabla^{\mathbf{X}}_{1}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\mathsf{f}_\mathrm{w}\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}+\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},0}\mathsf{f}_0\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},0}-\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathsf{N}-1}\mathsf{f}_{\mathsf{N}-1}\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathsf{N}-1}.$$

We clarify that all discrete gradients act only on the w*-variable. (Observe that the identities above are discrete versions of integration-by-parts, at least modulo the shift by* 1 *in the boundary points.)*

Proof. Note that the adjoint of $\nabla^{\mathbf{X}}_{\mathfrak{l}}$ with respect to Lebesgue measure on $\mathbb{Z}$ is $\nabla^{\mathbf{X}}_{-\mathfrak{l}}$. Moreover, for any $\mathsf{f},\mathsf{g}:\mathbb{Z}\to\mathbb{R}$, we have the discrete Leibniz rule $\nabla^{\mathbf{X}}_{\mathfrak{l}}(\mathsf{f}_\mathrm{x}\mathsf{g}_\mathrm{x})=\mathsf{f}_\mathrm{x}\nabla^{\mathbf{X}}_{\mathfrak{l}}\mathsf{g}_\mathrm{x}+\mathsf{g}_{\mathrm{x}+\mathfrak{l}}\nabla^{\mathbf{X}}_{\mathfrak{l}}\mathsf{f}_\mathrm{x}$. Using these, we have

$$\begin{aligned}[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\cdot\nabla^{\mathbf{X}}_{1}(\mathsf{f}_\cdot\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}&=\sum_{\mathrm{w}\in\mathbb{Z}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}\cdot\nabla^{\mathbf{X}}_{1}(\mathsf{f}_\mathrm{w}\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}})\\&=\sum_{\mathrm{w}\in\mathbb{Z}}\mathbf{1}_{\mathrm{w}-1\in[\![1,\mathsf{N}-1]\!]}\nabla^{\mathbf{X}}_{-1}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\cdot\mathsf{f}_\mathrm{w}\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}+\sum_{\mathrm{w}\in\mathbb{Z}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\cdot\nabla^{\mathbf{X}}_{-1}\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}\cdot\mathsf{f}_\mathrm{w}\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}.\end{aligned}$$

Observe that $\mathrm{w}-1\in[\![1,\mathsf{N}-1]\!]$ if and only if $\mathrm{w}\in[\![2,\mathsf{N}]\!]$. Moreover, we have $\nabla^{\mathbf{X}}_{-1}\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}=-\mathbf{1}_{\mathrm{w}=1}+\mathbf{1}_{\mathrm{w}=\mathsf{N}}$. We plug these two identities in the display above to show the first identity. For the second, the same argument works if we replace -1 by 1 in the second line above. So, we have $\mathbf{1}_{\mathrm{w}+1\in[\![1,\mathsf{N}-1]\!]}=\mathbf{1}_{\mathrm{w}\in[\![0,\mathsf{N}-2]\!]}$ instead of $\mathbf{1}_{\mathrm{w}-1\in[\![1,\mathsf{N}-1]\!]}$. Moreover, we have $\nabla^{\mathbf{X}}_{1}\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}=\mathbf{1}_{\mathrm{w}=0}-\mathbf{1}_{\mathrm{w}=\mathsf{N}-1}$. Finally, we have $\nabla^{\mathbf{X}}_{1}\mathbf{H}^{\mathsf{N}}$ instead of $\nabla^{\mathbf{X}}_{-1}\mathbf{H}^{\mathsf{N}}$. Putting these adjustments into the previous display now yields the second identity, so we are done. □

We will now apply Lemma 4.7 to identify the sum between the second line of (4.24) and the contribution from the last term in the first line therein which comes from (4.6). In particular:

{lemma:flux}

Lemma 4.8. *The quantity*

$$\begin{aligned}&[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot=0}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},0})]_\mathrm{x}-[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot=\mathsf{N}}\lambda\mathsf{N}^{\frac{3}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\mathsf{N}}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathsf{N}})]_\mathrm{x}\\&\qquad+\tfrac{1}{2}[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{1}(\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\cdot}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}\\&\qquad-\tfrac{1}{2}[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{-1}(\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\cdot+1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}\end{aligned}$$

has the form below, in which we use notation to be explained afterwards:

{eq:fluxI}
$$\sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_\mathsf{t}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}+\sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{b}_\ell[\boldsymbol{\phi}_\mathsf{t}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}+\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathsf{t},\mathrm{x}}.\tag{4.26}$$

The kernel $\boldsymbol{\mathcal{K}}^{\partial,\ell}$ *satisfies the estimate* (3.17) *and is deterministic. The functions* $\mathfrak{q}_{\ell,\mathrm{w}},\mathfrak{b}_\ell$ *are as in Proposition 3.7. Finally, the kernel* $\boldsymbol{\mathcal{S}}^{(\ell)}$ *satisfies the following upgraded version of* (3.16) *that incorporates regularity estimates (with respect to the just-below-macroscopic-scale* $\mathsf{N}^{1-\delta_\mathbf{S}}$*) as well:*

{eq:fluxII}
$$|(\nabla^{\mathbf{X}}_{\mathfrak{l}_1})^{\mathrm{a}}(\nabla^{\mathbf{X}}_{\mathfrak{l}_2})^{\mathrm{b}}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}|\lesssim\mathsf{N}^{-1-(\mathrm{a}+\mathrm{b})+(\mathrm{a}+\mathrm{b}+\mathrm{C})\delta_\mathbf{S}}.\tag{4.27}$$

Above, $\mathrm{a},\mathrm{b}\geqslant 0$ *are* $\mathrm{O}(1)$ *integers, as are* $|\mathfrak{l}_1|,|\mathfrak{l}_2|$*. The first iterated discrete gradient above acts on* x*, whereas the second acts on* y*. Finally, the constant* C *is fixed and* $\mathrm{O}(1)$*, and we only ever consider choices of* $\mathrm{a},\mathrm{b},\mathfrak{l}_1,\mathfrak{l}_2,\mathrm{x},\mathrm{w}$ *for which the* LHS *involves only values of* $\boldsymbol{\mathcal{S}}^{(\ell)}$ *on* $[\![0,\mathsf{N}]\!]\times[\![0,\mathsf{N}]\!]$*.*

Proof. By Lemma 4.7, we have

$$\begin{aligned}&\tfrac{1}{2}[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{1}(\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\cdot}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}\\&=\sum_{\mathrm{w}\in[\![2,\mathsf{N}]\!]}\mathsf{N}\nabla^{\mathbf{X}}_{-1}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\cdot\tfrac{1}{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\mathrm{w}}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}-\tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},1}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},1}+\tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathsf{N}}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\mathsf{N}}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathsf{N}},\end{aligned}$$

in which we use the first identity in Lemma 4.7, and by using the second identity therein, we have

$$\begin{aligned}&-\tfrac{1}{2}[\mathbf{e}^{\mathsf{t}_\mathsf{N}\mathscr{T}_\mathsf{N}}(\mathbf{1}_{\cdot\in[\![1,\mathsf{N}-1]\!]}\lambda\mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_{-1}(\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\cdot+1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\cdot}))]_\mathrm{x}\\&=-\sum_{\mathrm{w}\in[\![0,\mathsf{N}-2]\!]}\mathsf{N}\nabla^{\mathbf{X}}_{1}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathrm{w}}\cdot\tfrac{1}{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\mathrm{w}+1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathrm{w}}-\tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},0}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},1}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},0}+\tfrac{1}{2}\lambda\mathsf{N}^{\frac{3}{2}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_\mathsf{N},\mathrm{x},\mathsf{N}-1}\mathscr{V}'[\boldsymbol{\phi}_{\mathsf{t},\mathsf{N}}]\mathbf{Z}^{\mathsf{N}}_{\mathsf{t},\mathsf{N}-1}.\end{aligned}$$

By regularity for the heat kernel at time $\mathsf{t}_\mathsf{N}=\mathsf{N}^{-\delta_\mathbf{S}}$ in Proposition A.1, the first term on the RHS of each display has the form of the first term in (4.26). Indeed, using Lemma 4.20 in [33], the families $\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}]=$

$\mathscr{V}'[\phi_{\mathrm{w}}], \mathscr{V}'[\phi_{\mathrm{w}+1}]$ are both bulk admissible ones. Therefore, it suffices to show that

$$\begin{aligned}
&\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} - \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},1} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},1} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}-1} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}-1}
\end{aligned}$$

has form given by the last two terms in (4.26). First, let us modify the previous display by replacing $\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},1}$ in the second line by $\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0}$. The cost in doing so is given by $\mathsf{N} \nabla^{\mathbf{X}}_1 \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \cdot \mathsf{N}^{1/2} \mathfrak{f}[\phi_{\mathfrak{t}}]$ for some function $\mathfrak{f}$ which satisfies (4.19). Thus, the proof of Lemma 4.5 shows that this cost could be absorbed by the Err-term in (4.26). (We clarify that Lemma 4.5 does not have N times a gradient which acts on the heat kernel, but the effect of this additional operator is multiplication by $\mathsf{N}^{\mathrm{O}(\delta \mathrm{s})}$ because of Proposition A.1, so the reasoning that leads to Lemma 4.5 remains in effect.) By the same token, up to another contribution to Err in (4.26), we can change $\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}-1}$ in the last line above to $\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}}$. Thus, we are left with identifying

$$\begin{aligned}
&\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} - \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},1} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}-1}.
\end{aligned}$$

We now take the first term in the second line and the last term in the last line; then, we rewrite them as

$$\begin{aligned}
-\tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},1} &= -\tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \nabla^{\mathbf{X}}_{-1} \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} \\
&\approx -\tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},1} - \tfrac{1}{2} \lambda^2 \mathsf{N} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \phi_{\mathfrak{t},1} \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0}, \\
\tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}-1} &= \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \nabla^{\mathbf{X}}_{-1} \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&\approx \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} - \tfrac{1}{2} \lambda^2 \mathsf{N} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \phi_{\mathfrak{t},\mathsf{N}} \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}.
\end{aligned}$$

Above, the $\approx$ notation means up to terms that can be absorbed by the Err-term in (4.26) (similar to the proof of Lemma 4.4), and the $\approx$-claims follow by the formula (2.10) and Taylor expanding the exponential therein (again, similar to the proof of Lemma 4.4). Now, we cite Lemma 4.21 in [33] to get that $\mathscr{V}'[\phi_{\mathfrak{t},1}] \phi_{\mathfrak{t},1}$ is edge admissible; the same is true for $\mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \phi_{\mathfrak{t},\mathsf{N}}$, because the measure $\mathbb{P}^0$ (see Definition 2.1) has exchangeable one-dimensional marginals. So, the last term in each claim above can be absorbed by the second term in (4.26). We are left with

$$\begin{aligned}
&\mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} - \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \cdot \lambda \mathsf{N}^{\frac{3}{2}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}} \\
&- \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},0} \mathscr{V}'[\phi_{\mathfrak{t},1}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},0} + \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \mathbf{H}^{\mathsf{N}}_{0,\mathfrak{t}_{\mathsf{N}},\mathrm{x},\mathsf{N}} \mathscr{V}'[\phi_{\mathfrak{t},\mathsf{N}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathsf{N}}.
\end{aligned}$$

It now suffices to note that the sum of the first terms in each row cancel, as does the sum of the last terms. □

Finally, we address the remaining contributions from Notation 4.1 to the last term in (4.24).

{lemma:eleft}

Lemma 4.9. *Denote by* $\widetilde{\mathfrak{e}}[\tau_{\mathrm{x}} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{x}}$ *the term* $\mathfrak{e}[\tau_{\mathrm{x}} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{x}}$ *(see Notation 4.1) after subtracting the* RHS *of* (4.6)*:*

$$\widetilde{\mathfrak{e}}[\tau_{\mathrm{x}} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{x}} := \mathfrak{e}[\tau_{\mathrm{x}} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{x}} - \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \nabla^{\mathbf{X}}_1 (\mathscr{V}'[\phi_{\mathfrak{t},\mathrm{x}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{x}}) - \tfrac{1}{2} \lambda \mathsf{N}^{\frac{3}{2}} \nabla^{\mathbf{X}}_{-1} (\mathscr{V}'[\phi_{\mathfrak{t},\mathrm{x}+1}] \mathbf{Z}^{N}_{\mathfrak{t},\mathrm{x}}).$$

The quantity $[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}} \mathscr{T}_{\mathsf{N}}} (\mathbf{1}_{\cdot \in [\![1,\mathsf{N}-1]\!]} \widetilde{\mathfrak{e}}[\tau_{\cdot} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})]_{\mathrm{x}}$ *has the form* (4.26)*.*

Proof. This uses the same argument as the proof of Lemma 4.22 in [33], so we do not record the entire argument with details here, but rather the main ideas and highlights.

As noted in Lemma 4.22 in [33], each term in $\mathbf{1}_{\mathrm{w} \in [\![1,\mathsf{N}-1]\!]} \widetilde{\mathfrak{e}}[\tau_{\mathrm{w}} \phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{w}}$ has one of the following forms.

(1) We have terms of the form $\mathsf{N} \mathfrak{q}_{\mathrm{w}}[\phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\mathrm{w}}$, where $\mathfrak{q}_{\mathrm{w}}[\phi_{\mathfrak{t}}] = \mathbf{1}_{\mathrm{w} \in [\![1,\mathsf{N}-1]\!]} \mathfrak{q}[\tau_{\mathrm{w}} \phi]$ is bulk admissible with respect to the point w. Moreover, $\mathfrak{q}_{\mathrm{w}}$ is either forwards-oriented or backwards-oriented, and if it is forwards-oriented (resp. backwards-oriented) for some $\mathrm{w} \in [\![0, \mathsf{N}]\!]$, then it is forwards-oriented (resp. backwards-oriented) for all $\mathrm{w} \in [\![0, \mathsf{N}]\!]$, since the different $\mathfrak{q}_{\mathrm{w}}$ are spatial shifts of a common $\mathfrak{q}$. Thus, the corresponding contribution $\mathbf{e}^{\mathfrak{t}_{\mathsf{N}} \mathscr{T}_{\mathsf{N}}} (\mathsf{N} \mathfrak{q}_{\cdot}[\phi_{\mathfrak{t}}] \mathbf{Z}^{\mathsf{N}}_{\mathfrak{t},\cdot})$, has the form of the first term in (4.26).

(2) We have $\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}\mathsf{N}\nabla^{\mathbf{X}}_{\pm 1}(\mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}})$, in which $\mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}] = \mathfrak{q}[\tau_{\mathrm{w}}\boldsymbol{\phi}_{\mathrm{t}}]$ satisfies $\mathbb{E}^0\mathfrak{q}_{\mathrm{w}} = 0$ and is either forwards-oriented or backwards-oriented with respect to w. Thus, if we look at its image under heat semigroups, i.e. $\mathbf{e}^{\mathsf{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}\mathsf{N}\nabla^{\mathbf{X}}_{\pm 1}(\mathfrak{q}_{\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})$, we apply summation-by-parts to move the discrete gradient off $\mathfrak{q}_{\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot}$. By Lemma 4.7, this produces one term that can be absorbed by the first term in (4.26); said term corresponds to the first term on the RHS of the two identities in Lemma 4.8 (indeed, heat kernel regularity in Proposition A.1 shows that $\mathsf{N}\nabla^{\mathbf{X}}_{\mathfrak{l}}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_{\mathsf{N}},\mathrm{x},\mathrm{w}}$ satisfies the necessary a priori bound (3.16) if $\mathfrak{l} \lesssim 1$ and $\mathsf{t}_{\mathsf{N}} = \mathsf{N}^{-\delta_{\mathbf{S}}}$ with $\delta_{\mathbf{S}} > 0$ small). This summation-by-parts also produces two more terms absorbed by the edge-quantity given by the second term in (4.26); these terms correspond to the last two terms in the identities in Lemma 4.8.

(3) Finally, we have $\mathbf{1}_{\mathrm{w}\in[\![1,\mathsf{N}-1]\!]}\mathsf{N}^{1/2}\nabla^{\mathbf{X}}_{\pm 1}(\mathfrak{f}_{\mathrm{y}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{y}})$, in which for any $\mathrm{y} \in [\![1,\mathsf{N}-1]\!]$, the function $\mathfrak{f}_{\mathrm{y}}$ satisfies the estimate (4.19) if we replace $|\mathrm{w}|$ on the RHS by $|\mathrm{w}-\mathrm{y}|$. For this, we apply summation-by-parts as in Lemma 4.7 once again. The first term we obtain has the form of $\sum_{\mathrm{w}}\mathsf{N}^{1/2}\nabla^{\mathbf{X}}_{\mp 1}\mathbf{H}^{\mathsf{N}}_{0,\mathsf{t}_{\mathsf{N}},\mathrm{x},\mathrm{w}}\cdot\mathfrak{f}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{y}}$, in which the sum over w is over some subset of $[\![0,\mathsf{N}]\!]$. Since the gradient is scaled too weakly, heat kernel regularity estimates from Proposition A.1 imply that this term has the form of Err in (4.26). Other terms in our use of Lemma 4.7 have the form of the last line in (4.24) and thus are also of the form Err in (4.26) because of Lemma 4.5.

This completes the proof. □

4.4. **Introducing space- and time-averages.** By Lemmas 4.5, 4.6, 4.8, and 4.9 applied to (4.24), we have

{eq:ssde2}
$$\mathrm{d}\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathscr{T}_{\mathsf{N}}\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + [\mathbf{e}^{\mathsf{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathbf{S}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} + \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{b}_{\ell}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}. \tag{4.28}$$

We now proceed with introducing space- and time-averages into the admissible functions in the second line. We start by introducing a spatial-average into the $\mathfrak{q}_{\ell,\mathrm{w}}$-terms above.

{lemma:avxput}
Lemma 4.10. *Fix any $\ell \in [\![1,\mathrm{L}_1]\!]$ in (4.28). We have*

{eq:avxputI}
$$\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} = \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_{\ell}}_{\mathrm{t},\mathrm{w}} + \boldsymbol{\mathcal{E}}_{\mathrm{t},\mathrm{x}}. \tag{4.29}$$

The deterministic kernel $\boldsymbol{\mathcal{K}}^{(\ell)}$ satisfies (3.16). The term $\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_{\ell}}$ is given in Definition 3.4 with $\mathscr{Q}_{\ell}$ as in Proposition 3.7. The term $\boldsymbol{\mathcal{E}}_{\mathrm{t},\mathrm{x}}$ has the form of $\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}}$.

Proof. Recall $\tau_{\mathbf{Av}} = \mathsf{N}^{-2}\mathfrak{l}_{\mathbf{Av}} = \mathsf{N}^{-3\delta_{\mathbf{S}}}$ from Definition 3.4, and set $\mathscr{X} = \mathrm{Id} - \mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}$. We claim that

$$\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} = \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}[\mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}(\mathfrak{q}_{\ell,\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})]_{\mathrm{w}} + \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\mathscr{X}\boldsymbol{\mathcal{S}}^{(\ell)}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}.$$

The operator $\mathscr{X}$ acts on the w-variable in $\boldsymbol{\mathcal{S}}^{(\ell)}$ in the last term above. Indeed, this follows first by $\mathrm{Id} = \mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}} + \mathscr{X}$ applied to $\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}$ (viewed as a function of $\mathrm{w}\in[\![0,\mathsf{N}]\!]$). Then, we note that $\mathscr{X}$ is self-adjoint with respect to Lebesgue measure on $[\![0,\mathsf{N}]\!]$; see Lemma 3.6 in [28] for the self-adjoint property of $\mathscr{T}_{\mathsf{N},0,0}$ (and thus its semigroup). We now iterate the previous identity (but with $(\mathscr{X})^{\mathrm{m}}\boldsymbol{\mathcal{S}}^{(\ell)}$ for various integers $\mathrm{k}\geqslant 0$ instead of just $\boldsymbol{\mathcal{S}}^{(\ell)}$), so

$$\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} = \sum_{\mathrm{m}\in[\![0,\mathrm{M}]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}(\mathscr{X})^{\mathrm{m}}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}[\mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}(\mathfrak{q}_{\ell,\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})]_{\mathrm{w}} + \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}(\mathscr{X})^{\mathrm{M}+1}\boldsymbol{\mathcal{S}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\mathfrak{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}.$$

The second line of the previous display has the form of $\boldsymbol{\mathcal{E}}_{\mathrm{t},\mathrm{x}}$ from (4.29) by Lemma A.2; in words, the $\boldsymbol{\mathcal{S}}^{(\ell)}$-kernel has regularity at scale $\mathsf{N}^{1-\delta_{\mathbf{S}}}$ via (4.27), whereas $\mathscr{X}$ is essentially the average of discrete gradients at length-scale $\mathfrak{l}_{\mathbf{Av}} = \mathsf{N}^{1-3\delta_{\mathbf{S}}/2} \ll \mathsf{N}^{1-\delta_{\mathbf{S}}}$. On the other hand, if we set $\boldsymbol{\mathcal{K}}^{(\ell)} = \sum_{\mathrm{m}}(\mathscr{X})^{\mathrm{m}}\boldsymbol{\mathcal{S}}^{(\ell)}$, then the first term on the RHS has

the following form:

$$\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}[\mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}(\mathfrak{q}_{\ell,\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})]_{\mathrm{w}}$$

We note that the necessary estimate (3.16) for this choice of $\boldsymbol{\mathcal{K}}^{(\ell)}$ follows by the fact that the kernel $\boldsymbol{\mathcal{S}}^{(\ell)}$ satisfies the even stronger estimate (4.27), as well as the fact that $\mathscr{X}$ is a difference of averaging operators, and averaging operators preserve the estimate (3.16). Next, write the previous display as (see Definition 3.4 for notation)

$$\begin{aligned}&\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} + \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}[\mathbf{e}^{\tau_{\mathbf{Av}}\mathscr{T}_{\mathsf{N},0,0}}(\mathbf{1}_{|\mathrm{w}-\cdot|\gtrsim(\log\mathsf{N})^2\mathfrak{l}_{\mathbf{Av}}}\mathfrak{q}_{\ell,\cdot}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\cdot})]_{\mathrm{w}}\\ &= \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} + \sum_{\mathrm{y}\in[\![0,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \mathbf{H}^{\mathsf{N},0,0}_{0,\tau_{\mathbf{Av}},\mathrm{w},\mathrm{y}} \mathbf{1}_{|\mathrm{w}-\mathrm{y}|\gtrsim(\log\mathsf{N})^2\mathfrak{l}_{\mathbf{Av}}} \Big) \cdot \mathsf{N}\mathfrak{q}_{\ell,\mathrm{y}}[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{y}}.\end{aligned}$$

The term in parentheses above, which is a kernel depending on (x,y), is $\lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}$ for any $\mathrm{D}>0$. Indeed, we have $\boldsymbol{\mathcal{K}}^{(\ell)} \lesssim 1$ by (3.16). Moreover, by Proposition A.1 shows that $\mathbf{H}^{\mathsf{N},0,0}$ therein has exponential decay in space at scale $\mathsf{N}\tau^{1/2}_{\mathbf{Av}} = \mathfrak{l}_{\mathbf{Av}}$, and the indicator forces this to be $\lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}$. We now add up $\mathrm{O}(\mathsf{N})$-many of these (very) small terms to deduce the claim about the parenthetical term above. In particular, the second term in the last line above has the form of $\boldsymbol{\mathcal{E}}_{\mathrm{t},\mathrm{x}}$ in (4.29). This completes the proof. □

We now introduce time-averages into the first term on the RHS of (4.29) and the last term in (4.28).

{lemma:avtput}

Lemma 4.11. *Retain the notation of Proposition 3.7, (4.28), and of Lemma 4.10. For any $\ell \in [\![1,\mathrm{L}_1]\!]$, we have*

$$\boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} = \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} + \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{l}_{\mathbf{Av}}}(\boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}). \tag{4.30}$$

{eq:avtputI}

Moreover, for any $\ell \in [\![1,\mathrm{L}_2]\!]$, the following similar identity holds:

$$\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} = \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} + \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial}(\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}} \cdot \mathsf{N}\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}). \tag{4.31}$$

{eq:avtputII}

Proof. By construction (see Definitions 3.4 and 3.6), we have

$$\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} = \boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} + \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{l}_{\mathbf{Av}}}(\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}}).$$

Multiply both sides by $\boldsymbol{\mathcal{K}}^{(\ell)}$. Next, write $\boldsymbol{\Upsilon}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} = \mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} = \mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}$. We get from this the first term on the RHS of (4.30). Similarly, we note $\boldsymbol{\Upsilon}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} = \mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm} \mathbf{S}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}$ to get the last term in (4.30) from the last term above. (We clarify these rewritings of the RHS above are done by construction in Definition 3.4.) This shows (4.30). The proof of (4.31) is identical (but use Definition 3.5 instead of Definition 3.4). □

We are now in position to prove the main result Proposition 3.7 of this section.

Proof of Proposition 3.7. Combine (4.28) with Lemmas 4.10 and 4.11. □

5. Relative entropy analysis

{section:stochl

This is the first of two sections which contain the key technical aspects of this paper. In summary, this section is organized in the following manner:

(1) First, we will construct a spatially inhomogeneous perturbation of $\mathbb{P}^0$ from Definition 2.1, which interpolates between the two typical densities at the two edges (determined by $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$ in (2.6)-(2.7)). We will then estimate the relative entropy for the law of $\boldsymbol{\phi}_{\mathrm{t}}$ at fixed times t with respect to this inhomogeneous measure, ultimately by differentiating said entropy in time and comparison estimates between relative entropies with respect to $\mathbb{P}^0$ and its inhomogeneous perturbation, respectively.

(2) Second, we will construct an auxiliary dynamics which has the spatially inhomogeneous perturbation of $\mathbb{P}^0$ as an invariant measure (which makes the analysis of this auxiliary SDE much more accessible). At the same time, we show that the relative entropy of the path-space measure induced by $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$ with respect to that of this auxiliary SDE is controlled uniformly in N and locally uniformly in time. This second estimate, more or less, shows that the auxiliary dynamics and the $\boldsymbol{\phi}_{\mathrm{t}}$-process are "not too far" from each other, in some sense. So, properties of the auxiliary dynamics, which are easier to obtain with the existence of an explicit

invariant measure, will carry over to the ϕ_t-process as well. Additionally, we provide an L^{2p}-version of this path-wise density estimate.

(3) We conclude this section by recording the key technical inputs which follows from our entropy estimates (as alluded to in point (2) above).

5.1. **One-time estimates.** We start by introducing the reference measure with respect to which we will estimate relative entropies at fixed times. Said measure is an inhomogeneous version of the grand-canonical measures in Definition 2.1. In what follows, we also introduce notation for the relative entropy and Dirichlet form.

definition:gc2}

Definition 5.1. With $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$ from (2.6)-(2.7), we define the following measure on $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$:

{eq:gc2I}

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}[\mathrm{d}\boldsymbol{\phi}] := \prod_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \mathcal{Z}_{\mathrm{x}}^{-1}\mathbf{Exp}(-\mathscr{U}[\boldsymbol{\phi}_{\mathrm{x}}] + \upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}})\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}}. \tag{5.1}$$

Above, $\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}}$ denotes the Lebesgue measure on $\boldsymbol{\phi}_{\mathrm{x}} \in \mathbb{R}$. The function $\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} : [\![1,\mathsf{N}]\!] \to \mathbb{R}$ is chosen so that:

- We have $\upsilon_1^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} = -\mathsf{N}^{-\frac{1}{2}}\widetilde{\mathbf{A}}$ and $\upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} = \mathsf{N}^{-\frac{1}{2}}\widetilde{\mathbf{B}}$.
- The gradient $\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is independent of $\mathrm{x} \in [\![1,\mathsf{N}-1]\!]$ (and is thus $\mathrm{O}(\mathsf{N}^{-3/2})$).

Now, let $\mathfrak{P}_{\mathrm{t}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ denote the density with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ for the law of $\boldsymbol{\phi}_{\mathrm{t}}$ for any deterministic time $\mathrm{t} \geqslant 0$.

We also let $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ denote the expectation with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$.

efinition:kldf}

Definition 5.2. Fix a probability measure $\mathbb{P}$ on $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$ (think of $\mathbb{P} = \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}, \mathbb{P}^0$). Suppose that $\mathfrak{P}$ is a probability density with respect to $\mathbb{P}$. We define the *relative entropy* of $\mathfrak{P}$ (or $\mathfrak{P}\mathrm{d}\mathbb{P}$) with respect to $\mathbb{P}$ to be

{eq:kldfI}

$$\mathsf{H}[\mathfrak{P}\mathrm{d}\mathbb{P}|\mathbb{P}] := \mathbb{E}^{\mathbb{P}}\mathfrak{P}\log\mathfrak{P}. \tag{5.2}$$

Above, $\mathbb{E}^{\mathbb{P}}$ is the expectation with respect to $\mathbb{P}$. For any smooth $\mathsf{F} : \mathbb{R}^{[\![1,\mathsf{N}]\!]}$, let the associated *Dirichlet form* be

{eq:kldfII}

$$\mathbf{D}[\mathsf{F}|\mathbb{P}] := \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}^{\mathbb{P}}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathsf{F}[\boldsymbol{\phi}]|^2 + \mathbb{E}^{\mathbb{P}}|\partial_{\boldsymbol{\phi}_1}\mathsf{F}[\boldsymbol{\phi}]|^2 + \mathbb{E}^{\mathbb{P}}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathsf{F}[\boldsymbol{\phi}]|^2. \tag{5.3}$$

We also introduce the convenient notation

{eq:kldfIII}

$$\mathbf{H}_{\mathrm{t}} := \mathsf{H}[\mathfrak{P}_{\mathrm{t}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}|\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}] \quad \text{and} \quad \mathbf{D}_{\mathrm{t}} := \mathbf{D}[(\mathfrak{P}_{\mathrm{t}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})^{\frac{1}{2}}|\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}]. \tag{5.4}$$

We clarify that $\mathbb{P}$ in Definition 5.2 does *not* need to be invariant for the ϕ_{t}-process (and in fact will often not be). However, the relative entropy (5.2) is still a tool to compare $\mathfrak{P}\mathrm{d}\mathbb{P}$ and $\mathbb{P}$, and the Dirichlet form (5.3) gives a measure of "energy" coming from the spin-swap conservation law dynamics in (2.5)-(2.7), and the Langevin dynamics at the edges $1, \mathsf{N} \in [\![1,\mathsf{N}]\!]$. Moreover, the measure (5.1) is constructed to interpolate between the two typical ϕ-densities determined by (2.6)-(2.7) dynamics.

Before we continue, it will be important to establish the following comparison estimates for relative entropies with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ and $\mathbb{P}^0$ (as well as an L^p-estimate which will not be used until Section 6, but whose proof is very similar and thus recorded immediately below).

a:kldfcompare}

Lemma 5.3. *Suppose $\widetilde{\mathbb{P}}$ is a probability measure on $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$ whose densities with respect to $\mathbb{P}^0, \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ are denoted by $\mathfrak{P}^0, \mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, respectively. We have the following comparison of relative entropies:*

:kldfcompareI}

$$\mathsf{H}[\mathfrak{P}^0\mathrm{d}\mathbb{P}^0|\mathbb{P}^0] \lesssim \mathsf{H}[\mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}|\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}] + 1 \lesssim \mathsf{H}[\mathfrak{P}^0\mathrm{d}\mathbb{P}^0|\mathbb{P}^0] + 1. \tag{5.5}$$

Moreover, for any $p \geqslant 1$, we have the estimate

kldfcompareII}

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\tfrac{\mathrm{d}\mathbb{P}^0}{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}}}}\Big)^p + \mathbb{E}^0\Big(\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}\Big)^p \lesssim_p 1. \tag{5.6}$$

Proof. By construction in (5.2), we have

$$\mathsf{H}[\mathfrak{P}^0\mathrm{d}\mathbb{P}^0|\mathbb{P}^0] = \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\log\mathfrak{P}^0 = \mathsf{H}[\mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}|\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}] + \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\log\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}.$$

We again change measure and then use the standard entropy inequality in Lemma A.3 to get

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathfrak{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\log\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0} = \mathbb{E}^0\mathfrak{P}^0\log\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0} \leqslant \tfrac{1}{2}\mathsf{H}[\mathfrak{P}^0\mathrm{d}\mathbb{P}^0|\mathbb{P}^0] + \tfrac{1}{2}\log\mathbb{E}^0\mathbf{Exp}(2\log\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0})$$

$$\leqslant \tfrac{1}{2}\mathbf{H}[\mathfrak{P}^0 \mathrm{d}\mathbb{P}^0 | \mathbb{P}^0] + \tfrac{1}{2}\log \mathbb{E}^0 |\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}|^2.$$

By the previous two displays, it suffices to show that $\mathbb{E}^0|\frac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}|^2 \lesssim 1$ in order to obtain the first estimate in (5.5). To this end, we write (by construction in (2.9) and (5.1))

$$\mathbb{E}^0|\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}|^2 = \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^2 \boldsymbol{\mathcal{Z}}_{\mathrm{x}}^{-2} \cdot \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathbb{E}^0 \mathbf{Exp}(2\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}}).$$

By Assumption 2.2, we know that if $\boldsymbol{\phi} \sim \mathbb{P}^0$, then $\boldsymbol{\phi}_{\mathrm{x}}$ is sub-Gaussian and mean-zero, and thus each expectation on the RHS of the previous display is $\mathbf{Exp}(\mathrm{O}(\mathsf{N}^{-1}))$ since $\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} = \mathrm{O}(\mathsf{N}^{-1/2})$ as noted in Definition 5.1. Therefore, the second product over $\mathrm{x} \in [\![1,\mathsf{N}]\!]$ on the RHS above is $\lesssim 1$. In order to control the first product, we observe that $\boldsymbol{\mathcal{Z}}_{\mathrm{x}}\boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^{-1} = \boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^{-1}\int_{\mathbb{R}} \mathbf{Exp}(-\mathscr{U}[\boldsymbol{\phi}_{\mathrm{x}}] + \upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}})\mathrm{d}\boldsymbol{\phi}_{\mathrm{x}} = \mathbb{E}^0\mathbf{Exp}(\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}})$. As we argued, this term is $\mathbf{Exp}(\mathrm{O}(\mathsf{N}^{-1}))$, and thus so is its reciprocal squared. This shows that the first product on the RHS above is $\lesssim 1$, which completes the proof of the first bound in (5.5). The proof of the second is identical (one can formally swap the 0 superscript with the $\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}$ superscript). This completes the proof of (5.5).

We now prove (5.6). Similar to the previous display, we have

$$\mathbb{E}^0\Big(\tfrac{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}}{\mathrm{d}\mathbb{P}^0}\Big)^p = \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^p \boldsymbol{\mathcal{Z}}_{\mathrm{x}}^{-p} \cdot \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathbb{E}^0 \mathbf{Exp}(p\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}}),$$

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\tfrac{\mathrm{d}\mathbb{P}^0}{\mathrm{d}\mathbb{P}^{\widetilde{\mathbf{A}}}}\Big)^p = \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^p \boldsymbol{\mathcal{Z}}_{\mathrm{x}}^{-p} \cdot \prod_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \mathbf{Exp}(-p\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}}).$$

As shown above, we know that $\boldsymbol{\mathcal{Z}}_{\mathscr{U},0}^{-1}\boldsymbol{\mathcal{Z}}_{\mathrm{x}} = \mathbf{Exp}(\mathrm{O}(\mathsf{N}^{-1}))$. Thus, the first product on the RHS is $\lesssim_p 1$ in each line above. Moreover, because $\boldsymbol{\phi}_{\mathrm{x}}$ is sub-Gaussian of mean $\mathrm{O}(\mathsf{N}^{-1/2})$ with respect to either of $\mathbb{P}^0, \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, and since the υ_{x}-constants above are $\lesssim \mathsf{N}^{-1/2}$, as discussed earlier, the expectations in each line above are $\lesssim \mathbf{Exp}(\mathrm{O}_p(\mathsf{N}^{-1}))$. So, the second product in each line above is $\lesssim_p 1$. This completes the proof of (5.6). □

We now give an estimate for the entropy production for $\mathfrak{P}_{\mathrm{t}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. For this, and for a number of other stochastic estimates in this paper, we will require notation for the infinitesimal generator of $\boldsymbol{\phi}_{\mathrm{t}}$.

{notation:gener

Notation 5.4. Define the operator $\mathscr{L} = \mathscr{L}_{\mathrm{S}} + \mathscr{L}_{\mathrm{A}} + \mathscr{L}_1 + \mathscr{L}_{\mathsf{N}}$, where

$$\mathscr{L}_{\mathrm{S}} := \mathsf{N}^2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}] - \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}])(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}) + \mathsf{N}^2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} (\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})^2,$$

$$\mathscr{L}_{\mathrm{A}} := \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]\partial_{\boldsymbol{\phi}_{\mathrm{x}}},$$

$$\mathscr{L}_1 := (-\mathsf{N}^2\mathscr{U}'[\boldsymbol{\phi}_1] - \mathsf{N}^{\frac{3}{2}}\widetilde{\mathbf{A}})\partial_{\boldsymbol{\phi}_1} + \partial_{\boldsymbol{\phi}_1}^2 \quad \text{and} \quad \mathscr{L}_{\mathsf{N}} := (-\mathsf{N}^2\mathscr{U}'[\boldsymbol{\phi}_{\mathsf{N}}] + \mathsf{N}^{\frac{3}{2}}\widetilde{\mathbf{B}})\partial_{\boldsymbol{\phi}_{\mathsf{N}}} + \partial_{\boldsymbol{\phi}_{\mathsf{N}}}^2.$$

By (2.5)-(2.7) and standard Itô theory, $\mathscr{L}$ is the infinitesimal generator of $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$.

{lemma:eprod}

Lemma 5.5. *Fix any $\mathrm{t} \in [0, \mathrm{O}(1)]$. There exists a universal constant $\eta > 0$ such that*

$$\tfrac{\mathrm{d}}{\mathrm{dt}}\mathbf{H}_{\mathrm{t}} \leqslant -\eta\mathsf{N}^2\mathbf{D}_{\mathrm{t}} + \mathrm{O}(1) + \mathrm{O}(\mathbf{H}_{\mathrm{t}}). \tag{5.7}$$ {eq:eprodI}

From (5.7), we deduce the estimate (see (2.13) for γ_{N}):

$$\sup_{\mathrm{t}\in[0,\mathrm{O}(1)]} \mathbf{H}_{\mathrm{t}} + \mathsf{N}^2 \sup_{\mathrm{T}\in[0,\mathrm{O}(1)]} \int_0^{\mathrm{T}} \mathbf{D}_{\mathrm{t}}\mathrm{dt} \lesssim \gamma_{\mathsf{N}}\log\mathsf{N}. \tag{5.8}$$ {eq:eprodII}

Proof. We first deduce (5.8) from (5.7). Take (5.7) but drop $-\mathrm{c}\mathsf{N}^2\mathbf{D}_{\mathrm{t}}$ therein. We use the Gronwall inequality to deduce that $\mathbf{H}_{\mathrm{t}} \lesssim \mathbf{H}_0$ locally uniformly in $\mathrm{t} \in [0,\infty)$. On the other hand, again from (5.7), we obtain the estimate $\mathsf{N}^2\int_0^{\mathrm{T}} \mathbf{D}_{\mathrm{t}}\mathrm{dt} \lesssim \mathrm{T}(1 + \sup_{\mathrm{t}\in[0,\mathrm{T}]}\mathbf{H}_{\mathrm{t}}) \lesssim 1 + \mathbf{H}_0$ if $\mathrm{T} = \mathrm{O}(1)$, where the last bound follows by our estimate on relative entropies. In particular, we deduce that (5.8) holds if we replace the RHS therein with $\mathbf{H}_0$. However, by Lemma 5.3 and the relative entropy estimate (2.13), we know that $\mathbf{H}_0 \lesssim \gamma_{\mathsf{N}}\log\mathsf{N}$, so (5.8) follows.

We are left to establish (5.7). Recall the generator $\mathscr{L}$ for $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$. By the Kolmogorov backward equation and (5.4), we have the following, in which, just for this proof (for convenience), we let $\mathbb{E} = \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ be the expectation

with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, and we will use the notation $\mathfrak{P} := \mathfrak{P}_{\mathrm{t}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$:

$$\tfrac{\mathrm{d}}{\mathrm{dt}}\mathsf{H}_{\mathrm{t}} = \mathbb{E}\mathfrak{P}(\partial_{\mathrm{t}} + \mathscr{L})\log\mathfrak{P} = \mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{S}}\log\mathfrak{P} + \mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{A}}\log\mathfrak{P} + \mathbb{E}\mathfrak{P}\mathscr{L}_{1}\log\mathfrak{P} + \mathbb{E}\mathfrak{P}\mathscr{L}_{\mathsf{N}}\log\mathfrak{P}. \tag{5.9}$$ {eq:eprodI1}

We clarify that the second equality in (5.9) follows because $\mathbb{E}\mathfrak{P}\partial_{\mathrm{t}}\log\mathfrak{P} = \mathbb{E}\partial_{\mathrm{t}}\mathfrak{P} = \frac{\mathrm{d}}{\mathrm{dt}}\mathbb{E}\mathfrak{P} = \frac{\mathrm{d}}{\mathrm{dt}}1 = 0$. Now, let us note that by construction in Definition 5.1, the Langevin generator $\mathscr{L}_1$ (of speed N^2) is self-adjoint with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. Indeed, $\mathscr{L}_1$ is a Langevin generator for $\boldsymbol{\phi}_1$ with potential $-\mathscr{U}[\boldsymbol{\phi}_1] - \mathsf{N}^{-1/2}\widetilde{\mathbf{A}}\boldsymbol{\phi}_1$, while $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is a product measure on $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$ whose marginal at $\boldsymbol{\phi}_1$ has potential $-\mathscr{U}[\boldsymbol{\phi}_1] - \mathsf{N}^{-1/2}\widetilde{\mathbf{A}}\boldsymbol{\phi}_1$. Therefore, we have the following standard generator formula for Itô SDEs (which can be verified via direct integration-by-parts):

$$\mathbb{E}\mathfrak{P}\mathscr{L}_1\log\mathfrak{P} = -\mathrm{c}\mathsf{N}^2\mathbb{E}(\partial_{\boldsymbol{\phi}_1}\mathfrak{P}\cdot\partial_{\boldsymbol{\phi}_1}\log\mathfrak{P}) = -\mathrm{c}\mathsf{N}^2\mathbb{E}(\tfrac{|\partial_{\boldsymbol{\phi}_1}\mathfrak{P}|^2}{\mathfrak{P}}) = -\mathrm{c}_1\mathsf{N}^2\mathbb{E}(|\partial_{\boldsymbol{\phi}_1}\mathfrak{P}^{\frac{1}{2}}|^2). \tag{5.10}$$ {eq:eprodI2a}

Above, $\mathrm{c}, \mathrm{c}_1 > 0$ are fixed. By the same token, we have

$$\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathsf{N}}\mathfrak{P} = -\mathrm{c}_1\mathsf{N}^2\mathbb{E}(|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{\frac{1}{2}}|^2). \tag{5.11}$$ {eq:eprodI2b}

We are left to analyze the first two terms on the far RHS of (5.9). We start with the first; we write

$$\mathscr{L}_{\mathrm{S}} := \mathscr{L}_{\mathrm{S},1} + \mathscr{L}_{\mathrm{S},2} := \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathscr{L}_{\mathrm{S},1}^{\mathrm{x},\mathrm{x}+1} + \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathscr{L}_{\mathrm{S},2}^{\mathrm{x},\mathrm{x}+1},$$

where the bond operators are given by

$$\mathscr{L}_{\mathrm{S},1}^{\mathrm{x},\mathrm{x}+1} := \mathsf{N}^2(\{\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}] - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\} - \{\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}] - \upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\})(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}) + \mathsf{N}^2(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})^2,$$
$$\mathscr{L}_{\mathrm{S},2}^{\mathrm{x},\mathrm{x}+1} := \mathsf{N}^2(\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})(\partial_{\boldsymbol{\phi}_{\mathrm{x}}+1} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}).$$

The operator $\mathscr{L}_{\mathrm{S},1}^{\mathrm{x},\mathrm{x}+1}$ is self-adjoint with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ for any $\mathrm{x}, \mathrm{x}+1 \in [\![1,\mathsf{N}]\!]$; this follows by integration-by-parts (see Section 2 in [7], for instance). In particular, by the same token as (5.10)-(5.11), we have

$$\begin{aligned}\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{S},1}\log\mathfrak{P} &= -\mathrm{c}\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}\{(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}\cdot(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\log\mathfrak{P}\}\\ &= -\mathrm{c}_1\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}^{\frac{1}{2}}|^2.\end{aligned}$$

On the other hand, we have, for some $\mathrm{C} \lesssim 1$ and any $\varepsilon > 0$, that

$$\begin{aligned}\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{S},2}\log\mathfrak{P} &= \mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}(\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\cdot\mathbb{E}\{\mathfrak{P}\cdot(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\log\mathfrak{P}\}\\ &= \mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}(\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\cdot\mathbb{E}\Big(\mathfrak{P}^{\frac{1}{2}}\cdot\tfrac{(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}}{\mathfrak{P}^{\frac{1}{2}}}\Big)\\ &\leqslant \mathrm{C}\varepsilon^{-1}\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}|\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}|^2\cdot\mathbb{E}\mathfrak{P} + \varepsilon\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}^{\frac{1}{2}}|^2\\ &\lesssim \varepsilon^{-1} + \varepsilon\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}^{\frac{1}{2}}|^2.\end{aligned}$$

The third line uses the Schwarz inequality and $\partial_{\boldsymbol{\phi}_{\mathrm{x}}}\mathfrak{P} \propto \mathfrak{P}^{1/2}\partial_{\boldsymbol{\phi}_{\mathrm{x}}}\mathfrak{P}^{1/2}$. The last line uses $\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \lesssim \mathsf{N}^{-3/2}$ and $\mathbb{E}\mathfrak{P} = 1$ (since $\mathfrak{P}$ is a probability density). We will now combine the previous two displays (with $\varepsilon > 0$ small enough depending on $\mathrm{c}_1 > 0$ but fixed) to get the following for some $\eta > 0$ fixed:

$$\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{S}}\log\mathfrak{P} \leqslant -\eta\mathsf{N}^2\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})\mathfrak{P}^{\frac{1}{2}}|^2 + \mathrm{O}(1). \tag{5.12}$$ {eq:eprodI3}

We will now estimate the $\mathscr{L}_{\mathrm{A}}$-term in (5.9) and plug in the formula for $\mathscr{L}_{\mathrm{A}}$ (from Notation 5.4) therein. When we do so, we note that $\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]$ (see (2.8)) does *not* depend on $\boldsymbol{\phi}_{\mathrm{x}}$. Thus, we can integrate-by-parts (using the formula (5.1) to compute $\boldsymbol{\phi}_{\mathrm{x}}$-derivatives of the Lebesgue density therein) to get

$$\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{A}}\log\mathfrak{P} = \mathsf{N}^{\frac{3}{2}}\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]}\mathbb{E}(\mathfrak{P}\cdot\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]\cdot\partial_{\boldsymbol{\phi}_{\mathrm{x}}}\log\mathfrak{P})$$

$$= \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}(\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}] \cdot \partial_{\boldsymbol{\phi}_{\mathrm{x}}} \mathfrak{P})$$
$$= \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}\{\mathfrak{P} \cdot (\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}] - \upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}) \cdot \widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]\}.$$

We now plug in the formula (2.8) for $\widetilde{\mathbf{F}}$. When we do so, the contribution from $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}]$ vanishes because it is a telescoping sum. In particular, the previous display extends to the following (note that below, we can restrict to $\mathrm{x}, \mathrm{x}+1 \in [\![1, \mathsf{N}-1]\!]$ since $\mathscr{U}'[\boldsymbol{\phi}_{\mathsf{N}+1}] = 0$ by convention):

$$\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{A}} \log \mathfrak{P} = -\tfrac{1}{3}\beta \mathsf{N}^{\frac{3}{2}} \mathbb{E}\Big(\mathfrak{P} \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+2}]\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}-1}]\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}-2}]\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big)$$
$$= \mathrm{c}_2 \mathbb{E}\Big(\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+2}]\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big).$$

The last line holds for some $\mathrm{c}_2 \in \mathbb{R}$ fixed; on the RHS of the first line, we replace the first factor of $\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ by $\upsilon_{\mathrm{x}+3}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ to turn the resulting sum into a telescoping one that vanishes. The cost in doing so is given exactly by the second line above, because the spatial discrete gradient of $\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is constant of order $\mathsf{N}^{-3/2}$ by construction in Definition 5.1. Next, we replace $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+2}]$ by $\mathscr{U}'[\boldsymbol{\phi}_{\mathsf{N}}]$; that is, we write

$$\mathbb{E}\Big(\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+2}]\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) = \mathrm{I} + \mathrm{II}$$
$$:= \mathbb{E}\Big(\mathfrak{P}\mathscr{U}'[\boldsymbol{\phi}_{\mathsf{N}}] \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathbb{E}\Big(\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}] \sum_{\mathrm{w}\in[\![\mathrm{x}+2,\mathsf{N}-1]\!]} (\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}}] - \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}+1}])\Big)$$

For I, we replace $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}}] \mapsto \mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}}] - \upsilon_{\mathrm{w}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ for each w above. We then proceed with a number of manipulations to be explained afterwards to obtain the following, in which $\mathscr{Y}[\boldsymbol{\phi}_{\mathrm{w}}] := \mathscr{U}[\boldsymbol{\phi}_{\mathrm{w}}] - \upsilon_{\mathrm{w}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{w}}$ for convenience:

$$\mathrm{I} = \mathbb{E}\Big(\mathfrak{P} \cdot \mathscr{Y}'[\boldsymbol{\phi}_{\mathsf{N}}] \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathbb{E}\Big(\mathfrak{P} \cdot \upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathbb{E}\Big(\mathfrak{P} \cdot \mathscr{U}[\boldsymbol{\phi}_{\mathsf{N}}] \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big)$$
$$= \mathbb{E}\Big(\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathbb{E}\Big(\mathfrak{P} \cdot \upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathrm{O}(\mathsf{N}^{\frac{1}{2}})\mathbb{E}(\mathfrak{P} \cdot \mathscr{U}'[\boldsymbol{\phi}_{\mathsf{N}}])$$
$$= \mathbb{E}\Big(\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathbb{E}\Big(\mathfrak{P} \cdot \upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) + \mathrm{O}(\mathsf{N}^{\frac{1}{2}})\mathbb{E}(\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}) + \mathrm{O}(1).$$

The first line follows by the explanation given above. The second line follows from integration-by-parts to turn $\mathscr{Y}'[\boldsymbol{\phi}_{\mathsf{N}}]$ into $\partial_{\boldsymbol{\phi}_{\mathsf{N}}}$, noting that the sum over $\mathrm{x}, \mathrm{x}+1$ does not depend on $\boldsymbol{\phi}_{\mathsf{N}}$. We also estimate the last term in the first line by using $\upsilon \lesssim \mathsf{N}^{-1/2}$, which produces the final term in the second line. For the last line, we again replace $\mathscr{U} \mapsto \mathscr{Y}$ in the last term of the second line, which provides the last two terms in the previous display. Next, let us control the first term in the last line. To this end, we write $\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} = \partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} \cdot \mathfrak{P}^{-1/2} \cdot \mathfrak{P}^{1/2} \propto \partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{1/2} \cdot \mathfrak{P}^{1/2}$. After this, we apply the Schwarz inequality and the standard entropy inequality (see Lemma A.3) to get the following for any $\varepsilon, \delta > 0$ small but independent of N, where the last line is clarified afterwards:

$$\mathbb{E}\Big(\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) \leqslant \varepsilon\mathsf{N}^2\mathbb{E}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{\frac{1}{2}}|^2 + \varepsilon^{-1}\mathsf{N}^{-2}\mathbb{E}\Big\{\mathfrak{P}\Big(\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)^2\Big\}$$
$$\leqslant \varepsilon\mathsf{N}^2\mathbb{E}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{\frac{1}{2}}|^2 + \varepsilon^{-1}\delta^{-1}\mathbf{H}_{\mathrm{t}} + \varepsilon^{-1}\delta^{-1}\log\mathbb{E}\mathbf{Exp}\Big\{\delta\mathsf{N}^{-2}\Big(\sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)^2\Big\}$$
$$\leqslant \varepsilon\mathsf{N}^2\mathbb{E}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{\frac{1}{2}}|^2 + \mathrm{O}(\mathbf{H}_{\mathrm{t}}) + \mathrm{O}(1).$$

To obtain the last line above, we note that by Assumption 2.2, the random variables $\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]$ are (independent) sub-Gaussian mean-zero random variables, where the mean-zero property follows by a direct computation using (5.1) and integration-by-parts. (These properties are all with respect to the measure in $\mathbb{E}$.) Thus, the sum over $\mathrm{x}, \mathrm{x}+1$ in the second line above is sub-Gaussian with variance parameter $\lesssim \mathsf{N}$, so the moment generating

function of $\mathrm{O}(\mathsf{N}^{-2})$ times its square is uniformly bounded. By the same token, since υ-parameters herein are all $\lesssim \mathsf{N}^{-1/2}$,

$$\mathbb{E}\Big(\mathfrak{P}\cdot \upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) \leqslant \mathbf{H}_{\mathrm{t}} + \log \mathbb{E}\mathbf{Exp}\Big(\upsilon_{\mathsf{N}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\cdot \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) \lesssim \mathbf{H}_{\mathrm{t}}+1.$$

Now, we again use $\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P} \propto \partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{1/2}\cdot\mathfrak{P}^{1/2}$, so that by the Schwarz inequality, we have

$$\mathsf{N}^{\frac{1}{2}}\mathbb{E}(\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}) \lesssim \mathsf{N}^2\mathbb{E}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}}\mathfrak{P}^{\frac{1}{2}}|^2 + \mathsf{N}^{-\frac{3}{2}}.$$

This completes our analysis of I, which we summarize later in the proof. We are left with II, which we rewrite as

$$\begin{aligned}
\mathrm{II} &= \mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}+1}]-\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}}])\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&= \mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]-\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}])\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&+ \mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathrm{O}(\mathsf{N}^{-\frac{3}{2}})\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&+ \mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]-\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}])\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathrm{O}(\mathsf{N}^{-\frac{1}{2}})\Big)\\
&+ \mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathrm{O}(\mathsf{N}^{-\frac{3}{2}})\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathrm{O}(\mathsf{N}^{-\frac{1}{2}})\Big).
\end{aligned}$$

where the deterministic $\mathrm{c}_{\mathrm{x},\mathrm{w}}$-coefficients satisfy the following. First, we get $\mathrm{c}_{\mathrm{x},\mathrm{w}}=0$ if $\mathrm{x}+1\geqslant \mathrm{w}$. (Indeed, note that in II, the difference of $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{w}}]$-terms has no overlap with any of the $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{x}}]$-terms therein.) Second, we have $\mathrm{c}_{\mathrm{x},\mathrm{w}}\lesssim 1$. The second identity above follows from then replacing each $\mathscr{U}$ by $\mathscr{Y}$ and accounting for the errors in doing so. We clarify that all O-terms above are deterministic. By the same entropy inequality and sub-Gaussian argument as above, we have

$$\begin{aligned}
&\mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathrm{O}(\mathsf{N}^{-\frac{3}{2}})\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big) \leqslant \max_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbb{E}\Big(\mathfrak{P}\cdot \Big|\mathsf{N}^{-\frac{1}{2}}\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big|\Big)\\
&\lesssim \mathbf{H}_{\mathrm{t}} + \log \mathbb{E}\mathbf{Exp}\Big(\Big|\mathsf{N}^{-\frac{1}{2}}\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big|\Big) \lesssim \mathbf{H}_{\mathrm{t}}+1.
\end{aligned}$$

Next, we observe (based on the bound $\mathrm{c}_{\mathrm{x},\mathrm{w}}\lesssim 1$) that

$$\mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathrm{O}(\mathsf{N}^{-\frac{3}{2}})\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathrm{O}(\mathsf{N}^{-\frac{1}{2}})\Big) \lesssim 1.$$

Let us now deal with terms that have a difference of the form $\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]-\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}]$. By the same integration-by-parts argument (with the Schwarz inequality) as before, we get (for $\varepsilon,\delta>0$ small but independent of N)

$$\begin{aligned}
&\mathbb{E}\Big(\mathfrak{P}\cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]-\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}])\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&= \mathbb{E}\Big(\sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}}-\partial_{\boldsymbol{\phi}_{\mathrm{w}}})\mathfrak{P}\cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&\propto \mathbb{E}\Big(\sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}}-\partial_{\boldsymbol{\phi}_{\mathrm{w}}})\mathfrak{P}^{\frac{1}{2}}\cdot\mathfrak{P}^{\frac{1}{2}}\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big)\\
&\leqslant \varepsilon\mathsf{N}^2 \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}}-\partial_{\boldsymbol{\phi}_{\mathrm{w}}})\mathfrak{P}^{\frac{1}{2}}|^2 + \varepsilon^{-1}\mathsf{N}^{-2}\sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}\Big(\mathfrak{P}\cdot\Big|\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\Big|^2\Big)
\end{aligned}$$

$$\leqslant \varepsilon \mathsf{N}^2 \mathbf{D}_{\mathrm{t}} + \varepsilon^{-1} \max_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbb{E}\Big(\mathfrak{P} \cdot \mathsf{N}^{-1} \Big| \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}] \Big|^2 \Big)$$

$$\leqslant \varepsilon \mathsf{N}^2 \mathbf{D}_{\mathrm{t}} + \varepsilon^{-1}\delta^{-1} \mathbf{H}_{\mathrm{t}} + \varepsilon^{-1}\delta^{-1} \log \mathbb{E}\mathbf{Exp}\Big(\delta \mathsf{N}^{-1} \Big| \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}] \Big|^2 \Big) \leqslant \varepsilon \mathsf{N}^2 \mathbf{D}_{\mathrm{t}} + \varepsilon^{-1}\delta^{-1} \mathbf{H}_{\mathrm{t}} + \mathrm{O}(1).$$

By the same token (though without the need for an entropy inequality step), we have

$$\begin{aligned} &\mathbb{E}\Big(\mathfrak{P} \cdot \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}]) \cdot \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathrm{c}_{\mathrm{x},\mathrm{w}} \mathrm{O}(\mathsf{N}^{-\frac{1}{2}}) \Big) \\ &= \mathrm{O}(\mathsf{N}^{-\frac{1}{2}}) \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathbb{E}\Big((\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{w}}}) \mathfrak{P}^{\frac{1}{2}} \cdot \mathfrak{P}^{\frac{1}{2}} \Big) \\ &\leqslant \varepsilon \mathsf{N}^2 \sum_{\mathrm{w},\mathrm{w}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{w}}}) \mathfrak{P}^{\frac{1}{2}}|^2 + \mathrm{O}(1). \end{aligned}$$

We now combine the previous eleven displays to obtain the estimate below for $\varepsilon > 0$ small:

$$\mathbb{E}\mathfrak{P}\mathscr{L}_{\mathrm{A}} \log \mathfrak{P} \lesssim \varepsilon \mathsf{N}^2 \mathbb{E}|\partial_{\boldsymbol{\phi}_{\mathsf{N}}} \mathfrak{P}^{\frac{1}{2}}|^2 + \varepsilon \mathsf{N}^2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathbb{E}|(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}) \mathfrak{P}^{\frac{1}{2}}|^2 + \mathbf{H}_{\mathrm{t}} + 1. \tag{5.13}$$ {eq:eprodI4}

We now combine (5.9), (5.10), (5.11), (5.12), and (5.13) to obtain the following for some $\eta > 0$ fixed:

$$\tfrac{\mathrm{d}}{\mathrm{dt}} \mathbf{H}_{\mathrm{t}} \leqslant -\eta \mathsf{N}^2 \mathbf{D}_{\mathrm{t}} + \mathrm{O}(1) + \mathrm{O}(\mathbf{H}_{\mathrm{t}}),$$

which is exactly the desired estimate (5.7). This completes the proof. □

5.2. **Pathwise estimates.** We now turn the ideas in the proof of Lemma 5.5 into the construction of a perturbed dynamics (off of (2.5)-(2.7)), whose relative entropy with respect to the actual dynamics (2.5)-(2.7) is controlled uniformly in N and locally uniformly in time. The general idea is that the proof of Lemma 5.5 above is based on estimating terms that arise from the fact that $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is *not* invariant for $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$. However, a careful inspection of that proof shows that each such term is given by sums over $\mathrm{w}, \mathrm{w}+1 \in [\![1,\mathsf{N}]\!]$ of operators of the form $\partial_{\boldsymbol{\phi}_{\mathrm{w}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{w}}}$ times w-dependent coefficients whose (squared) $\ell^2([\![1,\mathsf{N}]\!])$-norms (in expectation) can be estimated in terms of $\mathbf{H}_{\mathrm{t}}$, which we have now shown to be $\lesssim 1$. Other terms which show up have the form $\partial_{\boldsymbol{\phi}_{\mathrm{w}}}$ for $\mathrm{w} = 1$ multiplied by terms whose second moments can also be estimated via $\mathbf{H}_{\mathrm{t}}$. *In particular, the topology in which we have control over these coefficients is exactly the one that allows us to apply the Girsanov theorem to construct a path-space measure which essentially "removes" these problematic terms which break the invariance of* $\widetilde{\mathbb{P}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$*, while at the same time having relative entropy with respect to the* $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$ *path-space measure which is bounded uniformly in* N *and locally uniformly in time.* With a similar procedure, we can also establish an L^{2p}-version of this path-wise law estimate, but only for short times.

Let us make the above precise. We begin with a (somewhat abstract) construction of the path-space measure.

{definition:pat

Definition 5.6. We construct a probability measure $\mathbb{P}^{\mathrm{path},\sim}$ on the path-space $\mathscr{C}([0,\infty), \mathbb{R}^{[\![1,\mathsf{N}]\!]})$ as follows. We let $\mathrm{t} \mapsto \boldsymbol{\psi}_{\mathrm{t}}$ be the $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$-valued Itô SDE whose infinitesimal generator is given by $\mathscr{L}^{\sim} := \mathscr{L} + \mathscr{L}_{\mathrm{S}}^{\sim} + \mathscr{L}_{\mathrm{A}}^{\sim} + \mathscr{L}_{\mathsf{N}}^{\sim}$, where $\mathscr{L}$ is from Notation 5.4, and

$$\begin{aligned} \mathscr{L}_{\mathrm{S}}^{\sim} &:= -\mathsf{N}^{\frac{1}{2}} \mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} (\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}), \\ \mathscr{L}_{\mathrm{A}}^{\sim} &:= \tfrac{2}{3} \mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \beta_2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}] \Big) (\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}) \\ &\quad + \tfrac{1}{3} \beta_2 \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}\leqslant\mathrm{x}} (\upsilon_{\mathrm{w}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \upsilon_{\mathrm{w}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{w}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \upsilon_{\mathrm{w}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}) \Big) (\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}), \\ \mathscr{L}_{\mathsf{N}}^{\sim} &:= -\tfrac{2}{3} \mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \beta_2 \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \cdot \partial_{\boldsymbol{\phi}_{\mathsf{N}}} - \tfrac{1}{3} \beta_2 \mathsf{N}^{\frac{3}{2}} \Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}) \Big) \partial_{\boldsymbol{\phi}_{\mathsf{N}}}. \end{aligned}$$

Above, we used the notation $\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} = \mathsf{N}^{3/2}(\upsilon_2^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_1^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}) \lesssim 1$ (though any $(\mathrm{x},\mathrm{x}+1)$ in place of $(1,2)$ would yield the same quantity because $\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is linear). We also recall the fixed constant $\beta_2 \neq 0$ from (2.4).

In particular, let $\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}$ be the law of this process restricted to the time-interval $[0,\mathrm{T}]$ (for any deterministic $\mathrm{T} > 0$, and $\mathbb{P}^{\mathrm{path},\sim}$ itself is a the limit of (the consistent family) $\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}$-measures as $\mathrm{T} \to \infty$. Finally, the law of the initial data of $\boldsymbol{\psi}_{\mathrm{t}}$ is chosen depending on the context.

We check that the measure $\mathbb{P}^{\mathrm{path},\sim}$ constructed above, driven by an Itô SDE, has $\widetilde{\mathbb{P}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ as an invariant measure. We clarify that the result below does not make any statements about symmetric and anti-symmetric parts of the generator of this Itô SDE, since it will not be important for us. However, computing these is readily available.

{lma:pathspace1}

Lemma 5.7. *The* SDE *constructed in Definition 5.6 has* $\widetilde{\mathbb{P}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ *as an invariant measure.*

Proof. By Echeverría's criterion [9], it suffices to obtain $\mathscr{L}^{\sim}\mathbf{1} = 0$, in which $\mathbf{1}$ is the constant function (equal to 1 with probability 1) and $*$ denotes the adjoint with respect to $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. As in the proof of Lemma 5.5, we claim

$$\begin{aligned}\mathscr{L}^{\sim,*}\mathbf{1} &= \mathscr{L}^{*}\mathbf{1} + \mathscr{L}_{\mathrm{S}}^{\sim,*}\mathbf{1} + \mathscr{L}_{\mathrm{A}}^{\sim} + \mathscr{L}_{\mathsf{N}}^{\sim} \\ &= \mathscr{L}_{\mathrm{S}}^{*}\mathbf{1} + \mathscr{L}_{\mathrm{A}}^{*}\mathbf{1} + \mathscr{L}_{\mathrm{S}}^{\sim,*}\mathbf{1} + \mathscr{L}_{\mathrm{A}}^{\sim,*}\mathbf{1} + \mathscr{L}_{\mathsf{N}}^{\sim,*}\mathbf{1}.\end{aligned}$$

Indeed, the $\mathscr{L}_1, \mathscr{L}_{\mathsf{N}}$ are generators for Langevin diffusions with potentials $\mathscr{U}[\boldsymbol{\phi}_{\mathrm{x}}] - \mathsf{N}^{-1/2}\upsilon_{\mathrm{x}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{x}}$, for $\mathrm{x} = 1, \mathsf{N}$, respectively. Thus, these generators are self-adjoint differential operators, hence their vanishing after acting on $\mathbf{1}$ and taking expectations. Now, recall Notation 5.4, and define

$$\mathscr{L}_{\mathrm{S},\mathscr{Y}} := \mathsf{N}^2 \sum_{\mathrm{x},\mathrm{x}+1 \in [\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}])(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}) + \mathsf{N}^2 \sum_{\mathrm{x},\mathrm{x}+1 \in [\![1,\mathsf{N}]\!]} (\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}})^2,$$

where $\mathscr{Y}[\boldsymbol{\phi}_{\mathrm{w}}] := \mathscr{U}[\boldsymbol{\phi}_{\mathrm{w}}] - \upsilon_{\mathrm{w}}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\boldsymbol{\phi}_{\mathrm{w}}$ as in the proof of Lemma 5.5. Now, by a standard integration-by-parts using the Lebesgue density for $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ from Definition 5.1 shows $\mathscr{L}_{\mathrm{S},\mathscr{Y}}$ is a self-adjoint differential operator with respect to $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. Thus, we have the following, in which we use notation from Definition 5.6:

$$\mathscr{L}_{\mathrm{S}}^{*}\mathbf{1} = (\mathscr{L}_{\mathrm{S}} - \mathscr{L}_{\mathrm{S},\mathscr{Y}})^{*}\mathbf{1} = -\mathscr{L}_{\mathrm{S}}^{\sim,*}\mathbf{1}.$$

Now, we use Notation 5.4 to compute $\mathscr{L}_{\mathrm{A}}^{*}\mathbf{1}$. As in the proof of Lemma 5.5, we integrate-by-parts to get

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathsf{F}\mathscr{L}_{\mathrm{A}}^{*}\mathbf{1} = \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(\mathsf{F}[\boldsymbol{\phi}] \cdot \widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}] \cdot \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}]) - \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]\partial_{\boldsymbol{\phi}_{\mathrm{x}}}\mathsf{F}[\boldsymbol{\phi}]).$$

Thus, modulo a differential operator, the adjoint $\mathscr{L}_{\mathrm{A}}^{*}$ is given by multiplication, so that

$$\mathscr{L}_{\mathrm{A}}^{*}\mathbf{1} = \mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} \widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}] \cdot \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}].$$

We now replace each $\mathscr{U}$ in $\widetilde{\mathbf{F}}[\tau_{\mathrm{x}}\boldsymbol{\phi}]$ (see (2.8)) by $\mathscr{Y}$. That is, we write

$$\begin{aligned}\mathscr{L}_{\mathrm{A}}^{*}\mathbf{1} &= \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+2}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}-1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}-2}]) \\ &+ \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+2}]\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] - \mathscr{Y}[\boldsymbol{\phi}_{\mathrm{x}}]\upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}-2}]) \\ &+ \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] - \mathscr{Y}[\boldsymbol{\phi}_{\mathrm{x}}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}-1}]\upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}) \\ &+ \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \\ &= \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] + \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x} \in [\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}].\end{aligned}$$

The last line follows via $\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+2}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] = \frac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] + \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}]\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}}$ and removing all telescoping sums. (Above, we extend $\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ from $[\![1,\mathsf{N}]\!]$ to $\mathbb{Z}$ by zero.) Now, we write

$$\begin{aligned}
\tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}]\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] &= \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}] \\
&= \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2\mathscr{Y}'[\boldsymbol{\phi}_{\mathsf{N}}] \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \\
&+ \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \sum_{\mathrm{w}\in[\![\mathrm{x}+1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]) \\
&= \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2\mathscr{Y}'[\boldsymbol{\phi}_{\mathsf{N}}] \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \\
&- \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}+1}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}]) \cdot \sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}] \\
&= \mathscr{P}_1^* + \mathscr{P}_2^*,
\end{aligned}$$

where, by integration-by-parts, the operators $\mathscr{P}_1, \mathscr{P}_2$ are given by

$$\begin{aligned}
\mathscr{P}_1 &:= \tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x}\in[\![1,\mathsf{N}-1]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \cdot \partial_{\boldsymbol{\phi}_{\mathsf{N}}}, \\
\mathscr{P}_2 &:= -\tfrac{2}{3}\mu^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\beta_2 \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}}\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}]\Big)(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}).
\end{aligned}$$

We clarify the important aspect that the coefficients for the differential operators in $\mathscr{P}_1, \mathscr{P}_2$ do not depend on the variable with respect to which said differential acts. Similarly, we write

$$\begin{aligned}
&\tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+2}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+1} - \upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-1}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-2})\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{x}}] \\
&= \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}}\Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+2}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+1} - \upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-1}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-2})\Big)\mathscr{Y}'[\boldsymbol{\phi}_{\mathsf{N}}] \\
&+ \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}}\Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+2}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+1} - \upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-1}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-2})\Big) \sum_{\mathrm{w}\in[\![\mathrm{x},\mathsf{N}-1]\!]} (\mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}}] - \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{w}+1}]) \\
&= \mathscr{P}_3^* + \mathscr{P}_4^*,
\end{aligned}$$

where, again by integration-by-parts, the operators $\mathscr{P}_3, \mathscr{P}_4$ are given by

$$\begin{aligned}
\mathscr{P}_3 &:= \tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}}\Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+2}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}+1} - \upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-1}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{x}-2})\Big)\partial_{\boldsymbol{\phi}_{\mathsf{N}}}, \\
\mathscr{P}_4 &:= -\tfrac{1}{3}\beta_2\mathsf{N}^{\frac{3}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}\leqslant\mathrm{x}}(\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{w}+2}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{w}+1} - \upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{w}-1}\upsilon^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\mathrm{w}-2})\Big)(\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}).
\end{aligned}$$

Ultimately, by gathering our computations thus far, we have

$$\mathscr{L}^{\sim,*}\mathbf{1} = \mathscr{P}_1^*\mathbf{1} + \mathscr{P}_2^*\mathbf{1} + \mathscr{P}_3^*\mathbf{1} + \mathscr{P}_4^*\mathbf{1} + \mathscr{L}_{\mathrm{A}}^{\sim,*}\mathbf{1} + \mathscr{L}_{\mathsf{N}}^{\sim,*}\mathbf{1}.$$

It now suffices to note that $\mathscr{P}_1 + \ldots + \mathscr{P}_4 + \mathscr{L}_{\mathrm{A}}^{\sim} + \mathscr{L}_{\mathsf{N}}^{\sim,*} = 0$ by construction in Definition 5.6. □

Now, we show below the desired relative entropy control on $\mathbb{P}^{\mathrm{path},\sim}$.

{lemma:pathspac

Lemma 5.8. *Fix any time $\mathrm{T} \in [0, \mathrm{O}(1)]$, and let $\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}$ denote the path-space measure on $\mathscr{C}([0,\mathrm{T}],\mathbb{R}^{[\![1,\mathsf{N}]\!]})$ induced by the* SDE (2.5)-(2.7) *for $\mathrm{t} \mapsto \boldsymbol{\phi}_{\mathrm{t}}$. Moreover, suppose that the* SDEs *defining $\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}, \mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}$ have the same initial data distributed via $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ from Definition 5.1. Letting $\mathbb{E}^{\mathrm{path},\sim}$ be the expectation with respect to $\mathbb{P}^{\mathrm{path},\sim}$, we have*

$$\mathbf{H}[\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}|\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}] := \mathbb{E}^{\mathrm{path},\sim}\Big(\frac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}} \log \frac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}}\Big) \lesssim_{\mathrm{T}} 1. \tag{5.14}$$

{eq:pathspace2

Proof. As noted earlier, the key to the proof of (5.14) is that we can compute the Radon-Nikodym derivatives in (5.14) via the Girsanov theorem and the SDE in Definition 5.6. We claim, with explanation after, that

pathspace2I1a}
$$\frac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}} = \mathbf{Exp}\Big(\mathrm{c}_1\mathsf{N}^{-\frac{1}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathbf{m}_{\mathrm{T},\mathrm{x},\mathrm{x}+1} - \tfrac{1}{2}\mathrm{c}_1^2\mathsf{N}^{-1} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \mathrm{T}\Big) \tag{5.15}$$

pathspace2I1b}
$$\times \mathbf{Exp}\Big(\mathrm{c}_2\mathsf{N}^{-1} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \int_0^{\mathrm{T}} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{w}}]\Big)\mathrm{d}\mathbf{m}_{\mathrm{t},\mathrm{x},\mathrm{x}+1}\Big) \tag{5.16}$$

pathspace2I1c}
$$\times \mathbf{Exp}\Big(-\tfrac{1}{2}\mathrm{c}_2^2\mathsf{N}^{-2} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \int_0^{\mathrm{T}} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{w}}]\Big)^2 \mathrm{dt}\Big) \tag{5.17}$$

pathspace2I1d}
$$\times \mathbf{Exp}\Big(\mathrm{c}_3\mathsf{N}^{\frac{1}{2}} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}\leqslant\mathrm{x}} (\upsilon_{\mathrm{w}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{w}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{w}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{w}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\Big)\mathbf{m}_{\mathrm{T},\mathrm{x},\mathrm{x}+1}\Big) \tag{5.18}$$

pathspace2I1e}
$$\times \mathbf{Exp}\Big(-\tfrac{1}{2}\mathrm{c}_3^2\mathsf{N} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}\leqslant\mathrm{x}} (\upsilon_{\mathrm{w}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{w}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{w}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{w}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\Big)^2 \mathrm{T}\Big) \tag{5.19}$$

pathspace2I1f}
$$\times \mathbf{Exp}\Big(\mathrm{c}_4\mathsf{N}^{-1} \int_0^{\mathrm{T}} \Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]\Big)\mathrm{d}\mathbf{m}_{\mathrm{t},\mathsf{N}} - \tfrac{1}{2}\mathrm{c}_4^2\mathsf{N}^{-2} \int_0^{\mathrm{T}} \Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]\Big)^2 \mathrm{dt}\Big) \tag{5.20}$$

pathspace2I1g}
$$\times \mathbf{Exp}\Big(\mathrm{c}_5\mathsf{N}^{\frac{1}{2}} \Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\Big)\mathbf{m}_{\mathrm{T},\mathsf{N}}\Big) \tag{5.21}$$

pathspace2I1h}
$$\times \mathbf{Exp}\Big(-\tfrac{1}{2}\mathrm{c}_5^2\mathsf{N}\Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} (\upsilon_{\mathrm{x}+2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}+1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} - \upsilon_{\mathrm{x}-1}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\upsilon_{\mathrm{x}-2}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})\Big)^2 \mathrm{T}\Big). \tag{5.22}$$

The notation used above is explained as follows:

- The processes $\mathrm{t}\mapsto \mathbf{m}_{\mathrm{t},\mathrm{x},\mathrm{x}+1}$ and $\mathrm{t}\mapsto \mathbf{m}_{\mathrm{t},\mathsf{N}}$ are independent standard Brownian motions.
- The constants $\mathrm{c}_1,\dots,\mathrm{c}_5\in\mathbb{R}$ are all $\mathrm{O}(1)$ and deterministic.

The formula (5.15)-(5.22) is an immediate consequence of Girsanov's formula, but for a more intuitive presentation, let us explain it as follows:

- The generator in Definition 5.6 is a perturbation of the $\mathbb{P}^{\mathrm{path}}$-process by drift terms, hence the applicability of Girsanov's theorem.
- There are two types of additional drifts in Definition 5.6. The first, coming from $\mathscr{L}_{\mathrm{S}}^{\sim}, \mathrm{L}_{\mathrm{A}}^{\sim}$, comes from "bond" differentials $\partial_{\boldsymbol{\phi}_{\mathrm{x}+1}} - \partial_{\boldsymbol{\phi}_{\mathrm{x}}}$. Note that the diffusion component of the generator $\mathscr{L}_{\mathrm{S}}$ from Notation 5.4 consists of sums over $\mathrm{x},\mathrm{x}+1$ of these bond differentials. So, Girsanov's theorem tells us to invert this bond differential acting on every such drift in $\mathscr{L}_{\mathrm{S}}^{\sim}, \mathrm{L}_{\mathrm{A}}^{\sim}$ to generate the corresponding exponential martingale, then to sum over all bonds $\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]$. This leads to (5.15)-(5.19). (For the power-counting, note that the strength of the bond differential in $\mathscr{L}_{\mathrm{S}}$ from Notation 5.4 is order N, so we divide by N in this inversion procedure.)
- The other type of drift in Definition 5.6 comes from $\mathscr{L}_{\mathsf{N}}^{\sim}$, each of which corresponds to a site differential $\partial_{\boldsymbol{\phi}_{\mathsf{N}}}$. Because $\mathscr{L}_{\mathsf{N}}$ has a site differential of speed $\asymp\mathsf{N}$, we invert every site differential in $\mathscr{L}_{\mathsf{N},\sim}$ with an extra power of N^{-1} to generate the corresponding exponential martingale. This gives (5.20)-(5.22).

Now, when we take logarithms then expectations of (5.15)-(5.22), all Itô integrals vanish. Now, consider drift terms with deterministic integrands, i.e. terms inside of the exponential in (5.15), (5.19), (5.22). The drift term in (5.15) is $\lesssim_{\mathrm{T}} 1$. The drift term in (5.19) is $\lesssim_{\mathrm{T}} 1$, because the sum over w in (5.19) is telescoping with summands $\lesssim \mathsf{N}^{-1}$ (see Definition 5.1). The drift term in (5.22) is $\lesssim_{\mathrm{T}} 1$ for the same reason. We leave the other drifts alone for now. Thus, we have

$$\mathbf{H}[\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}|\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}] \lesssim \mathsf{N}^{-2} \sum_{\mathrm{x},\mathrm{x}+1\in[\![1,\mathsf{N}]\!]} \int_0^{\mathrm{T}} \mathbb{E}\Big(\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]} \mathbf{1}_{\mathrm{w}<\mathrm{x}} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{w}}]\Big)^2 \mathrm{dt} + \mathsf{N}^{-2} \int_0^{\mathrm{T}} \mathbb{E}\Big(\sum_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} \mathscr{Y}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]\Big)^2 \mathrm{dt},$$

where the expectation is with respect to the (2.5)-(2.7) process. Now, observe that the sum over $\mathrm{w}\in[\![1,\mathsf{N}]\!]$, with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, is sub-Gaussian of mean 0 and variance parameter $\lesssim\mathsf{N}$ by the uniform Lipschitz property of $\mathscr{U}'$ (and thus of $\mathscr{Y}'$) from Assumption 2.2. The same holds for the last squared term above. Therefore by

the entropy inequality of Lemma A.3 and the entropy estimate from Lemma 5.5, we have (for small $\varepsilon > 0$) the following, in which $\mathbf{z}_x$ is a sub-Gaussian random variable of mean 0 and variance parameter $\lesssim \mathsf{N}^{-1}$ with respect to $\mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$:

$$\mathbf{H}[\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}|\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}] \lesssim \mathsf{N}^{-2} \sum_{x,x+1\in[\![1,\mathsf{N}]\!]} \int_0^{\mathrm{T}} \Big\{ \varepsilon^{-1}\mathsf{N}\mathbf{H}_{\mathrm{t}} + \varepsilon^{-1}\mathsf{N} \log \mathbb{E}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}} \mathbf{Exp}\Big[\varepsilon \mathsf{N}^{-1}\mathbf{z}_x^2\Big] \Big\} \mathrm{dt}$$
$$\lesssim \mathsf{N}^{-2} \sum_{x,x+1\in[\![1,\mathsf{N}]\!]} \int_0^{\mathrm{T}} \Big\{ \varepsilon^{-1}\mathsf{N}\mathbf{H}_{\mathrm{t}} + \varepsilon^{-1}\mathsf{N} \Big\} \mathrm{dt} \lesssim_{\mathrm{T}} 1.$$

We clarify that the last line follows since $\mathbb{E}\mathbf{Exp}(\varepsilon\sigma^2\mathsf{z}^2) \lesssim 1$ if z is sub-Gaussian of mean 0 and variance parameter $\lesssim \sigma^2$ if $\varepsilon > 0$ is small enough. Moreover, we have used the estimate (5.7) in Lemma 5.5 with $\mathbf{H}_0 = 0$, since in this lemma, we have assumed that the initial data of $\boldsymbol{\phi}_{\mathrm{t}}$ is $\sim \mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$. This completes the proof. □

Lastly, we record a short-time L^{2p}-version of the estimate in (5.14). We clarify that by "short-time", we mean for a time-horizon that does not depend on N (but rather other parameters such as β_2 in (2.4)).

{lemma:pathspac

Lemma 5.9. *Fix any $p \geqslant 1$. There exists $\mathrm{T}_p > 0$ independent of N such that under the setting of Lemma 5.8,*

$$\sup_{\mathrm{T}\in[0,\mathrm{T}_p]} \mathbb{E}^{\mathrm{path},\sim}\Big(\tfrac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}}\Big)^{2p} \lesssim_p 1. \tag{5.23}$$

{eq:pathspace3

In particular, for any (deterministic) $\mathrm{t} \in [0,\mathrm{T}_p]$, if we recall $\mathfrak{P}_{\mathrm{t}}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$ from Definition 5.1, then

$$\mathbb{E}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}|\mathfrak{P}_{\mathrm{t}}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}|^{2p} \lesssim_p 1. \tag{5.24}$$

{eq:pathspace3

Proof. The estimate (5.24) since L^{2p}-norms are contractive under marginals, and $\mathfrak{P}_{\mathrm{t}}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$ is a one-time marginal (at time t) of $\mathrm{d}\mathbb{P}^{\mathrm{path}}/\mathrm{d}\mathbb{P}^{\mathrm{path},\sim}$. Indeed, for any deterministic time t, the law of the $\mathbb{P}^{\mathrm{path},\sim}$-process is $\mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$ because this is the case at $\mathrm{t} = 0$, and we have Lemma 5.7. Thus, it suffices to show (5.23). We also assume that $\mathrm{T} > 0$; if $\mathrm{T} = 0$, then the result follows because the initial data for both path-space measures are the same by assumption.

As shown in the proof of Lemma 5.8, the formula (5.15)-(5.22) gives

$$\tfrac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}} = \mathbf{Exp}\Big(\int_0^{\mathrm{T}} \mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}] \mathrm{d}\mathbf{m}_{\mathrm{t}} - \tfrac{1}{2}\int_0^{\mathrm{T}} |\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2 \mathrm{dt} \Big),$$

in which $\mathfrak{d} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ is sub-Gaussian of mean 0 and variance parameter $\lesssim 1$ with respect to $\mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$, and $\mathbf{m}_{\mathrm{t}}$ is a standard Brownian motion. Thus, we have

$$\mathbb{E}^{\mathrm{path},\sim}\Big(\tfrac{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path}}}{\mathrm{d}\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}}\Big)^{2p} = \mathbb{E}^{\mathrm{path},\sim}\mathbf{Exp}\Big(2p\int_0^{\mathrm{T}} \mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{d}\mathbf{m}_{\mathrm{t}} - p\int_0^{\mathrm{T}} |\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2\mathrm{dt}\Big)$$
$$\lesssim \Big\{ \mathbb{E}^{\mathrm{path},\sim}\mathbf{Exp}\Big(4p\int_0^{\mathrm{T}} \mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]\mathrm{dt} - 8p^2\int_0^{\mathrm{T}} |\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2\mathrm{dt}\Big)\Big\}^{\frac{1}{2}} \Big\{ \mathbb{E}^{\mathrm{path},\sim}\mathbf{Exp}\Big(\mathrm{C}_p\int_0^{\mathrm{T}} |\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2\mathrm{dt}\Big)\Big\}^{\frac{1}{2}}$$

for some $\mathrm{C}_p \lesssim_p 1$. The first curly-brackets-factor is $= 1$ because it has the form of an exponential martingale. For the second, we have the following (to be explained afterwards):

$$\mathbb{E}^{\mathrm{path},\sim}\mathbf{Exp}\Big(\mathrm{C}_p\int_0^{\mathrm{T}} |\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2\mathrm{dt}\Big) = \mathbb{E}^{\mathrm{path},\sim}\mathbf{Exp}\Big(\mathrm{C}_p\mathrm{T}^{-1}\int_0^{\mathrm{T}} \mathrm{T}|\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2\mathrm{dt}\Big)$$
$$\leqslant \mathrm{T}^{-1}\int_0^{\mathrm{T}} \mathbb{E}^{\mathrm{path},\sim} \mathrm{e}^{\mathrm{C}_p\mathrm{T}|\mathfrak{d}[\boldsymbol{\phi}_{\mathrm{t}}]|^2} = \mathrm{T}^{-1}\int_0^{\mathrm{T}} \mathbb{E}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}} \mathrm{e}^{\mathrm{C}_p\mathrm{T}|\mathfrak{d}[\boldsymbol{\phi}]|^2}.$$

The first identity is just multiplication by $\mathrm{T}^{-1}\mathrm{T} = 1$, since $\mathrm{T} > 0$. The second line follows from convexity of the exponential function, and because $\mathbb{P}^{\mathrm{path},\sim}$ has its $\mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$-initial law as an invariant measure by Lemma 5.7. Now, since $\mathfrak{d} : \mathbb{R}^{[\![1,\mathsf{N}]\!]} \to \mathbb{R}$ is sub-Gaussian of mean 0 and variance parameter $\lesssim 1$ with respect to $\mathbb{P}^{\tilde{\mathbf{A}},\tilde{\mathbf{B}}}$, if $\mathrm{C}_p\mathrm{T}$ is small enough (but independent of N), then the last expectation above is $\lesssim_{p,\mathrm{T}} 1$. It now suffices to combine the previous two displays in order to complete the proof. □

5.3. **Some consequences.** We record estimates that result from our entropy analysis and that will be important later in the paper. The first is an a priori regularity estimate for $\mathbf{j}^{\mathsf{N}}$, which ultimately amounts to estimates for the $\boldsymbol{\phi}_{\mathrm{t}}$-process (i.e. the gradient process).

{lemma:phiap}

Lemma 5.10. *Fix* $\mathrm{D}, \delta, \mathrm{T} > 0$ *independent of* N. *Fix any intervals* $\mathbb{I}_1, \dots, \mathbb{I}_{\mathsf{N}^{\mathrm{D}}} \subseteq [\![1, \mathsf{N}]\!]$. *With high probability,*

{eq:phiapI}
$$\sup_{\mathrm{t}\in[0,\mathrm{T}]} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} \Big| \lesssim \mathsf{N}^{\delta}. \tag{5.25}$$

By taking $\mathbb{I}_{\mathrm{J}} = \{\mathrm{J}\}$ *and* $\mathrm{D} = 1$, *we get the estimate below with high probability:*

{eq:phiapII}
$$\sup_{\mathrm{t}\in[0,\mathrm{T}]} \sup_{\mathrm{x}\in[\![1,\mathsf{N}]\!]} |\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{\delta}. \tag{5.26}$$

Proof. (5.26) follows from (5.25) as explained above, so we focus on (5.25). We let $\mathrm{t} \mapsto \boldsymbol{\psi}_{\mathrm{t}}$ be the $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$-valued process corresponding the path-space measure $\mathbb{P}_{\mathrm{T}}^{\mathrm{path},\sim}$ in Definition 5.6, and we assume that its initial data has law $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. In particular, for any deterministic time $\mathrm{t} \in [0,\mathrm{T}]$, we have $\boldsymbol{\psi}_{\mathrm{t}} \sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. Thus, for any $0 < \mathrm{K} \lesssim 1$, we have the following where $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ on the LHS indicates an expectation with respect to the path-space law of $\boldsymbol{\psi}_{\mathrm{t}}$ with initial data $\sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$:

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{\mathrm{t}\in[0,\mathrm{T}]\cap\mathsf{N}^{-\mathrm{K}}\mathbb{Z}} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\psi}_{\mathrm{t},\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \lesssim \mathsf{N}^{\mathrm{K}+\mathrm{D}} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(|\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\psi}_{\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big).$$

With respect to the measure $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, the random variables $\boldsymbol{\psi}_{\mathrm{x}}$ are independent, mean $\mathrm{O}(\mathsf{N}^{-1/2})$, and sub-Gaussian with variance parameter $\lesssim 1$ (see Assumption 2.2). Moreover, because $\mathbb{I}_{\mathrm{J}} \subseteq [\![1,\mathsf{N}]\!]$, we know that $|\mathbb{I}_{\mathrm{J}}|^{-1/2}\mathsf{N}^{-1/2} \lesssim |\mathbb{I}_{\mathrm{J}}|^{-1}$; this implies that in each probability on the RHS, we can replace $\boldsymbol{\psi}_{\mathrm{x}}$ by its centered version up to changing the implied constant in $\gtrsim$ therein. Then, by a standard Chernoff bound for independent, mean-zero, sub-Gaussian random variables, we have the following estimate for some $\mathrm{c} > 0$ fixed:

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(|\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\psi}_{\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \lesssim \mathbf{Exp}(-\mathrm{c}\mathsf{N}^{2\delta}).$$

Combining the previous two displays then yields

{eq:phiapII0}
$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{\mathrm{t}\in[0,\mathrm{T}]\cap\mathsf{N}^{-\mathrm{K}}\mathbb{Z}} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\psi}_{\mathrm{t},\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \lesssim \mathbf{Exp}(-\tfrac{1}{2}\mathrm{c}\mathsf{N}^{2\delta}). \tag{5.27}$$

A standard short-time continuity estimate allows us to bootstrap and remove the intersection with $\mathsf{N}^{-\mathrm{K}}\mathbb{Z}$ on the LHS, if we change the exponent on the RHS to $-\mathsf{N}^{\kappa}$ for some $\kappa > 0$; this is as long as $\mathrm{K} > 0$ is large enough (but still $\mathrm{O}(1)$), so that $\mathsf{N}^{-\mathrm{K}}$ is much smaller than the microscopic time-scale. Next, we claim that

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{\mathrm{t}\in[0,\mathrm{T}]} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \lesssim \Big\{ \log \Big[\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{\mathrm{t}\in[0,\mathrm{T}]} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\psi}_{\mathrm{t},\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \Big]^{-1} \Big\}^{-1} \lesssim \mathsf{N}^{-2\delta}.$$

Here, we used the entropy inequality from Lemma A.3, where we change measure from the law of the $\boldsymbol{\phi}$-process to that of the $\boldsymbol{\psi}$-process and use Lemma 5.8 to control the relative entropy. This gives the first bound above, and the second is what we just showed. Now, let us again use the entropy inequality in Lemma A.3 to change the law of the initial data, and we use (2.13) to estimate the corresponding relative entropy. This gives

$$\mathbb{P}\Big(\sup_{\mathrm{t}\in[0,\mathrm{T}]} \sup_{\mathrm{J}\in[\![1,\mathsf{N}^{\mathrm{D}}]\!]} |\mathbb{I}_{\mathrm{J}}|^{-\frac{1}{2}} \Big| \sum_{\mathrm{x}\in\mathbb{I}_{\mathrm{J}}} \boldsymbol{\phi}_{\mathrm{t},\mathrm{x}} \Big| \gtrsim \mathsf{N}^{\delta} \Big) \lesssim \gamma_{\mathsf{N}} \log \mathsf{N} \cdot (\log \mathsf{N})^{-1} \lesssim \gamma_{\mathsf{N}} \to 0.$$

This completes the proof. □

As a result of Lemma 5.10, we get the following estimate on $\mathbf{R}^{\mathsf{N}}$ from (3.5), which is exactly the first desired estimate (3.19) in Proposition 3.9.

{lemma:rwedge}

Lemma 5.11. *The estimate* (3.19) *is true. In particular, if we define*

$$\mathbf{R}_{\mathrm{t},\mathrm{x}}^{\mathsf{N},\wedge} := \mathbf{R}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}} \cdot \mathbf{1}[|\mathbf{R}_{\mathrm{t},\mathrm{x}}^{\mathsf{N}}| \lesssim \mathsf{N}^{-\frac{1}{6}\delta_{\mathbf{S}}}],$$

then with high probability, we have $\mathbf{R}^{\mathsf{N},\wedge} = \mathbf{R}^{\mathsf{N}}$ *on* $[0,\mathrm{T}] \times [\![0,\mathsf{N}]\!]$, *where* $\mathrm{T} > 0$ *is any fixed time-horizon.*

Proof. The same proof of Lemma 6.2 in [33] yields the result here as well. Indeed, in a nutshell, recall that $\mathbf{S}^{\mathsf{N}}$ is a smoothing of $\mathbf{Z}^{\mathsf{N}}$ on length-scale $\lesssim (\log \mathsf{N})^2 \mathsf{N}^{-\delta_{\mathbf{S}}/2}$ (see (3.5)). Lemma 5.10 shows that $\mathbf{j}^{\mathsf{N}}$ (and thus $\mathbf{Z}^{\mathsf{N}}$) has

Hölder regularity of exponent $1/2$, with a norm of $\lesssim \mathsf{N}^{\delta}$. Thus, we have $|\mathbf{R}^{\mathsf{N}} - 1| \lesssim \mathsf{N}^{\delta}[(\log \mathsf{N})^2 \mathsf{N}^{-\delta \mathrm{s}/2}]^{1/2} \lesssim \mathsf{N}^{\delta \mathrm{s}/6}$ with high probability. This completes the proof. □

We observe that the proof of Lemma 5.10 is based on large deviations bounds for sums of independent, mean-zero random variables, with respect to the $\mathbb{P}^{\mathrm{path},\sim}$-measure from Definition 5.6. We will use the same philosophy to control more general spatial-averages of mean-zero local functions, namely the $\mathbf{Av}^{\mathbf{X}}$-terms in Proposition 3.7.

{lemma:avx}

Lemma 5.12. *Take any $\mathrm{D}, \delta, \mathrm{T} > 0$ independent of N. Fix any $\mathcal{Q}_{\ell}$ as in Proposition 3.7. With high probability,*

$$\sup_{\mathrm{t} \in [0,\mathrm{T}]} \sup_{\mathrm{x} \in [\![0,\mathsf{N}]\!]} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}}| \lesssim \mathsf{N}^{-\frac{1}{2}+\frac{3}{4}\delta_{\mathrm{S}}+\delta}. \tag{5.28}$$

{eq:avxI}

Proof. We use the same argument as in the proof of Lemma 6.3 in [33]; let us provide an intuitive account of this argument (which is complete and rigorous outside of several claims whose verifications are the same as in [33]). We clarify, however, that the estimate (5.28) is essentially a CLT-type estimate, because $\mathbf{Av}^{\mathbf{X},\mathcal{Q}_{\ell}}$ is an average of $\sim \mathfrak{l}_{\mathbf{Av}} = \mathsf{N}^{1-3\delta \mathrm{s}/2}$ bulk fluctuations (see Definition 3.4), hence the power-saving of $\mathsf{N}^{-1/2+3\delta \mathrm{s}/4} = \mathfrak{l}_{\mathbf{Av}}^{-1/2}$ on the RHS. The extra $\delta > 0$ is to ensure that the bound holds with high probability.

First, by the same union-bound-plus-change-of-measure argument as in the proof of Lemma 5.10, it is enough to show the following one-point estimate for some $\kappa > 0$:

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}}| \gtrsim \mathsf{N}^{-\frac{1}{2}+\frac{3}{4}\delta_{\mathrm{S}}+\delta}) \lesssim \mathbf{Exp}(-\mathsf{N}^{\kappa}).$$

We clarify that the $\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}}$ is a function of only $\boldsymbol{\phi}_{\mathrm{t}}$, which we are sampling according to the measure $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ above. Now, by Definition 3.4, we can write the following (in which we recall all notation below from Definition 3.4):

$$\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}} = \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \mathbf{H}_{0,\tau_{\mathbf{Av}},\mathrm{x},\mathrm{w}}^{\mathsf{N},0,0} \cdot \mathbf{1}_{|\mathrm{x}-\mathrm{w}| \lesssim (\log \mathsf{N})^2 \mathfrak{l}_{\mathbf{Av}}} \mathsf{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}] \mathbf{Z}_{\mathrm{t},\mathrm{w}}^{\mathsf{N}} (\mathbf{Z}_{\mathrm{t},\mathrm{x}_{\pm}}^{\mathsf{N}})^{-1}.$$

Recall $\mathsf{q}_{\ell,\mathrm{w}}$ are either all forwards-oriented or all backwards-oriented, and moreover that $\pm = -$ in the former case while $\pm = +$ in the latter. We now perform the following modifications to the previous display:

(1) By (2.10), write $\mathbf{Z}_{\mathrm{t},\mathrm{w}}^{\mathsf{N}} (\mathbf{Z}_{\mathrm{t},\mathrm{x}_{\pm}}^{\mathsf{N}})^{-1} = \mathbf{Exp}\{\lambda(\mathbf{j}_{\mathrm{t},\mathrm{w}}^{\mathsf{N}} - \mathbf{j}_{\mathrm{t},\mathrm{x}_{\pm}}^{\mathsf{N}})\}$. Recall $\mathrm{x}_{\pm} = \mathrm{x} + \mathrm{O}((\log \mathsf{N})^3 \mathfrak{l}_{\mathbf{Av}})$ in Definition 3.4. As $\mathfrak{l}_{\mathbf{Av}} \lesssim \mathsf{N}^{1-3\delta \mathrm{s}/2}$, the arguments in the proof of Lemma 5.10, give $|\mathbf{j}_{\mathrm{t},\mathrm{w}}^{\mathsf{N}} - \mathbf{j}_{\mathrm{t},\mathrm{x}_{\pm}}^{\mathsf{N}}| \lesssim \mathsf{N}^{-1/2} \mathsf{N}^{\delta} \mathsf{N}^{1/2} \mathsf{N}^{-3\delta \mathrm{s}/4} \lesssim \mathsf{N}^{-\delta \mathrm{s}/2}$ with exponentially high probability if $\delta > 0$ is small enough. So, with exponentially high probability, we can replace $\mathbf{Z}_{\mathrm{t},\mathrm{w}}^{\mathsf{N}} (\mathbf{Z}_{\mathrm{t},\mathrm{x}_{\pm}}^{\mathsf{N}})^{-1}$ by a function of $\boldsymbol{\phi}_{\mathrm{t}}$ that is $1 + \mathrm{O}(\mathsf{N}^{-\rho})$ and has disjoint support from $\mathsf{q}_{\ell,\mathrm{w}}$.

(2) Since $\mathsf{q}_{\ell,\mathrm{w}}$ is bulk-admissible (see Proposition 3.7), it satisfies (3.2) and $\mathbb{E}^0 \mathsf{q}_{\ell,\mathrm{w}} = 0$. Thus, $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \mathsf{q}_{\ell,\mathrm{w}} \lesssim \mathsf{N}^{-1/2}$, since $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ is a perturbation of $\mathbb{E}^0$ at all $\mathrm{x} \in [\![1,\mathsf{N}]\!]$ by $\mathrm{O}(\mathsf{N}^{-1/2})$ (see Definition 5.1), and $\mathsf{q}_{\ell,\mathrm{w}}$ has uniformly bounded support. Let us now center $\mathsf{q}_{\ell,\mathrm{w}}$ above with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ and thus subtract from it $\mathrm{O}(\mathsf{N}^{-1/2})$; the error in this modification $\lesssim \mathsf{N}^{-1/2}$, since $\mathbf{H}^{\mathsf{N},0,0}$ is a probability kernel on $[\![0,\mathsf{N}]\!]$ with respect to either spatial variable. Finally, we replace this mean-zero version of $\mathsf{q}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]$ by a function which is also mean-zero with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ but which is also $\lesssim \mathsf{N}^{\varepsilon}$ deterministically (where $\varepsilon > 0$ is small). We may do this because $\mathsf{q}_{\ell,\mathrm{w}}$ has the deterministic estimate (3.2), and the upper bound therein is $\lesssim \mathsf{N}^{\varepsilon}$ with exponentially high probability because of (5.26).

(3) As a consequence of the previous two bullet points, with exponentially high probability, we have

$$\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}} = \sum_{\mathrm{w} \in [\![0,\mathsf{N}]\!]} \mathbf{H}_{0,\tau_{\mathbf{Av}},\mathrm{x},\mathrm{w}}^{\mathsf{N},0,0} \cdot \mathbf{1}_{|\mathrm{x}-\mathrm{w}| \lesssim (\log \mathsf{N})^2 \mathfrak{l}_{\mathbf{Av}}} \widetilde{\mathsf{q}}_{\ell,\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}] \mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}] + \mathrm{O}(\mathsf{N}^{-\frac{1}{2}}),$$

where $\widetilde{\mathsf{q}}_{\ell,\mathrm{w}}$ is a local function that is mean-zero with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ and $\lesssim \mathsf{N}^{\varepsilon}$ deterministically, and $\mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}}]$ has disjoint support from $\widetilde{\mathsf{q}}_{\ell,\mathrm{w}}$ (and is $\mathrm{O}(1)$ deterministically). Thus we can use the same martingale argument as in the proof of Lemma 6.3 in [33] to show that the sum on the RHS above exhibits square-root cancellations at exponential scale. That is, ultimately by an application of the Azuma-Hoeffding inequality, we obtain

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{X},\mathcal{Q}_{\ell}}| \gtrsim \mathsf{N}^{2\varepsilon} \|\mathbf{H}_{0,\tau_{\mathbf{Av}},\mathrm{x},\cdot}^{\mathsf{N},0,0}\|_{\ell^2([\![0,\mathsf{N}]\!])} + \mathsf{N}^{-\frac{1}{2}}) \lesssim \mathbf{Exp}(-\mathrm{c}\mathsf{N}^{2\varepsilon})$$

for some $\mathrm{c} > 0$.

The previous display now completes the proof once we combine it with Proposition A.1, which shows that the ℓ^2-norm above is $\lesssim \mathsf{N}^{-1/2}\tau_{\mathbf{Av}}^{-1/4}$ (per standard heat kernel estimates); now, recall that $\tau_{\mathbf{Av}} = \mathsf{N}^{-3\delta\mathfrak{s}}$ from Definition 3.4. So, the RHS of the inequality in the probability above is $\lesssim \mathsf{N}^{2\varepsilon}\mathsf{N}^{-1/2+3\delta\mathfrak{s}/4} + \mathsf{N}^{-1/2}$; now take $2\varepsilon = \delta$. □

We conclude this section with time-regularity estimates for $\mathbf{j}^{\mathsf{N}}$, which ultimately lead to similar estimates for its exponential $\mathbf{Z}^{\mathsf{N}}$, and that will be important in our analysis of $\mathbf{Z}^{\mathsf{N}}$-factors in $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}$-terms and $\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}$-terms in Proposition 3.7.

lemma:currtreg}

Lemma 5.13. *Fix any* $\mathrm{D},\mathrm{T}>0$ *independent of* N. *Recall the time-scale* $\mathfrak{t}_{\mathbf{Av}} \leqslant \mathsf{N}^{-2/3}$ *from Definition 3.4. There exists* $\rho>0$ *fixed such that with high probability, we have*

{eq:currtregI}
$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}}-\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\rho}. \tag{5.29}$$

We clarify that we drop all (t,s)*-pairs in the previous display for which* $\mathrm{t}+\mathrm{s}<0$ *(as in Definition 3.4).*

Proof. Suppose that $\boldsymbol{\mathcal{G}} : [\![0,\mathsf{N}]\!]\times[\![0,\mathsf{N}]\!]\to\mathbb{R}$ satisfies the following probability kernel properties:

(1) We have $\boldsymbol{\mathcal{G}}_{\mathrm{x},\mathrm{w}}\geqslant 0$ and $\boldsymbol{\mathcal{G}}_{\mathrm{x},\mathrm{w}}\lesssim \mathfrak{l}_{\mathsf{N}}^{-1}\mathbf{1}[|\mathrm{x}-\mathrm{w}|\lesssim\mathfrak{l}_{\mathsf{N}}]$, with a just-below-macroscopic-scale $\mathfrak{l}_{\mathsf{N}}=\mathsf{N}^{1-\kappa}$ for some $\kappa>0$ small but fixed. Moreover, we have $\|\boldsymbol{\mathcal{G}}_{\mathrm{x},\cdot}\|_{\ell^1([\![0,\mathsf{N}]\!])}=1$.

(2) We have $|(\nabla_1^{\mathbf{X}})^{\mathrm{a}}\boldsymbol{\mathcal{G}}_{\mathrm{x},\mathrm{w}}|\lesssim\mathfrak{l}_{\mathsf{N}}^{-1-\mathrm{a}}\mathbf{1}[|\mathrm{x}-\mathrm{w}|\lesssim\mathfrak{l}_{\mathsf{N}}]$ for any $0\leqslant\mathrm{a}\lesssim 1$, where the gradients act on w, and where we only consider $\mathrm{x},\mathrm{w},\mathrm{a}$ for which the LHS of this bound depends only on pairs of points in $[\![0,\mathsf{N}]\!]$.

We now define the operator

$$\boldsymbol{\mathcal{G}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\cdot}]_{\mathrm{x}} := \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]}\boldsymbol{\mathcal{G}}_{\mathrm{x},\mathrm{w}}\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}.$$

We note that with high probability, we have $|\boldsymbol{\mathcal{G}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\cdot}]_{\mathrm{x}}-\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}|\lesssim\mathsf{N}^{-\omega}$ for some $\omega>0$ uniformly over $[0,2\mathrm{T}]\times[\![0,\mathsf{N}]\!]$. Indeed, the LHS is bounded above by an average of discrete gradients of $\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\cdot}$ at scale $\lesssim\mathfrak{l}_{\mathsf{N}}$; now use Lemma 5.10 to control said gradients. Using this and the triangle inequality, we obtain

$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}}-\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}|\lesssim\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\boldsymbol{\mathcal{G}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\cdot}]_{\mathrm{x}}-\boldsymbol{\mathcal{G}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\cdot}]_{\mathrm{x}}|+\mathsf{N}^{-\omega}$$

with high probability. Thus, it suffices to control the first term on the RHS above. By the SDE (2.1)-(2.3),

$$\mathrm{d}\boldsymbol{\mathcal{G}}[\mathbf{j}^{\mathsf{N}}_{\mathrm{t},\cdot}]_{\mathrm{x}} = \boldsymbol{\mathcal{G}}[\mathsf{N}^{\frac{3}{2}}\nabla_1^{\mathbf{X}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\cdot}]]_{\mathrm{x}}\mathrm{d}\mathrm{t}+\boldsymbol{\mathcal{G}}[\mathsf{N}\mathbf{F}[\tau_{\cdot}\boldsymbol{\phi}_{\mathrm{t}}]]_{\mathrm{x}}\mathrm{d}\mathrm{t}+\boldsymbol{\mathcal{G}}[\sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{b}_{\mathrm{t},\cdot}]_{\mathrm{x}}+\boldsymbol{\mathcal{G}}[\mathbf{1}_{\cdot=0}\mathsf{N}\widetilde{\mathbf{A}}+\mathbf{1}_{\cdot=\mathsf{N}}\mathsf{N}\widetilde{\mathbf{B}}]\mathrm{d}\mathrm{t}.$$

We recall that we extended $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{x}}]=0$ for $\mathrm{x}\notin[\![0,\mathsf{N}]\!]$. Thus, we must show that with high probability,

{eq:currtregI1}
$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}\int_0^{\mathrm{s}}|\boldsymbol{\mathcal{G}}[\mathsf{N}^{\frac{3}{2}}\nabla_1^{\mathbf{X}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{r},\cdot}]]_{\mathrm{x}}|\mathrm{d}\mathrm{r}\lesssim\mathsf{N}^{-\rho}, \tag{5.30}$$

{eq:currtregI2}
$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}\int_0^{\mathrm{s}}|\boldsymbol{\mathcal{G}}[\mathsf{N}\mathbf{F}[\tau_{\cdot}\boldsymbol{\phi}_{\mathrm{t}+\mathrm{r}}]]_{\mathrm{x}}|\mathrm{d}\mathrm{r}\lesssim\mathsf{N}^{-\rho}, \tag{5.31}$$

{eq:currtregI3}
$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}\mathsf{N}^{\frac{1}{2}}|\boldsymbol{\mathcal{G}}[\mathbf{b}_{\mathrm{t}+\mathrm{s},\cdot}]-\boldsymbol{\mathcal{G}}[\mathbf{b}_{\mathrm{t},\cdot}]|\lesssim\mathsf{N}^{-\rho}, \tag{5.32}$$

{eq:currtregI4}
$$\sup_{\mathrm{s}\in[0,\mathfrak{t}_{\mathbf{Av}}]}\sup_{\mathrm{t}\in[0,\mathrm{T}]}\sup_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}\mathsf{N}\mathrm{s}\|\boldsymbol{\mathcal{G}}_{\mathrm{x},\cdot}\|_{\ell^\infty([\![0,\mathsf{N}]\!])}\lesssim\mathsf{N}^{-\rho}. \tag{5.33}$$

We start with (5.30). We can extend the summation in the $\boldsymbol{\mathcal{G}}$-operator therein from $\mathrm{w}\in[\![0,\mathsf{N}]\!]$ to $\mathrm{w}\in\mathbb{Z}$, because $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t},\mathrm{w}}]=0$ for all $\mathrm{w}\in\mathbb{Z}\setminus[\![0,\mathsf{N}]\!]$. Then, use the fact that $\nabla_1^{\mathbf{X}}$ has adjoint $\nabla_{-1}^{\mathbf{X}}$ with respect to Lebesgue measure on $\mathbb{Z}$ to write the following

$$\boldsymbol{\mathcal{G}}[\mathsf{N}^{\frac{3}{2}}\nabla_1^{\mathbf{X}}\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{r},\cdot}]]_{\mathrm{x}}=\sum_{\mathrm{w}\in[\![1,\mathsf{N}]\!]}\mathsf{N}^{\frac{3}{2}}\nabla_{-1}^{\mathbf{X}}\boldsymbol{\mathcal{G}}_{\mathrm{x},\mathrm{w}}\cdot\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{r},\mathrm{w}}].$$

The RHS is restricted to $\mathrm{w}\in[\![1,\mathsf{N}]\!]$ because this is where $\mathscr{U}'[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{r},\mathrm{w}}]$ is supported. Moreover, the gradient acts on w on the RHS. By Lemma 5.10 and potential bounds in Assumption 2.2, as well as $\boldsymbol{\mathcal{G}}$-estimates from the beginning of the proof, the RHS of the previous display is $\lesssim\mathsf{N}^{\varepsilon}\mathsf{N}^{3/2}\mathfrak{l}_{\mathsf{N}}^{-2}\mathfrak{l}_{\mathsf{N}}\lesssim\mathsf{N}^{\varepsilon}\mathsf{N}^{3/2}\mathfrak{l}_{\mathsf{N}}^{-1}\lesssim\mathsf{N}^{1/2+\kappa+\varepsilon}$, in which $\varepsilon>0$ is small but fixed. Integrating this over $\mathrm{r}\in[0,\mathrm{s}]$ then gives a contribution of $\lesssim\mathsf{N}^{1/2+\kappa+\varepsilon}\mathrm{s}\lesssim\mathsf{N}^{-\rho}$. This shows (5.30).

Now take (5.32). First, note that $\boldsymbol{\mathcal{G}}[\mathbf{b}_{t+s,\cdot}]_x - \boldsymbol{\mathcal{G}}[\mathbf{b}_t]_x$ itself has the law of a standard Brownian motion at time s multiplied by $\|\boldsymbol{\mathcal{G}}_{x,\cdot}\|_{\ell^2([0,N])} \lesssim \mathfrak{l}_N^{-1/2} \lesssim N^{-1/2+\kappa/2}$. Thus, $N^{1/2-\kappa/2}|\boldsymbol{\mathcal{G}}[\mathbf{b}_{t+s,\cdot}]_x - \boldsymbol{\mathcal{G}}[\mathbf{b}_{t,\cdot}]_x| \lesssim \mathfrak{t}_{\mathbf{Av}}^{1/2}\mathbf{w}_s$ for a standard Brownian motion $\mathbf{w}_s$. This implies that

$$\mathbb{P}\Big(\sup_{s\in[0,\mathfrak{t}_{\mathbf{Av}}]} |\boldsymbol{\mathcal{G}}[\mathbf{b}_{t+s,\cdot}]_x - \boldsymbol{\mathcal{G}}[\mathbf{b}_{t,\cdot}]_x| \gtrsim N^{\omega} N^{-\frac{1}{2}+\frac{1}{2}\kappa} \mathfrak{t}_{\mathbf{Av}}^{\frac{1}{2}} \Big) \lesssim \mathbb{P}\Big(\sup_{s\in[0,1]} |\mathbf{w}_s| \gtrsim N^{\omega} \Big) \lesssim \mathbf{Exp}(-\upsilon N^{-\omega}),$$

where $\upsilon, \omega > 0$ are small but fixed, and the last bound follows by, say, the reflection principle. We clarify that the first bound above follows by time-changing by a factor $\mathfrak{t}_{\mathbf{Av}}$. Taking a union bound over $x \in [\![0, N]\!]$ then provides (5.32). Next, we note that (5.33) follows since $\|\boldsymbol{\mathcal{G}}_{x,\cdot}\|_{\ell^\infty([0,N])} \lesssim N^{-1+\kappa}$ for some $\kappa > 0$ small.

This now leaves (5.31). Like in the proof of Lemma 5.10, let $\boldsymbol{\psi}_{t,x}$ be the process corresponding to the $\mathbb{P}_{2T}^{\mathrm{path},\sim}$-measure from Definition 5.6. Moreover, we assume its initial data (and thus its law at all deterministic times) is distributed with respect to its invariant measure $\sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ from Definition 5.1. The operation below does nothing with probability $1 - \mathrm{O}(\mathbf{Exp}(-\upsilon N^{\omega}))$ for some $\omega > 0$:

- We have $\mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}] = \mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]\mathbf{1}(|\mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]| \lesssim N^{\varepsilon})$ for $\varepsilon > 0$ small but fixed for all $r \in [0, s]$ and all (s, t, x) on the LHS of (5.31). This holds by (5.27) (but without the intersection with $N^{-K}\mathbb{Z}$ as explained after (5.27)) to control $\boldsymbol{\psi}_{\cdot,\cdot}$ uniformly by $N^{\varepsilon'}$ for small $\varepsilon' > 0$, the formula (2.4), and the potential bound in Assumption 2.2.

So, let us use the notation $\mathbf{F}^{\mathrm{cut}}[\tau.\boldsymbol{\psi}_{t+r}] := \mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]\mathbf{1}(|\mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]| \lesssim N^{\varepsilon})$. Note that $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathbf{F}[\tau.\boldsymbol{\psi}] = \mathrm{O}(N^{-1/2})$. Indeed, we note that $\mathbb{E}^0\mathbf{F}[\tau.\boldsymbol{\psi}] = 0$ by (2.4), the fact that $\mathbb{E}^0\mathscr{U}'[\boldsymbol{\phi}_\cdot] = 0$, and the fact that $\mathbb{P}^0$ is a product measure on $\mathbb{R}^{[\![1,N]\!]}$ (see Definition 2.1). Moreover, by (2.4), the function $\mathbf{F}[\tau.\boldsymbol{\psi}]$ has support that is contained in an interval of length $\mathrm{O}(1)$, and the marginals of $\mathbb{P}^0, \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ on such an interval are $\mathrm{O}(N^{-1/2})$-perturbations of each other. Thus, by the same reasoning as in the bullet point above, we get $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathbf{F}^{\mathrm{cut}}[\tau.\boldsymbol{\psi}] \lesssim N^{-1/2}$. Lastly, if we use the notation $\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}] = \mathbf{F}^{\mathrm{cut}}[\tau.\boldsymbol{\psi}] - \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\mathbf{F}^{\mathrm{cut}}[\tau.\boldsymbol{\psi}]$, then with probability $1 - \mathrm{O}(\mathbf{Exp}(-\upsilon N^{\omega}))$, we get

$$\boldsymbol{\mathcal{G}}[N\mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]]_x = \boldsymbol{\mathcal{G}}[N\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]]_x + \mathrm{O}(N^{-\frac{1}{2}}).$$

Note that $w \mapsto \mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]$, since $\boldsymbol{\psi}_{t+r} \sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, is such that if $|w_1 - w_2| \geqslant 6$, say, then $\mathbf{F}^{\circ}[\tau_{w_1}\boldsymbol{\psi}_{t+r}], \mathbf{F}^{\circ}[\tau_{w_2}\boldsymbol{\psi}_{t+r}]$ are $\mathrm{O}(N^{\varepsilon})$-bounded independent and mean-zero random variables. Thus, $\boldsymbol{\mathcal{G}}[N\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]]_x$ is a sum of $\mathrm{O}(1)$-many smaller sums of martingale increments bounded by $\mathrm{O}(N^{\varepsilon})$. The Azuma-Hoeffding inequality then gives

$$\boldsymbol{\mathcal{G}}[N\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]]_x = \boldsymbol{\mathcal{G}}[N\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]]_x\mathbf{1}(|\boldsymbol{\mathcal{G}}[N\mathbf{F}^{\circ}[\tau.\boldsymbol{\psi}_{t+r}]]_x| \lesssim N^{\omega}N^{\varepsilon}\|\boldsymbol{\mathcal{G}}_{x,\cdot}\|_{\ell^2([0,N])}) \lesssim N^{\omega+\varepsilon}\mathfrak{l}_N^{-\frac{1}{2}} \lesssim N^{-\frac{1}{2}+\omega+\varepsilon+\kappa}$$

with probability $1 - \mathrm{O}(\mathbf{Exp}(-\upsilon N^{\omega}))$. Integrating this over $r \in [0, s]$ and multiplying by N gives a contribution of $\lesssim N^{-1/2+\omega+\varepsilon+\kappa}Ns \lesssim N^{-\rho}$. In particular, we ultimately deduce that

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{s\in[0,\mathfrak{t}_{\mathbf{Av}}]} \sup_{t\in[0,T]} \sup_{x\in[\![0,N]\!]} \int_0^s |\boldsymbol{\mathcal{G}}[N\mathbf{F}[\tau.\boldsymbol{\psi}_{t+r}]]_x|\mathrm{d}r \gtrsim N^{-\rho} \Big) \lesssim \mathbf{Exp}(-\upsilon N^{\omega})$$

for some $\upsilon, \omega > 0$. Like in the proof of Lemma 5.10, we can use the entropy inequality in Lemma A.3 and Lemma 5.8 to change measure from the law of $\boldsymbol{\psi}$ to that of the $\boldsymbol{\phi}$-process, so that

$$\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\sup_{s\in[0,\mathfrak{t}_{\mathbf{Av}}]} \sup_{t\in[0,T]} \sup_{x\in[\![0,N]\!]} \int_0^s |\boldsymbol{\mathcal{G}}[N\mathbf{F}[\tau.\boldsymbol{\phi}_{t+r}]]_x|\mathrm{d}r \gtrsim N^{-\rho} \Big) \lesssim N^{-\omega}.$$

Again as in the proof of Lemma 5.10, we may now use the entropy inequality in Lemma A.3 to change the law of the initial data above, with (2.13) to control the corresponding entropy. In particular, we can remove $\widetilde{\mathbf{A}}, \widetilde{\mathbf{B}}$ from the superscript above if we change the RHS of $\lesssim \gamma_N \log N \cdot (\log N)^{-1} \ll 1$. This completes the proof. □

6. Boltzmann-Gibbs estimates

{section:stochl

We now establish estimates for $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}$- and $\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}$-terms from Proposition 3.7. These are the *Boltzmann-Gibbs*-type estimates. We begin by stating the corresponding estimate for $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}$-terms.

{prop:bgbulk}

Proposition 6.1. *Fix $T \geqslant 0$ independent of N, and fix any $\mathscr{Q}_\ell$ in Proposition 3.7. There exists a constant $\beta_{\mathrm{CLT}} > 0$ such that with high probability, we have the following estimate:*

$$\int_0^T N^{-1} \sum_{x\in[\![0,N]\!]} |\mathbf{Av}_{t,x}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|\mathrm{d}t \lesssim_T N^{-1-\beta_{\mathrm{CLT}}}. \tag{6.1}$$

{eq:bgbulkI}

We now present the corresponding estimate for $\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}$-terms.

{prop:bgedge}

Proposition 6.2. *Fix any $\mathrm{T} \geqslant 0$ independent of N, and fix any edge admissible $\mathfrak{b}_\ell$ as in Proposition 3.7. There exists a constant $\beta_\partial > 0$ such that with high probability, we have the following estimate:*

{eq:bgedgeI}

$$\max_{\mathrm{dist}(\mathrm{x},\{0,\mathsf{N}\}) \lesssim 1} \int_0^{\mathrm{T}} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathfrak{b}_\ell}| \mathrm{dt} \lesssim_{\mathrm{T}} \mathsf{N}^{-\beta_\partial}. \tag{6.2}$$

The rest of this section is dedicated to proofs of Propositions 6.1 and 6.2. These proofs have technical bearing on the arguments in the rest of the paper, so the reader is invited to skip to Section 7.

6.1. **Argument for Proposition 6.1.** We start with Proposition 6.1, as the proof of Proposition 6.2 uses essentially the same ideas. Moreover, we recall from Section 1 that the main schematic of this argument is to localize the LHS of (6.1) into pieces for which we can compare to the half-space setting and inherit estimates from there (namely, from [33]). In a nutshell, the architecture of this subsection is the following:

- First, we introduce some preliminaries. This includes the reduction to the case where $\mathrm{T} = \mathrm{T}_{\mathsf{N}} \to 0$ (slowly). Indeed, by the triangle inequality, if we can control contributions on intervals of length T_{N}, then we can control intervals of length $\mathrm{O}(1)$ as long as $\mathrm{T}_{\mathsf{N}} \to 0$ sufficiently slowly. Then, we use the various entropy and moment estimates in Section 5 to reduce to the case where the initial data of $\boldsymbol{\phi}_{\mathrm{t}}$ is $\sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ from Definition 5.1. (It does not matter so much if we choose $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ or $\mathbb{P}^0$ from Definition 2.1, but the former is more convenient.) Finally, we decompose the space-time average $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathfrak{Q}_\ell}$ in (6.1) into an average of smaller-scale space-time averages defined with respect to different space-time scales (which are ultimately determined by distance between the averaging domain and the edge (see Notation 6.3). We also make a few technical modifications (see Notation 6.4). The conclusion here is the reduction to Propositions 6.5 and 6.6 (estimates on smaller-scale averages).
- Second, we take each smaller-scale average from above, and we compare them to their half-space analogues. The basic principle is that if a spatial domain on which we average is separated from one edge (say, the right edge), then on a short-enough time-scale we can compare the dynamics on this block to those in the half-space.
- Finally, we reduce the proof of Propositions 6.5 and 6.6 to estimates in [33] for the half-space dynamics.

6.1.1. *Preliminaries.* Fix a time-scale T_{N} which vanishes slowly in the large-N limit. Now we decompose $[0,\mathrm{T}]$ into disjoint time-intervals $\{\mathbb{T}_{\mathrm{J}}\}_{\mathrm{J} \geqslant 1}$ of length $\asymp \mathrm{T}_{\mathsf{N}}$; there are $\asymp \mathrm{T}_{\mathsf{N}}^{-1}$-many such intervals. Next, we write

$$\int_0^{\mathrm{T}} \mathsf{N}^{-1} \sum_{\mathrm{x} \in [\![0,\mathsf{N}]\!]} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathfrak{Q}_\ell}| \mathrm{dt} \lesssim \mathrm{T}_{\mathsf{N}}^{-1} \sup_{\mathrm{J} \geqslant 1} \int_{\mathbb{T}_{\mathrm{J}}} \mathsf{N}^{-1} \sum_{\mathrm{x} \in [\![0,\mathsf{N}]\!]} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathfrak{Q}_\ell}| \mathrm{dt}.$$

We claim that to prove Proposition 6.1, it suffices to show that for some $\varepsilon_{\mathsf{N}} \to 0$ independently of T_{N}, we have

{eq:bgbulkII}

$$\sup_{\mathrm{J} \geqslant 1} \mathbb{P}\Big(\int_{\mathbb{T}_{\mathrm{J}}} \mathsf{N}^{-1} \sum_{\mathrm{x} \in [\![0,\mathsf{N}]\!]} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathfrak{Q}_\ell}| \mathrm{dt} \gtrsim \mathsf{N}^{-1-\beta_{\mathrm{CLT}}} \Big) \leqslant \varepsilon_{\mathsf{N}}, \tag{6.3}$$

where $\beta_{\mathrm{CLT}} > 0$ is a fixed constant. Indeed, we can place the supremum inside the probability if we multiply the RHS above by $\mathrm{T}_{\mathsf{N}}^{-1}$ because of a union bound. Then, we choose $\mathrm{T}_{\mathsf{N}} \to 0$ slow enough so that $\varepsilon_{\mathsf{N}} \mathrm{T}_{\mathsf{N}}^{-1} \to 0$. So, the rest of this subsection is dedicated to the proof of (6.3). Without loss of generality, we will restrict to $\mathrm{J} = 1$, and thus $\mathbb{T}_{\mathrm{J}} = [0, \mathrm{T}_{\mathsf{N}}]$, say. Indeed, the only aspect of the argument if $\mathrm{J} = 1$ which uses the exact choice of initial data is the validity of the initial entropy estimate (2.13). However, this estimate is valid for the law of the $\boldsymbol{\phi}_{\mathrm{t}}$-process at $\mathrm{t} = \inf \mathbb{T}_{\mathrm{J}}$ for any J, not just $\mathrm{J} = 1$, because of Lemma 5.5. In particular, it will suffice to prove

{eq:bgbulkIII}

$$\mathbb{P}\Big(\mathbf{1}_{\mathscr{E}} \int_0^{\mathrm{T}_{\mathsf{N}}} \mathsf{N}^{-1} \sum_{\mathrm{x} \in [\![0,\mathsf{N}]\!]} |\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathfrak{Q}_\ell}| \mathrm{dt} \gtrsim \mathsf{N}^{-1-\beta_{\mathrm{CLT}}} \Big) \leqslant \varepsilon_{\mathsf{N}} \tag{6.4}$$

given any sequence of times T_{N} which vanishes in the large-N limit; above, the event $\mathscr{E}$ is a path-space event that holds with high probability and will be determined. (Although the indicator of $\mathscr{E}$ is redundant, we will eventually take an expectation of the quantity on the LHS, and this indicator will be helpful for said expectation.) First, by the entropy inequality from Lemma A.3, we can change the law of the $\boldsymbol{\phi}_{\mathrm{t}}$-initial data to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ by paying a

relative entropy cost and taking a log:

{eq:bgbulkIII1}
$$\mathbb{P}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathsf{T}_\mathsf{N}}\mathsf{N}^{-1}\sum_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|\mathrm{dt}\gtrsim\mathsf{N}^{-1-\beta_{\mathrm{CLT}}}\Big) \tag{6.5}$$
$$\lesssim\gamma_\mathsf{N}\log\mathsf{N}\Big\{\log\Big[\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathsf{T}_\mathsf{N}}\mathsf{N}^{-1}\sum_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|\mathrm{dt}\gtrsim\mathsf{N}^{-1-\beta_{\mathrm{CLT}}}\Big)\Big]^{-1}\Big\}^{-1}.$$

We note that $\gamma_\mathsf{N}\log\mathsf{N}$ comes from an estimate on the relative entropy of the initial data law with respect to $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, which comes from Lemma 5.5. We claim that it now suffices to show the following estimate, where $\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ means expectation with initial data distributed via $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$:

{eq:bgbulkIII2}
$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathsf{T}_\mathsf{N}}\mathsf{N}^{-1}\sum_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|\mathrm{dt}\Big)\lesssim\sup_{\mathrm{t}\in[0,\mathsf{T}_\mathsf{N}]}\max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]}\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(\mathbf{1}_{\mathscr{E}}|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|)\lesssim\mathsf{N}^{-1-2\beta_{\mathrm{CLT}}}. \tag{6.6}$$

Indeed, (6.6) and the Markov inequality shows that the probability on the RHS of (6.4) is $\lesssim\mathsf{N}^{-\beta_{\mathrm{CLT}}}$, and thus the RHS of (6.4) itself is $\lesssim\gamma_\mathsf{N}\log\mathsf{N}\cdot\beta_{\mathrm{CLT}}^{-1}(\log\mathsf{N})^{-1}\ll 1$. We now estimate the expectations in the middle of (6.6). For simplicity, we restrict to $\mathrm{x}\in[\![0,\mathsf{N}]\!]$ such that $\mathrm{x}\leqslant\mathsf{N}/2$; the case $\mathrm{x}\geqslant\mathsf{N}/2$ follows by using the same argument upon the reflection on $[\![0,\mathsf{N}]\!]$ which swaps $0\leftrightarrow\mathsf{N}$. Additionally, assume that $\mathfrak{q}_{\ell,\mathrm{w}}$ in $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}$ (see Proposition 3.7 and Definition 3.4) is backwards-oriented; again, for forwards-oriented functions, a similar reflection about w applied to the argument below works. We again recall Definition 3.4 to write

{eq:bgbulkpf1a}
$$\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}=\mathfrak{t}_{\mathbf{Av}}^{-1}\int_0^{\mathfrak{t}_{\mathbf{Av}}}\sum_{\mathrm{w}\in\mathbb{I}_\mathrm{x}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\mathfrak{q}_\mathrm{w}[\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}]\cdot\mathbf{Z}_{\mathrm{t}-\mathrm{s},\mathrm{w}}^\mathsf{N}(\mathbf{Z}_{\mathrm{t},\mathrm{x}_+}^\mathsf{N})^{-1}\mathrm{ds}, \tag{6.7}$$

where the notation above is described below.

- The function $\mathfrak{q}_\mathrm{w}$ stands for $\mathfrak{q}_{\ell,\mathrm{w}}$; we drop the ℓ-subscript and use it for another index below.
- The interval $\mathbb{I}_\mathrm{x}\subseteq[\![0,\mathsf{N}]\!]$ contains x and has length $\lesssim\mathsf{N}^{1-3\delta_\mathbf{S}/2}(\log\mathsf{N})^{\mathrm{O}(1)}$.
- We know that x_+ is to the right of $\mathbb{I}_\mathrm{x}$, and for any $\mathrm{w}\in\mathbb{I}_\mathrm{x}$, we have $|\mathrm{x}_+-\mathrm{w}|\lesssim(\log\mathsf{N})^3\mathsf{N}^{1-3\delta_\mathbf{S}/2}$.
- We have $\mathfrak{t}_{\mathbf{Av}}=\mathsf{N}^{-2/3-\mathrm{C}_0\delta_\mathbf{S}}$ for a large but fixed constant $\mathrm{C}_0>0$.
- The kernel $\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}$ is the $\mathbf{H}^{\mathsf{N},0,0}$-kernel (see Definition 3.3) at time $\mathfrak{t}_\mathsf{N}=\mathsf{N}^{-\delta_\mathbf{S}}$. Therefore, from Proposition A.1, we have the following for any integers $0\leqslant\mathrm{a}\lesssim 1$ and $\mathfrak{l}_1,\ldots,\mathfrak{l}_\mathrm{a}\lesssim 1$, where gradients act on the second variable:

$$\Big\|\Big(\prod_{\ell\in[\![1,\mathrm{a}]\!]}\nabla_{\mathfrak{l}_\ell}^\mathbf{X}\Big)\boldsymbol{\mathcal{H}}_{\cdot,\cdot}\Big\|_{\ell^\infty([\![0,\mathsf{N}]\!]\times[\![0,\mathsf{N}]\!])}\lesssim\mathsf{N}^{-1-\mathrm{a}}\mathsf{N}^{\mathrm{O}(\mathrm{a}\delta_\mathbf{S})}. \tag{6.8}$$

In what follows, we perform a finite number of high probability manipulations, with each manipulation restricting $\mathscr{E}$ in (6.6) further. However, since there are only finitely many manipulations, the final conclusion will hold on a high probability event (which will be $\mathscr{E}$). First, we write $(\mathbf{Z}_{\mathrm{t},\mathrm{x}_+}^\mathsf{N})^{-1}=(\mathbf{Z}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N})^{-1}\cdot\mathbf{Z}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N}(\mathbf{Z}_{\mathrm{t},\mathrm{x}_+}^\mathsf{N})^{-1}$. Then, we use (2.10) to write $\mathbf{Z}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N}(\mathbf{Z}_{\mathrm{t},\mathrm{x}_+}^\mathsf{N})^{-1}=\mathbf{Exp}\{\lambda(\mathbf{j}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N}-\mathbf{j}_{\mathrm{t},\mathrm{x}_+}^\mathsf{N})+\lambda\mathscr{R}_\lambda\mathfrak{t}_\mathbf{Av}\}$. By Lemma 5.13 with the bound $\lambda\mathscr{R}_\lambda\mathfrak{t}_\mathbf{Av}\lesssim\mathsf{N}^{-2/3}$, we deduce that this exponential is $\mathbf{Exp}(\mathrm{O}(\mathsf{N}^{-\rho}))\lesssim 1$ for some $\rho>0$. So, with high probability, we have the following bound, where the last line follows by changing variables in the ds-integral:

$$|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}|\lesssim\Big|\mathfrak{t}_\mathbf{Av}^{-1}\int_0^{\mathfrak{t}_\mathbf{Av}}\sum_{\mathrm{w}\in\mathbb{I}_\mathrm{x}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\mathfrak{q}_\mathrm{w}[\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}]\cdot\mathbf{Z}_{\mathrm{t}-\mathrm{s},\mathrm{w}}^\mathsf{N}(\mathbf{Z}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N})^{-1}\mathrm{ds}\Big|$$
$$=\Big|\mathfrak{t}_\mathbf{Av}^{-1}\int_{-\mathfrak{t}_\mathbf{Av}}^0\sum_{\mathrm{w}\in\mathbb{I}_\mathrm{x}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\mathfrak{q}_\mathrm{w}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\cdot\mathbf{Z}_{\mathrm{t}+\mathrm{s},\mathrm{w}}^\mathsf{N}(\mathbf{Z}_{\mathrm{t}-\mathfrak{t}_\mathbf{Av},\mathrm{x}_+}^\mathsf{N})^{-1}\mathrm{ds}\Big|.$$

Throughout the following, we will freely change the time-variable in the $(\mathbf{Z}^\mathsf{N})^{-1}$-factor by a shift of $\leqslant\mathfrak{t}_\mathbf{Av}$ for the sake of $\lesssim$-statements, justified by the same reasoning as above. We will now introduce a multi-scale space-time decomposition of the quantity above, which will allow us to localize this space-time average into pieces that we can compare to half-space dynamics. In particular, the decomposition below is exactly the one that is used in [33] in the half-space setting. We refer the reader to Section 7 therein for a further discussion.

{notation:ms}
Notation 6.3. Decompose $\mathbb{I}_\mathrm{x}=\mathbb{I}_{\mathrm{x},1}\cup\ldots\cup\mathbb{I}_{\mathrm{x},\mathrm{K}}$ into disjoint intervals with $\mathrm{K}\lesssim 1$ such that:

- We have $\max \mathbb{I}_{\mathrm{x},\mathrm{k}} + 1 = \min \mathbb{I}_{\mathrm{x},\mathrm{k}+1}$ for all $\mathrm{k} = 1, \ldots, \mathrm{K}-1$.
- We have $|\mathbb{I}_{\mathrm{x},\mathrm{k}}| \asymp \mathsf{N}^{\gamma_{\mathrm{k}}}$, where $\gamma_1 > 0$ is small but fixed and $\gamma_{\mathrm{k}+1} = \gamma_{\mathrm{k}} + \varepsilon$ with $\varepsilon > 0$ small for $\mathrm{k} = 1, \ldots, \mathrm{K}-2$. The index K is the smallest such index for which $\gamma_{\mathrm{K}-1} \geqslant \frac{1}{2} + \gamma$, where $\gamma > 0$ is small but fixed as well. Note that $|\mathbb{I}_{\mathrm{x},\mathrm{K}}| \asymp |\mathbb{I}_{\mathrm{x}}|$, since the union $\mathbb{I}_{\mathrm{x},1} \cup \ldots \cup \mathbb{I}_{\mathrm{x},\mathrm{K}-1}$ is of $\mathrm{O}(1)$-many intervals whose lengths are $\ll |\mathbb{I}_{\mathrm{x}}|$.

Next, for any k-index, decompose $[-\mathfrak{t}_{\mathbf{Av}}, 0] = \mathbb{U}_{\mathrm{k},1} \cup \ldots \cup \mathbb{U}_{\mathrm{k},\mathrm{L}}$ into disjoint time-intervals such that:

- We have $\sup \mathbb{U}_{\mathrm{k},\ell} = \inf \mathbb{U}_{\mathrm{k},\ell+1}$ for any $\ell = 1, \ldots, \mathrm{L}-1$.
- We have that $|\mathbb{U}_{\mathrm{k},\ell}| \asymp \mathsf{N}^{-2}\mathsf{N}^{2\gamma_{\mathrm{k}-1}}\mathsf{N}^{-\varepsilon} + \mathsf{N}^{-4/3}\mathsf{N}^{2\gamma_{\mathrm{k}-1}/3}\mathsf{N}^{-\varepsilon}$, for $\varepsilon > 0$ small and $\mathrm{k} \geqslant 2$, is independent of ℓ.
- Suppose now that $\mathrm{k} = 1$. If $\min \mathbb{I}_{\mathrm{x},1} \leqslant \mathsf{N}^{\gamma_1}$, then let $|\mathbb{U}_{1,\ell}| \asymp \mathsf{N}^{-3/2}$ by independent of ℓ. If $\min \mathbb{I}_{\mathrm{x},1} > \mathsf{N}^{\gamma_1}$, then let $|\mathbb{U}_{1,\ell}| = |\mathbb{U}_{2,\ell}|$ for each ℓ.

We note that L can depend on k, though we omit this dependence as it will not play a role.

Equipped with Notation 6.3, we can now write

$$\begin{aligned}|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}| &\lesssim \sum_{\mathrm{k}\in[\![1,\mathrm{K}]\!]} \mathrm{L}^{-1} \sum_{\ell\in[\![1,\mathrm{L}]\!]} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathfrak{t}_{\mathbf{Av}},\mathrm{x}_+})^{-1} \mathrm{ds} \Big| \\ &\lesssim \sum_{\mathrm{k}\in[\![1,\mathrm{K}]\!]} \mathrm{L}^{-1} \sum_{\ell\in[\![1,\mathrm{L}]\!]} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_+})^{-1} \mathrm{ds} \Big|.\end{aligned}$$

Write $(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_+})^{-1} = (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}})^{-1} \cdot \mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_+})^{-1}$, where $\mathrm{x}_\mathrm{k} = \mathbb{I}_{\mathrm{x},\mathrm{k}}$. By (2.10), we have $\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_+})^{-1} = \mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_+})\}$, in which the difference on the RHS of this identity is a sum over $\mathrm{y} \in [\![\mathrm{x}_\mathrm{k}, \mathrm{x}_+]\!]$ of $\mathsf{N}^{-1/2}\boldsymbol{\phi}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{y}}$. By Lemma 5.10, this sum is $\lesssim \mathsf{N}^{\delta}\mathsf{N}^{-1/2}|\mathrm{x}_\mathrm{k} - \mathrm{x}_+|^{1/2} \lesssim \mathsf{N}^{2\delta}\mathsf{N}^{-3\delta\mathfrak{s}/4} \lesssim \mathsf{N}^{-\rho}$ for $\delta > 0$ small and $\rho > 0$ fixed; the last bound follows because $\mathrm{x}_\mathrm{k} \in \mathbb{I}_{\mathrm{x},\mathrm{k}} \subseteq \mathbb{I}_{\mathrm{x}}$. Thus,

$$|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}| \lesssim \sum_{\mathrm{k}\in[\![1,\mathrm{K}]\!]} \mathrm{L}^{-1} \sum_{\ell\in[\![1,\mathrm{L}]\!]} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}})^{-1} \mathrm{ds} \Big|$$

with high probability. Next, let us apply (2.10) and and write

$$\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} (\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}})^{-1} = \mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}})\} \cdot \mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}}) - \lambda \mathscr{R}_\lambda(\mathrm{s} - \inf \mathbb{U}_{\mathrm{k},\ell})\}.$$

As we explained above, each exponential above is $\mathbf{Exp}(\mathrm{O}(\mathsf{N}^{-\rho})) \lesssim 1$. Moreover, by the bulk-admissible bound (3.2) for $\mathfrak{q}_{\mathrm{w}}$, we know that $|\mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]| \lesssim \mathsf{N}^{\varepsilon}$ for $\varepsilon > 0$ small with high probability. Thus, with high probability,

{eq:bgbulkpf2}
$$|\mathbf{Av}_{\mathrm{t},\mathrm{x}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}| \lesssim \sum_{\mathrm{k}\in[\![1,\mathrm{K}]\!]} \mathrm{L}^{-1} \sum_{\ell\in[\![1,\mathrm{L}]\!]} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}} \mathrm{ds} \Big|, \tag{6.9}$$

where the notation $\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}$ is for technical benefits and is explained in the following construction.

{notation:cut}
Notation 6.4. Fix $\varepsilon > 0$ small. We let $\boldsymbol{\eta} : \mathbb{R} \to \mathbb{R}$ satisfy the following properties:

- We have $\boldsymbol{\eta} \equiv 1$ on $[-\mathsf{N}^{\varepsilon}, \mathsf{N}^{\varepsilon}]$ and $\boldsymbol{\eta} \equiv 0$ on $[-2\mathsf{N}^{\varepsilon}, \mathsf{N}^{\varepsilon}]$.
- We have $\|\boldsymbol{\eta}\|_{\mathscr{C}^{\mathrm{m}}(\mathbb{R})} \lesssim_{\mathrm{m}} 1$ for any $\mathrm{m} \geqslant 0$.

Next, for $\rho > 0$ fixed, let $\boldsymbol{\nu} : \mathbb{R} \to \mathbb{R}$ satisfy the following properties:

- We have $\boldsymbol{\nu} \equiv 1$ on $1 + [-\mathsf{N}^{-\rho}, \mathsf{N}^{-\rho}]$ and $\boldsymbol{\nu} \equiv 0$ on $1 + [-2\mathsf{N}^{-\rho}, 2\mathsf{N}^{-\rho}]$.
- We have $\|\boldsymbol{\nu}\|_{\mathscr{C}^{\mathrm{m}}(\mathbb{R})} \lesssim_{\mathrm{m}} 1$ for any $\mathrm{m} \geqslant 0$.

We now define

$$\begin{aligned}\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}} &:= \widetilde{\mathfrak{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \mathbf{J}^{\mathrm{k},\ell,\mathrm{x}}_{\mathrm{s}}, \\ \widetilde{\mathfrak{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] &:= \mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] \cdot \boldsymbol{\eta}(\mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]), \\ \mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}] &:= \mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}})\} \cdot \boldsymbol{\eta}(\mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{w}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}})\}), \\ \mathbf{J}^{\mathrm{k},\ell,\mathrm{x}}_{\mathrm{s}} &:= \mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}}) - \lambda \mathscr{R}_\lambda(\mathrm{s} - \inf \mathbb{U}_{\mathrm{k},\ell})\} \\ &\quad \times \boldsymbol{\nu}(\mathbf{Exp}\{\lambda(\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}_\mathrm{k}} - \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell},\mathrm{x}_\mathrm{k}}) - \lambda \mathscr{R}_\lambda(\mathrm{s} - \inf \mathbb{U}_{\mathrm{k},\ell})\}).\end{aligned}$$

We will now estimate the expectation of the summands in (6.9). For $\mathrm{k} \geqslant 2$, we have the following.

{prop:k2}

Proposition 6.5. *Recall $\beta_{\mathrm{CLT}} > 0$ from (6.6), and let $\varepsilon > 0$ be small. There exists a high probability path-space event $\mathscr{G}$ such that the following estimate holds for any $\mathrm{k} \in [\![2, \mathrm{K}]\!]$ and $\ell \in [\![1, \mathrm{L}]\!]$:*

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|}\int_{\mathbb{U}_{\ell,\mathrm{k}}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},\mathrm{k}}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}}\mathrm{d}\mathrm{s}\Big|\Big) \lesssim \mathsf{N}^{\mathrm{O}(\varepsilon)}\mathsf{N}^{-1}|\mathbb{U}_{\mathrm{k},\ell}|^{-\frac{1}{2}}\|\boldsymbol{\mathcal{H}}_{\mathrm{x},\cdot}\|_{\ell^\infty(\mathbb{I}_{\mathrm{x},\mathrm{k}})}|\mathbb{I}_{\mathrm{x},\mathrm{k}}|^{\frac{1}{2}} + \mathsf{N}^{-1-2\beta_{\mathrm{CLT}}}. \tag{6.10}$$

{eq:k2I}

We clarify that $\mathscr{G}$ is another high probability event, not necessarily the one on which (6.9) holds, for instance. However, the intersection of finitely-many high probability events is also high probability, so we can freely insert the indicator of any other high probability event (i.e. $\mathscr{G}$).

Let us address the case $\mathrm{k} = 1$, which requires a bit of a different analysis. First, we claim that deterministically, we have the following estimate (which we explain afterwards and which uses Notation 6.4):

$$\tfrac{1}{|\mathbb{U}_{\ell,1}|}\int_{\mathbb{U}_{\ell,1}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},1}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\boldsymbol{\omega}}^{1,\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}}\mathrm{d}\mathrm{s} = \tfrac{1}{|\mathbb{U}_{\ell,1}|}\int_{\mathbb{U}_{\ell,1}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},1}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\mathsf{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathrm{d}\mathrm{s} + \mathrm{O}(\mathsf{N}^{-1-\frac{1}{2}\rho}). \tag{6.11}$$

{eq:bgbulkpf3}

Indeed, we have replaced $\mathbf{J}^{\mathrm{k},\ell,\mathrm{x}}_{\mathrm{s}}$ by 1, whose difference is deterministically $\lesssim \mathsf{N}^{-\rho}$ by construction in Notation 6.4. We multiply this by $\mathsf{N}^{\mathrm{O}(\varepsilon)}$ coming from the size of $\widetilde{\mathsf{q}}_{\mathrm{w}}\mathbf{G}_{\mathrm{w}}$, as well as $\mathsf{N}^{-1+\mathrm{O}(\delta_{\mathrm{S}})}$ coming from (6.8). Finally, we sum over $\mathbb{I}_{\mathrm{x},1}$, whose size is $\lesssim \mathsf{N}^{\gamma_1}$ for some small $\gamma_1 > 0$. Ultimately, we get the last term above. Now,:

{prop:k1}

Proposition 6.6. *Recall $\beta_{\mathrm{CLT}} > 0$ from (6.6). There is a high probability event $\mathscr{G}$ so that for all $\ell \in [\![1, \mathrm{L}]\!]$,*

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,1}|}\int_{\mathbb{U}_{\ell,1}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},1}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\mathsf{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|\Big) \lesssim \mathsf{N}^{-1-2\beta_{\mathrm{CLT}}}. \tag{6.12}$$

{eq:k1I}

Assuming the validity of Propositions 6.5 and 6.6, we can now show Proposition 6.1.

Proof of Proposition 6.1. As noted before (and explained after) (6.6), it suffices to show (the last bound in) (6.6). This follows by combining (6.9), Proposition 6.5, (6.11), and Proposition 6.6 (while noting that each restricts to a possibly different high probability event, though because there are only finitely many of these events, the final intersection holds with high probability). Indeed, as shown in the proof of Proposition 7.1 in Section 7.1.5 of [33], the first term on the RHS of (6.10) is $\lesssim \mathsf{N}^{-1-2\beta_{\mathrm{CLT}}}$ (for some small but fixed $\beta_{\mathrm{CLT}} > 0$). □

We will now dedicate the rest of this subsection to Propositions 6.5 and 6.6.

6.1.2. *Reductions to the half-space.* Our goal now is to compare the $\mathbf{j}^{\mathsf{N}}$-process to some half-space modification. For example, instead of studying $\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}$ in (6.10), we will consider the same function but of a half-space version of the $\mathbf{j}^{\mathsf{N}}$-process. Thus, we start by introducing this half-space model.

{definition:ha

Definition 6.7. Define $\mathbf{h}^{\mathsf{N}} : [0,\infty) \times [\![0,\infty]\!]$ as the solution to the following SDE, which we explain after:

$$\begin{aligned}\mathrm{d}\mathbf{h}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} &= \mathsf{N}^{\frac{3}{2}}\nabla^{\mathbf{X}}_1\mathscr{U}'[\boldsymbol{\kappa}_{\mathrm{t},\mathrm{x}}]\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\tau_{\mathrm{x}}\boldsymbol{\kappa}_{\mathrm{t}}]\mathrm{d}\mathrm{t} + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{w}_{\mathrm{t},\mathrm{x}},\\ \mathrm{d}\mathbf{h}^{\mathsf{N}}_{\mathrm{t},0} &= \mathsf{N}^{\frac{3}{2}}\mathscr{U}'[\boldsymbol{\kappa}_{\mathrm{t},1}]\mathrm{d}\mathrm{t} + \mathsf{N}\mathbf{F}[\boldsymbol{\kappa}_{\mathrm{t}}] + \sqrt{2}\mathsf{N}^{\frac{1}{2}}\mathrm{d}\mathbf{w}_{\mathrm{t},0}.\end{aligned}$$

We restrict to $\mathrm{x} \in [\![1,\infty]\!]$ in the first line. We also used the notation $\boldsymbol{\kappa}_{\mathrm{t},\mathrm{x}} := -\mathsf{N}^{1/2}\nabla^{\mathbf{X}}_{-1}\mathbf{h}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathsf{N}^{1/2}(\mathbf{h}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{h}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}-1})$ for all $\mathrm{x} \in [\![1,\infty]\!]$. In particular, $\boldsymbol{\kappa}_{\mathrm{t}} := (\boldsymbol{\kappa}_{\mathrm{t},\mathrm{x}})_{\mathrm{x}\in[\![1,\infty]\!]} \in \mathbb{R}^{[\![1,\infty]\!]}$, we extend it to $\mathbb{R}^{\mathbb{Z}}$ using the rule $\mathscr{U}'[\boldsymbol{\kappa}_{\mathrm{t},\mathrm{x}}] = 0$ for $\mathrm{x} \in [\![-\infty, 0]\!]$. The $\mathrm{t} \mapsto \mathbf{w}_{\mathrm{t},\mathrm{x}}$ are independent standard Brownian motions, whose relation to the Brownian motions in (2.1)-(2.3) will be clarified when relevant. Finally, the initial data above will also be specified depending on the context. We emphasize that there is no boundary parameter (such as $\widetilde{\mathbf{A}}$ in (2.2)).

Let us now present the desired comparisons, first in the case of $\mathrm{k} = 2$ (i.e. the setting of interest in Proposition 6.5), then in the case of $\mathrm{k} = 1$ (i.e. the setting of interest in Proposition 6.6).

{lemma:compare

Lemma 6.8. *Fix any $\mathrm{k} \geqslant 2$ and $\ell \in [\![1, \mathrm{L}]\!]$ as in Proposition 6.5. Suppose that $\mathbf{h}^{\mathsf{N}}_{0,\mathrm{x}} = \mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf\mathbb{U}_{\mathrm{k},\ell},\mathrm{x}}$ for all $\mathrm{x} \in [\![0, \mathsf{N}]\!]$. Suppose also that $\mathbf{h}^{\mathsf{N}}_{0,\mathrm{x}}$, for $\mathrm{x} \in [\![\mathsf{N}+1,\infty]\!]$, is chosen so that the following is satisfied:*

- *The restriction of $\boldsymbol{\kappa}_{\mathrm{t}}$ to $[\![\mathsf{N}+1,\infty]\!]$ is distributed according to the $\mathbb{R}^{[\![\mathsf{N}+1,\infty]\!]}$-marginal of $\mathbb{P}^0$.*
- *The restriction of $\boldsymbol{\kappa}_{\mathrm{t}}$ to $[\![\mathsf{N}+1,\infty]\!]$ is independent of the restriction of $\mathbf{h}^{\mathsf{N}}_{0,\cdot}$ to $[\![0,\infty]\!]$.*

Finally, suppose $\mathbf{w}_{r,x} = \mathbf{b}_{t+\inf \mathbb{U}_{k,\ell}+r,x}$ *for all* $x \in [\![0, N]\!]$. *There exists a high probability event* $\mathscr{G}$ *as in Proposition 6.5 (that is thus independent of* (k, ℓ, t, x, s, w)*-variables on the* LHS *of (6.10)) such that for any* $D > 0$,

$$\mathbf{1}_{\mathscr{G}} \sup_{s\in[0,|\mathbb{U}_{k,\ell}|]} \max_{\operatorname{dist}(y,\mathbb{I}_{x,k})\lesssim 1} |\mathbf{j}^N_{t+\inf \mathbb{U}_{k,\ell}+s,y} - \mathbf{h}^N_{s,y}| \lesssim_D N^{-D}. \tag{6.13}$$ {eq:comparek2I}

Proof. For notational convenience, assume $t + \inf \mathbb{U}_{k,\ell} = 0$, and write $t := |\mathbb{U}_{k,\ell}|$. We claim that $\min \mathbb{I}_{x,k} \gtrsim N^{\gamma_1} + N^{\varepsilon} N t^{1/2} + N^{\varepsilon} N^2 t^{3/2}$; this follows by construction in Notation 6.3, while noting therein that $\min \mathbb{I}_{x,k} \geqslant \max \mathbb{I}_{x,k-1}$ with $\mathbb{I}_{x,k-1} \subseteq [\![0, N]\!]$ having length $N^{\gamma_{k-1}} = N^{\gamma_k} N^{\varepsilon}$ for a possibly different but still small $\varepsilon > 0$. Now, we let $\mathbb{L}$ be interval containing $\mathbb{I}_{x,k}$ such that the distance between $\mathbb{L}$ and the edge $\{0, N\}$ is also $\gtrsim N^{\gamma_1} + N^{\varepsilon} N t^{1/2} + N^{\varepsilon} N^2 t^{3/2}$. We can do this by our lower bound on $\min \mathbb{I}_{x,k}$, as well as the fact that $\mathbb{I}_{x,k} \subseteq \mathbb{I}_x$, the latter of which is contained in $[\![0, \frac{1}{2}N + o(N)]\!]$ by assumption after (6.6). We also arrange for $\mathbb{L}$ so that the distance between $\min \mathbb{L}$ and $\mathbb{I}_{x,k}$, as well as the distance between $\max \mathbb{L}$ and $\mathbb{I}_{x,k}$, is $\gtrsim N^{\gamma_1} + N^{\varepsilon} N t^{1/2} + N^{\varepsilon} N^2 t^{3/2}$. In words, $\mathbb{L}$ is a proportionally-scaled neighborhood of $\mathbb{I}_{x,k}$ which is similarly separated from the edge $\{0, N\}$.

Now, we set $\mathbf{k}^N_{s,y} := \mathbf{1}_{y\in\mathbb{L}} \cdot (\mathbf{j}^N_{s,y} - \mathbf{h}^N_{s,y})$. With notation to be explained afterwards, we have

$$\partial_s \mathbf{k}^N_{s,y} = N^2 \nabla_1^{\mathbf{X}} (\mathbf{a}_{s,y} \nabla_{-1}^{\mathbf{X}} \mathbf{k}^N_{s,y}) + N^{\frac{3}{2}} \sum_{|w|\lesssim 1} \mathfrak{f}_{s,w,y} \nabla_w^{\mathbf{X}} \mathbf{k}^N_{s,y} + O(N^{O(1)} \{ \|\mathbf{j}^N_{s,\cdot}\|_{\ell^\infty([\![0,N]\!])} + \|\mathbf{k}^N_{s,\cdot}\|_{\ell^\infty([\![0,N]\!])} \}) \cdot \mathbf{1}_{y\approx\partial\mathbb{L}}.$$

Above, we set $\mathbf{a}_{s,y} = (\mathscr{U}'[\boldsymbol{\phi}_{s,y}] - \mathscr{U}'[\boldsymbol{\kappa}_{s,y}]) \cdot (\boldsymbol{\phi}_{s,y} - \boldsymbol{\kappa}_{s,y})^{-1}$, which is valued in $[\Lambda^{-1}, \Lambda]$ for some fixed $\Lambda \geqslant 1$ by the mean-value theorem and Assumption 2.2. The term $\mathfrak{f}_{s,w,y}$, which also comes from the mean-value theorem applied to the difference of (2.4) evaluated at $\boldsymbol{\phi}_s, \boldsymbol{\kappa}_s$, satisfies $|\mathfrak{f}_{s,w,y}| \lesssim 1 + \max_{|z-y|\lesssim 1} |\boldsymbol{\phi}_{s,z}|^C + \max_{|z-y|} |\boldsymbol{\kappa}_{s,z}|^C$ deterministically for some $0 \leqslant C \lesssim 1$; this follows from the bounds on the potential in Assumption 2.2. Finally, $y \approx \partial\mathbb{L}$ is short-hand for the distance between y and the boundary of $\mathbb{L}$ being $\lesssim 1$. We clarify that the Brownian motions in the $\mathbf{j}^N, \mathbf{h}^N$ SDEs cancelled since we restrict to $\mathbb{L} \subseteq [\![0, N]\!]$. Next, let $\boldsymbol{\Gamma}_{r,s,y,z}$ be a function of $(r, s, y, z) \in [0, \infty) \times [0, \infty) \times \mathbb{Z} \times \mathbb{Z}$, restricted to $r \leqslant s$, which satisfies

$$\partial_s \boldsymbol{\Gamma}_{r,s,y,z} = N^2 \nabla_1^{\mathbf{X}} (\mathbf{a}_{s,y} \nabla_{-1}^{\mathbf{X}} \boldsymbol{\Gamma}_{r,s,y,z}) + N^{\frac{3}{2}} \sum_{|w|\lesssim 1} \mathfrak{f}_{s,w,y} \nabla_w^{\mathbf{X}} \boldsymbol{\Gamma}_{r,s,y,z} \quad \text{and } \boldsymbol{\Gamma}_{r,r,y,z} = \mathbf{1}_{y=z},$$

in which all gradients act on y. In particular, $\boldsymbol{\Gamma}$ is the heat kernel for the linear part of the $\mathbf{k}^N$-equation. Thus, for any y such that $\operatorname{dist}(y, \mathbb{I}_{x,k}) \lesssim 1$, we have the estimate

$$|\mathbf{k}^N_{s,y}| \lesssim N^{O(1)} \int_0^s \sum_{z\approx\partial\mathbb{L}} \boldsymbol{\Gamma}_{r,\tau,y,z} \cdot (\|\mathbf{j}^N_{\tau,\cdot}\|_{\ell^\infty([\![0,N]\!])} + \|\mathbf{k}^N_{\tau,\cdot}\|_{\ell^\infty([\![0,N]\!])}) d\tau \tag{6.14}$$ {eq:comparek2I0}

by the Duhamel formula and the fact that $\mathbf{k}^N$ has vanishing initial data by construction. We now claim that for some high probability $\mathscr{G}$, which is independent of all space-time variables (t, x, s, w, y) at hand, we have

$$\mathbf{1}_{\mathscr{G}} \sup_{s\in[0,|\mathbb{U}_{k,\ell}|]} \max_{y\in\mathbb{L}} \max_{|w|\lesssim 1} |\mathfrak{f}_{s,w,y}| \lesssim N^{\varepsilon_0}, \tag{6.15}$$ {:comparek2I1a}

$$\mathbf{1}_{\mathscr{G}} \sup_{s\in[0,|\mathbb{U}_{k,\ell}|]} \|\mathbf{j}^N_{s,\cdot}\|_{\ell^\infty([\![0,N]\!])} + \|\mathbf{k}^N_{s,\cdot}\|_{\ell^\infty([\![0,N]\!])} \lesssim N^{O(1)}. \tag{6.16}$$ {:comparek2I1b}

where $\varepsilon_0 > 0$ is small but fixed. Let us assume this for now and prove it at the end of this argument. By the first bound (6.15), we can use heat kernel estimates from Proposition 7.9 in [33] to obtain that the $\boldsymbol{\Gamma}$-factor in (6.14) is $\lesssim_D N^{-D}$ for any $D > 0$. Indeed, Proposition 7.9 in [33] shows $\boldsymbol{\Gamma}_{r,\tau,y,z} \lesssim \mathbf{Exp}(N^{1+O(\varepsilon_0)} t) \mathbf{Exp}(-\upsilon N^{-1} t^{-1/2} |y - z|)$; if $\operatorname{dist}(y, \mathbb{I}_{x,k}) \lesssim 1$ and $z \approx \partial\mathbb{L}$, we get $|y - z| \gtrsim N^{\gamma_1} + N^{\varepsilon} N t^{1/2} + N^{\varepsilon} N^2 t^{3/2}$; now take $\varepsilon_0 > 0$ small enough. This ultimately implies that on $\mathscr{G}$, we have $|\mathbf{k}^N_{s,y}| \lesssim_D N^{-D}$ for any $D > 0$ uniformly over $(s, y) \in [0, |\mathbb{U}_{k,\ell}|] \times \mathbb{I}_{x,k}$. This completes the proof of (6.13), so we are left to show (6.15)-(6.16).

The estimate (6.15) follows by the $\mathfrak{f}_{s,w,y}$-bound in terms of $\boldsymbol{\phi}$-variables, which we then bound using Lemma 5.10. As for the second bound, note that $\|\mathbf{j}^N_{0,\cdot}\|_{\ell^\infty([\![0,N]\!])} \lesssim N^{O(1)}$ with high probability; indeed, we have $|\mathbf{j}^N_{0,x}| \lesssim 1$ for some $x \in [\![0, N]\!]$, by assumption (2) in Theorem 2.4, and values at other $x \in [\![0, N]\!]$ are controlled with Lemma 5.10 for $\boldsymbol{\phi}_{0,\cdot}$-variables. The $\mathbf{j}^N$-estimate in (6.16) now follows as the SDE satisfied by $\mathbf{j}^N$ involves only functions of the $\boldsymbol{\phi}_t$-variables that are polynomially bounded in said variables (i.e. satisfy an estimate like (3.2) for various points x therein).

The estimate (6.16) for $\mathbf{k}^N$ was shown in the proof of Proposition 7.9 in [33] (which addressed the half-space setting). In a nutshell, the $\boldsymbol{\kappa}_{t,x}$-values are uniformly $\lesssim_\delta N^\delta$ on $[0, O(1)] \times [\![0, N^{O(1)}]\!]$ with high probability for

any $\delta > 0$. This holds by essentially the same reasoning as the proof of Lemma 5.10; it is shown precisely in Lemma 6.1 of [33]. (Note that Lemma 6.1 of [33] requires the initial data of $\boldsymbol{\kappa}_t$ to have law whose density with respect to $\mathbb{P}^0$ is $\lesssim \mathsf{N}^{\delta}$ for small enough $\delta > 0$, but this follows by construction in the statement of Lemma 6.8 and the estimate (5.6).) With this $\boldsymbol{\kappa}_t$-bound, we can control the RHS of the SDE for $\mathbf{k}^{\mathsf{N}}$ from Definition 6.7, so we are done. □

We now present a corresponding half-space comparison for $\mathrm{k} = 1$. Observe that the following restricts to the case where $\widetilde{\mathbf{A}} = 0$ in (2.2); indeed, we will eventually reduce our analysis for the LHS of (6.12) to this setting.

{lemma:compare}

Lemma 6.9. *Take the setting of Lemma 6.8 but assume that* $\mathrm{k} = 1$*, as well as that* $\widetilde{\mathbf{A}} = 0$ *in* (2.2) *and* (2.6)*. Then the same estimate* (6.13) *remains true (for some high probability event* $\mathscr{G}$ *which does not depend on* $(\ell, \mathrm{t}, \mathrm{x}, \mathrm{s}, \mathrm{w})$*-variables on the* LHS *of* (6.12)*, and for any* $\mathrm{D} > 0$*).*

Proof. We proceed with two cases. First, assume that $\min \mathbb{I}_{\mathrm{x},1} > \mathsf{N}^{\gamma_1}$. In this case, the proof of Lemma 6.8 (for $\mathrm{k} = 2$ specifically) applies here, since the gap between $\min \mathbb{I}_{\mathrm{x},1}$ and the edge 0 is again, by assumption, $\gtrsim \mathsf{N}^{\gamma_1}$, and the time-scale $|\mathbb{U}_{1,\ell}|$ is the same as $|\mathbb{U}_{2,\ell}|$ by assumption in Notation 6.3.

Thus it is enough to assume that $\min \mathbb{I}_{\mathrm{x},1} \leqslant \mathsf{N}^{\gamma_1}$, and thus $\mathbb{I}_{\mathrm{x},1} \subseteq [\![0, \mathrm{O}(\mathsf{N}^{\gamma_1})]\!]$, because $\mathbb{I}_{\mathrm{x},1}$ is an interval of size $\lesssim \mathsf{N}^{\gamma_1}$ (see Notation 6.3). We also recall from Notation 6.3 that in this case, we have $|\mathbb{U}_{1,\ell}| \lesssim \mathsf{N}^{-3/2}$. Moreover, as in the proof of Lemma 6.8, assume that $\mathrm{t} + \inf \mathbb{U}_{1,\ell} = 1$ and write $\mathrm{t} := |\mathbb{U}_{1,\ell}|$, just for notational convenience. Let $\mathbb{L}$ be an interval containing $\{0\}, \mathbb{I}_{\mathrm{x},1}$ such that $\max \mathbb{L} \leqslant \frac{1}{2}\mathsf{N}$, and the distance between $\max \mathbb{L}$ and $\mathbb{I}_{\mathrm{x},1}$ is $\gtrsim \mathsf{N}^{\frac{2}{3}}$. (There is nothing special about $\frac{2}{3}$, it is just strictly less than 1 and "large enough" in a sense to be explained soon.) As in the proof of Lemma 6.8, we set $\mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}} := \mathbf{1}_{\mathrm{y} \in \mathbb{L}}(\mathbf{j}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}} - \mathbf{h}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}})$, and we have

$$\partial_{\mathrm{s}} \mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}} = \mathsf{N}^2 \nabla_1^{\mathbf{X}} (\mathbf{a}_{\mathrm{s},\mathrm{y}} \nabla_{-1}^{\mathbf{X}} \mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}}) + \mathsf{N}^{\frac{3}{2}} \sum_{|\mathrm{w}| \lesssim 1} \mathfrak{f}_{\mathrm{s},\mathrm{w},\mathrm{y}} \nabla^{\mathbf{X}}_{\mathrm{w}} \mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}} + \mathrm{O}(\mathsf{N}^{\mathrm{O}(1)} \{ \|\mathbf{j}^{\mathsf{N}}_{\mathrm{s},\cdot}\|_{\ell^\infty([\![0,\mathsf{N}]\!])} + \|\mathbf{k}^{\mathsf{N}}_{\mathrm{s},\cdot}\|_{\ell^\infty([\![0,\mathsf{N}]\!])} \}) \cdot \mathbf{1}_{\mathrm{y} \approx \partial \mathbb{L}}.$$

By construction and Assumption 2.2, we know that $|\mathbf{a}| \lesssim 1$. Thus, if we combine this with (6.15)-(6.16), then

$$|\partial_{\mathrm{s}} \mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}}| \lesssim \mathsf{N}^2 \sum_{|\mathrm{w}| \lesssim 1} |\mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}+\mathrm{w}}| + \mathsf{N}^{\mathrm{O}(1)} \mathbf{1}_{\mathrm{y} \approx \partial \mathbb{L}}$$

on some high probability event $\mathscr{G}$. Recall also that $\mathbf{k}^{\mathsf{N}}$ has vanishing initial data. We now fix any $\mathrm{y} \in \mathbb{L}$ for which $\mathrm{dist}(\mathrm{y}, \partial \mathbb{L}) \gtrsim \log \mathsf{N}$, say (so that $\mathrm{y} \not\approx \partial \mathbb{L}$). We integrate the previous inequality to obtain

$$\sup_{\mathrm{s} \in [0,\mathrm{t}]} |\mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}}| \lesssim \int_0^{\mathrm{t}} \mathsf{N}^2 \sup_{|\mathrm{w}| \lesssim 1} |\mathbf{k}^{\mathsf{N}}_{\mathrm{r},\mathrm{y}+\mathrm{w}}| \mathrm{dr}.$$

Now assume that $\mathrm{dist}(\mathrm{y}, \mathbb{I}_{\mathrm{x},1}) \lesssim 1$. In this case, we know $\mathrm{y} + \mathrm{w}_1 + \ldots + \mathrm{w}_{\mathrm{M}} \in \mathbb{L}$ has distance $\gtrsim \log \mathsf{N}$ from $\partial \mathbb{L}$ for $|\mathrm{w}_1|, \ldots, |\mathrm{w}_{\mathrm{M}}| \lesssim 1$ if $\mathrm{M} = \lfloor \mathsf{N}^{7/12} \rfloor$. (There is nothing special about $\frac{7}{12}$ except that it is in $(\frac{1}{2}, \frac{2}{3})$; that this interval is non-empty is what we meant earlier by $\frac{2}{3}$ being large enough.) So, we can iterate the previous estimate above M-times to get the following (in which we also use that on $\mathscr{G}$, we have $|\mathbf{k}^{\mathsf{N}}| \lesssim \mathsf{N}^{\mathrm{O}(1)}$ by (6.16)):

$$\sup_{\mathrm{s} \in [0,\mathrm{t}]} |\mathbf{k}^{\mathsf{N}}_{\mathrm{s},\mathrm{y}}| \lesssim (\mathrm{C}\mathsf{N}^2)^{\mathrm{M}} \cdot \mathsf{N}^{\mathrm{O}(1)} \cdot \tfrac{\mathrm{t}^{\mathrm{M}}}{\mathrm{M}!}.$$

Above, $\mathrm{C} = \mathrm{O}(1)$. Now, use $\mathrm{t} \lesssim \mathsf{N}^{-3/2}$ and $\mathrm{M} \asymp \mathsf{N}^{7/12}$ to establish that the RHS above is $\lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}$ uniformly over $\mathrm{y} \in \mathbb{I}_{\mathrm{x},1}$ on the same high probability event, for any $\mathrm{D} > 0$. (For this, we use Stirling's approximation to obtain $\mathrm{M}! \gtrsim \mathrm{M}^{\mathrm{M}} \mathrm{C}^{-\mathrm{M}}$, so the RHS above is $\lesssim \mathrm{C}_1^{\mathrm{M}} \mathsf{N}^{\mathrm{M}/2} \mathrm{M}^{-\mathrm{M}} \lesssim (\mathrm{C}_2 \mathsf{N}^{1/2} \cdot \mathsf{N}^{-7/12})^{\mathrm{M}}$.) This completes the proof. □

6.1.3. *Wrapping it up.* We will now deduce Propositions 6.5 and 6.6 using Lemmas 6.8 and 6.9 to reduce to the half-space setting. We start with Proposition 6.5 and move to Proposition 6.6.

Proof of Proposition 6.5. By the bulk-admissible estimate in (3.2) and the $\boldsymbol{\phi}$-estimate in Lemma 5.10, there is a high probability event on which we have the estimate $\|\mathbf{D}\mathfrak{q}_{\mathrm{w}}[\boldsymbol{\phi}_\tau]\|_\infty \lesssim \mathsf{N}^{\mathrm{O}(1)}$ for each $\mathrm{w} \in [\![0, \mathsf{N}]\!]$ and $\tau \in [0, \mathrm{O}(1)]$, where $\mathbf{D}\mathfrak{q}_{\mathrm{w}}$ is the gradient with respect to $\boldsymbol{\phi}_{\mathrm{x}}$-variables (for all $\mathrm{x} \in [\![0, \mathsf{N}]\!]$). Thus, if we replace $\boldsymbol{\phi}$ and $\mathbf{j}^{\mathsf{N}}$ by $\boldsymbol{\kappa}$ and $\mathbf{h}^{\mathsf{N}}$ (see Definition 6.7 for notation) in $\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}}$ in (6.10), the cost in doing so is $\lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}$ for any $\mathrm{D} > 0$, as long as we restrict to a high probability event $\mathscr{G}$ as in Lemma 6.8. Indeed, this follows by Lemma 6.8, and ultimately,

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \Big(\mathbf{1}_{\mathscr{G}} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \sum_{\mathrm{w} \in \mathbb{I}_{\mathrm{x},\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}} \mathrm{ds} \Big| \Big) \lesssim \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}} \Big(\mathbf{1}_{\mathscr{G}} \Big| \tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|} \int_{\mathbb{U}_{\ell,\mathrm{k}}} \sum_{\mathrm{w} \in \mathbb{I}_{\mathrm{x},\mathrm{k}}} \boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}} \cdot \widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell,\mathrm{half}}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}} \mathrm{ds} \Big| \Big) + \mathsf{N}^{-\mathrm{D}}$$

where, with $\mathbf{J}^{\mathrm{k},\ell,\mathrm{x},\mathrm{half}}$ given by the formula from Notation 6.4 but replacing $\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell}+\cdot,\cdot}$ by $\mathbf{h}^{\mathsf{N}}_{\cdot,\cdot}$,

$$\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell,\mathrm{half}}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}} := \widetilde{\mathrm{q}}_{\mathrm{w}}[\boldsymbol{\kappa}_{\mathrm{s}}] \cdot \mathbf{G}_{\mathrm{w}}[\boldsymbol{\kappa}_{\mathrm{s}}] \cdot \mathbf{J}^{\mathrm{k},\ell,\mathrm{x},\mathrm{half}}_{\mathrm{s}}.$$

(We clarify that like in Lemma 6.8, the initial data for $\mathrm{s} \mapsto \mathbf{h}^{\mathsf{N}}$ is constructed to coincide with that of $\mathbf{j}^{\mathsf{N}}_{\mathrm{t}+\inf \mathbb{U}_{\mathrm{k},\ell}+\cdot,\cdot}$.) In the estimate above, we take $\mathrm{D} > 0$ large but fixed. Finally, by construction (see the statement of Lemma 6.8), the initial data of $\boldsymbol{\kappa}$ has law given by the product of $\mathbb{P}^0$ on $\mathbb{R}^{[\![\mathsf{N}+1,\infty]\!]}$ and the law of $\boldsymbol{\phi}$ at time $\mathrm{t} + \inf \mathbb{U}_{\mathrm{k},\ell} \in [0, \mathrm{T}_{\mathsf{N}}]$ as a probability measure on $\mathbb{R}^{[\![1,\mathsf{N}]\!]}$. By (5.24), the law of $\boldsymbol{\phi}$ at any time in $[0, \mathrm{T}_{\mathsf{N}}]$ has a density with respect to the measure $\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$ with $\mathrm{L}^p(\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}})$-norm that is $\lesssim_p 1$; the validity of this estimate comes because $\mathrm{T}_{\mathsf{N}} \ll 1$. Therefore, the density of this law with respect to $\mathbb{P}^0$ has $\mathrm{L}^p(\mathbb{P}^0)$-norm that is $\lesssim_p 1$ by (5.6) and the Hölder inequality. Thus, the law of the initial data for $\boldsymbol{\kappa}$ has a density with respect to $\mathbb{P}^0$ which is $\lesssim_p 1$ in $\mathrm{L}^p(\mathbb{P}^0)$-norm as well. So, by the Cauchy-Schwarz inequality, we have

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|}\int_{\mathbb{U}_{\ell,\mathrm{k}}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},\mathrm{k}}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell,\mathrm{half}}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}}\mathrm{d}\mathrm{s}\Big|\Big) \lesssim \Big\{\mathbb{E}^0\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,\mathrm{k}}|}\int_{\mathbb{U}_{\ell,\mathrm{k}}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},\mathrm{k}}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\boldsymbol{\omega}}^{\mathrm{k},\ell,\mathrm{half}}_{\mathrm{s},\mathrm{t},\mathrm{w},\mathrm{x}}\mathrm{d}\mathrm{s}\Big|^2\Big)\Big\}^{\frac{1}{2}}$$

by changing measure for the law of the $\boldsymbol{\kappa}$ initial data to $\mathbb{P}^0$. (We can actually apply the Hölder inequality instead of Cauchy-Schwarz to have an exponent of $1+\nu$ for any $\nu > 0$ instead of just 2 on the RHS.) The previous two estimates place us in the setting of the half-space problem in [33]; that is, we combine them with Proposition 7.5 of [33] to complete the proof. □

Proof of Proposition 6.6. We first assume that $\min \mathbb{I}_{\mathrm{x},1} > \mathsf{N}^{\gamma_1}$. For this case, the desired estimate (6.12) follows by the same argument as in the proof of Proposition 6.6 above for $\mathrm{k} = 2$. That is, we use Lemma 6.9 to reduce to the half-space setting of [33]. Then, we cite Proposition 7.6 in [33], i.e. the analogous result to Proposition 6.6 but in the half-space. Thus, it suffices to assume that $\min \mathbb{I}_{\mathrm{x},1} \leqslant \mathsf{N}^{\gamma_1}$. The only difference between this case and the previous one is that the half-space setting of [33] fixes $\widetilde{\mathbf{A}} = 0$ in (2.2) (out of convenience). In particular, we will first proceed by reducing our analysis of the LHS of (6.12) to the case of $\widetilde{\mathbf{A}}$. We claim that

$$\begin{aligned}&\mathbb{E}\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,1}|}\int_{\mathbb{U}_{\ell,1}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},1}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\mathrm{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|\Big)\\ &\lesssim \mathbb{E}_{\widetilde{\mathbf{A}}=0}\Big(\mathbf{1}_{\mathscr{G}}\Big|\tfrac{1}{|\mathbb{U}_{\ell,1}|}\int_{\mathbb{U}_{\ell,1}}\sum_{\mathrm{w}\in\mathbb{I}_{\mathrm{x},1}}\boldsymbol{\mathcal{H}}_{\mathrm{x},\mathrm{w}}\cdot\widetilde{\mathrm{q}}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathbf{G}_{\mathrm{w}}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big),\end{aligned}$$

where the subscript $\widetilde{\mathbf{A}} = 0$ indicates taking $\widetilde{\mathbf{A}} = 0$ in the law of the $\mathbf{j}^{\mathsf{N}}, \boldsymbol{\phi}$ processes inside the expectation. Above, $\mathscr{G}$ can be any (path-space) event. To show this, we note by (2.1)-(2.3), the density of the general $\widetilde{\mathbf{A}} \in \mathbb{R}$-dynamics to the $\widetilde{\mathbf{A}} = 0$-dynamics has the form of $\mathbf{Exp}(\mathrm{c}\mathsf{N}^{1/2}\widetilde{\mathbf{A}}\mathbf{b}_{\mathrm{t},0} + \frac{1}{2}\mathrm{c}^2\mathsf{N}\widetilde{\mathbf{A}}^2\mathrm{t})$, in which $\mathrm{t} = |\mathbb{U}_{1,\ell}| \lesssim \mathsf{N}^{-3/2}$ and $\mathrm{c} \in \mathbb{R}$ is fixed. Indeed, this follows by Girsanov's theorem, since the role of $\widetilde{\mathbf{A}}$ is in an additional drift in (2.2). Thus, the L^p-norm (with respect to the law of the $\widetilde{\mathbf{A}} = 0$-dynamics) is $\lesssim \mathbf{Exp}(\mathrm{O}_p(\mathsf{N}\mathrm{t})) \lesssim 1$; we emphasize the importance of a short time-scale on which we apply Girsanov's theorem. At this point, we can, as in the proof of Proposition 6.5, compare to the half-space setting of Definition 6.7 via Lemma 6.9, and then use the analogous half-space estimate in Proposition 7.6 of [33]. We do not reproduce the same details here. This completes the proof. □

6.2. **Argument for Proposition 6.2.** The architecture of this argument is similar to that of the proof of Proposition 6.1, though it turns out to also be much simpler. First, we write $\mathfrak{b} = \mathfrak{b}_{\ell}$ for convenience. Second, we assume that the support of $\mathfrak{b}$ is a finite neighborhood containing 0, not N. (The case of N follows by a reflection on $[\![0,\mathsf{N}]\!]$ that swaps $0 \leftrightarrow \mathsf{N}$.) As in (6.4), it suffices to assume that $\mathrm{T} = \mathrm{T}_{\mathsf{N}} \to 0$ (slowly). As in (6.5), we have

$$\mathbb{P}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathrm{T}_{\mathsf{N}}}|\mathbf{A}\mathbf{v}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|\mathrm{d}\mathrm{t} \gtrsim \mathsf{N}^{-\beta_{\mathrm{CLT}}}\Big) \lesssim \gamma_{\mathsf{N}}\log\mathsf{N}\Big\{\log\Big[\mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathrm{T}_{\mathsf{N}}}|\mathbf{A}\mathbf{v}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|\mathrm{d}\mathrm{t} \gtrsim \mathsf{N}^{-\beta_{\mathrm{CLT}}}\Big)\Big]^{-1}\Big\}^{-1},$$

where the superscript $\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}$ above denotes that the initial data for the $\boldsymbol{\phi}_{\mathrm{t}}$-process implicit on the RHS has initial data $\sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$. Thus, as in (6.6), it suffices to prove that for some high-probability event, we have

{eq:bgedgeI1}
$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\int_0^{\mathrm{T}_{\mathsf{N}}}|\mathbf{A}\mathbf{v}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|\mathrm{d}\mathrm{t}\Big) \lesssim \sup_{\mathrm{t}\in[0,\mathrm{T}_{\mathsf{N}}]}\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(\mathbf{1}_{\mathscr{E}}|\mathbf{A}\mathbf{v}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|) \lesssim \mathsf{N}^{-2\beta_{\mathrm{CLT}}}. \tag{6.17}$$

With these reductions, we are now in position to prove Proposition 6.2.

Proof of Proposition 6.2. We first claim that for any event $\mathscr{E}$, we have

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}(\mathbf{1}_{\mathscr{E}}|\mathbf{Av}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|) \lesssim \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\widetilde{\mathbf{A}}=0}(\mathbf{1}_{\mathscr{E}}|\mathbf{Av}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|^{\frac{3}{2}})^{\frac{2}{3}},$$

in which the superscript $\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}$ indicates an initial data for the $\boldsymbol{\phi}$-process that is $\sim \mathbb{P}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}$, and the subscript $\widetilde{\mathbf{A}}=0$ means that we set $\widetilde{\mathbf{A}}=0$ in (2.2). This follows by the same Girsanov trick in the proof of Proposition 6.6 above. Indeed, $\mathbf{Av}^{\mathbf{T},\mathfrak{b}}$ depends only on the $\boldsymbol{\phi}$-process on a time-window of length $\lesssim \mathsf{N}^{-3/2}$ (see Definition 3.5) just as in the proof of Proposition 6.6. Now, as in the proof of Proposition 6.1, we have (for an appropriate high probability event $\mathscr{E}$) that the following set of estimates holds (with explanation given afterwards):

$$\begin{aligned}
\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\widetilde{\mathbf{A}}=0}(\mathbf{1}_{\mathscr{E}}|\mathbf{Av}^{\mathbf{T},\mathfrak{b}}_{\mathrm{t},\mathrm{x}}|^{\frac{3}{2}}) &= \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\widetilde{\mathbf{A}}=0}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_0^{\mathrm{t}_{\partial}}\mathfrak{b}[\boldsymbol{\phi}_{\mathrm{t}-\mathrm{s}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{s},\mathrm{x}}(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}})^{-1}\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big)\\
&\lesssim \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\widetilde{\mathbf{A}}=0}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_{-\mathrm{t}_{\partial}}^{0}\mathfrak{b}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}+\mathrm{s},\mathrm{x}}(\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{t}_{\partial},\mathrm{x}})^{-1}\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big)\\
&\lesssim \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}_{\widetilde{\mathbf{A}}=0}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_{-\mathrm{t}_{\partial}}^{0}\mathfrak{b}[\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big)+\mathsf{N}^{-\rho}\\
&\lesssim \mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_{0}^{\mathrm{t}_{\partial}}\mathfrak{b}[\boldsymbol{\kappa}_{\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big)+\mathsf{N}^{-\rho}.
\end{aligned}$$

The first line follows by construction in Definition 3.5. The second line follows from changing variables in the $\mathrm{d}\mathrm{s}$-integration and the high probability replacement of $\mathbf{Z}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}$ by $\mathbf{Z}^{\mathsf{N}}_{\mathrm{t}-\mathrm{t}_{\partial},\mathrm{x}}$ as explained prior to Notation 6.3. The third line follows since, by the proof of (6.11), there is a high probability event $\mathscr{E}$ for which the ratio of $\mathbf{Z}^{\mathsf{N}}$-factors is $=1+\mathrm{O}(\mathsf{N}^{-\rho})$ for some $\rho>0$ (by time-regularity of $\mathbf{j}^{\mathsf{N}}$ via Lemma 5.13). The fourth line uses the notation of Definition 6.7, and it follows because of Lemma 6.9. Indeed, the function $\mathfrak{b}$ is supported in a finite neighborhood of 0 with $\boldsymbol{\phi}_{\mathrm{x}}$-derivatives (for all $\mathrm{x}\in[\![0,\mathsf{N}]\!]$) bounded by N^{ε} on some high probability event $\mathscr{E}$ by (3.2) and Lemma 5.10. Moreover, Lemma 6.9 (specifically the case where $\min\mathbb{I}_{\mathrm{x},1}\leqslant\mathsf{N}^{\gamma_1}$) shows that $|\boldsymbol{\phi}_{\mathrm{t}+\mathrm{s},\mathrm{x}}-\boldsymbol{\kappa}_{\mathrm{s},\mathrm{x}}|\lesssim_{\mathrm{D}}\mathsf{N}^{-\mathrm{D}}$ for any $\mathrm{D}>0$ (on some high probability event $\mathscr{E}$) for all $\mathrm{x}\lesssim\mathsf{N}^{\gamma_1}$, and thus all x in the support of $\mathfrak{b}$.

Now, as in the proof of Proposition 6.5, we can change measure for the initial data of $\boldsymbol{\kappa}$, because its law with respect to $\mathbb{P}^0$ has $\mathrm{L}^p(\mathbb{P}^0)$-norm which is $\lesssim_p 1$ for any $p\geqslant 1$. (Again, we refer to the proof of Proposition 6.5 for an explantation.) Thus, by the Hölder inequality, we have

$$\mathbb{E}^{\widetilde{\mathbf{A}},\widetilde{\mathbf{B}}}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_0^{\mathrm{t}_{\partial}}\mathfrak{b}[\boldsymbol{\kappa}_{\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|^{\frac{3}{2}}\Big)\lesssim\mathbb{E}^{0}\Big(\mathbf{1}_{\mathscr{E}}\Big|\mathrm{t}_{\partial}^{-1}\int_0^{\mathrm{t}_{\partial}}\mathfrak{b}[\boldsymbol{\kappa}_{\mathrm{s}}]\mathrm{d}\mathrm{s}\Big|^{2}\Big)^{\frac{3}{4}},$$

where $\mathbb{E}^0$ indicates that the initial data of $\boldsymbol{\kappa}$ is $\sim\mathbb{P}^0$. It is now shown in the proof of Proposition 7.2 in [33] (which addresses the half-space version of the problem) that the second moment above is $\lesssim\mathsf{N}^{-\rho}$ for some $\rho>0$. □

7. Proof of Proposition 3.9

{section:final}

The first estimate (3.19) follows immediately by Lemma 5.11. Thus, this section is dedicated to the proof of (3.20). The architecture of this section is outlined as follows:

(1) The goal of (3.20) is to compare $\mathbf{S}^{\mathsf{N}}$ and $\mathbf{Q}^{\mathsf{N}}$. First, we introduce a couple of intermediate (or "auxiliary") processes which facilitate this comparison, which mostly consist of including high probability a priori estimates to help with a standard SHE-type analysis of these equations.
(2) Then, our comparison of $\mathbf{S}^{\mathsf{N}},\mathbf{Q}^{\mathsf{N}}$ consists of SHE-type analyses to compare $\mathbf{S}^{\mathsf{N}}$ to the first auxiliary SDE, then the first auxiliary SDE to the next, until we arrive at $\mathbf{Q}^{\mathsf{N}}$ (after only a couple steps in this procedure).

7.1. Auxiliary SDEs.

Let us introduce the following stopping time, whose role we explain below:

$$\begin{aligned}
\mathrm{t}_{\mathrm{st}} := 1 &\wedge \inf\{\mathrm{s}\in[0,1]:\|\boldsymbol{\phi}_{\mathrm{s},\cdot}\|_{\ell^\infty([\![1,\mathsf{N}]\!])}\geqslant\mathsf{N}^{\delta_{\mathbf{S}}}\}\\
&\wedge\inf\Big\{\mathrm{s}\in[0,1]:\max_{\ell\in[\![1,\mathrm{L}_1]\!]}\|\mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{s},\cdot}\|_{\ell^\infty([\![0,\mathsf{N}]\!])}\geqslant\mathsf{N}^{-\frac{1}{2}+\delta_{\mathbf{S}}}\Big\}\\
&\wedge\inf\{\mathrm{s}\in[0,1]:\|\mathbf{R}^{\mathsf{N}}_{\mathrm{s},\cdot}-\mathbf{R}^{\mathsf{N},\wedge}_{\mathrm{s},\cdot}\|_{\ell^\infty([\![0,\mathsf{N}]\!])}\neq 0\}.
\end{aligned}\tag{7.1}$$

{eq:tst}

For the relevant notation above, we refer to Definitions 3.4, 3.5, Proposition 3.7, and Lemma (5.11). Let us recall that $\delta_{\mathbf{S}} > 0$ is the small constant from Definition 3.3 as well. Moreover, by Lemmas 5.10, 5.11, and 5.12,

$$\mathbb{P}(\mathfrak{t}_{\mathrm{st}} = 1) = 1 - \mathrm{o}(1). \tag{7.2}$$ {eq:tst2}

We clarify that the role of $\mathfrak{t}_{\mathrm{st}}$ is to provide a priori estimates for the coefficients and terms appearing on the RHS of the SDE (3.10)-(3.13), thereby facilitating our analysis of this SDE. Next, we introduce the following auxiliary SDE, which can be seen as (3.10)-(3.13) after removing $\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}$ and $\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}$, and inserting $\mathfrak{t}_{\mathrm{st}}$:

{eq:ysde}
$$\begin{aligned}
\mathrm{d}\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} &= \mathscr{T}_{\mathsf{N}}\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{Y}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathbf{t}_{\mathbf{Av}}}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\cdot\mathbf{Av}^{\mathbf{X},\mathcal{Q}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm})\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\cdot\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}})\mathrm{d}\mathrm{t}.
\end{aligned}\tag{7.3}$$

Above, the term $\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{Y}^{\mathsf{N}}]$ is given by replacing $\mathbf{S}^{\mathsf{N}}$ with $\mathbf{Y}^{\mathsf{N}}$ in (3.14). We remark that by (7.2), the insertion of the $\mathfrak{t}_{\mathrm{st}}$-indicators is harmless with high probability. To further illustrate this, take the following modified version of (3.10)-(3.13) with similar stopping time insertions:

{eq:sstsde}
$$\begin{aligned}
\mathrm{d}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{x}} &= \mathscr{T}_{\mathsf{N}}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\cdot}\mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{S}^{\mathsf{N},\mathrm{st}}]_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{w}_\pm}\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathbf{t}_{\mathbf{Av}}}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{X},\mathcal{Q}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{w}_\pm})\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{w}}\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}\mathbf{S}^{\mathsf{N},\mathrm{st}}_{\mathrm{t},\mathrm{w}})\mathrm{d}\mathrm{t}.
\end{aligned}\tag{7.4}$$

We clarify that the initial data for $\mathbf{Y}^{\mathsf{N}}, \mathbf{S}^{\mathsf{N},\mathrm{st}}, \mathbf{S}^{\mathsf{N}}$ are all the same.

{lemma:sst}
Lemma 7.1. *We have $\mathbf{S}^{\mathsf{N}} = \mathbf{S}^{\mathsf{N},\mathrm{st}}$ on $[0,1]\times[\![0,\mathsf{N}]\!]$ with high probability.*

Proof. The processes $\mathbf{S}^{\mathsf{N}}, \mathbf{S}^{\mathsf{N},\mathrm{st}}$ satisfy the same (linear) SDE with the same initial data until the stopping time $\mathfrak{t}_{\mathrm{st}}$. Thus, by Itô calculus, they agree on $[0,\mathfrak{t}_{\mathrm{st}}]\times[\![0,\mathsf{N}]\!]$ with probability 1. Now use (7.2). □

We will now introduce a further modification of the SDE (7.4), in which we introduce an a priori estimate for the $\mathbf{S}^{\mathsf{N},\mathrm{st}}$-terms hitting the space-time and time-averages therein. First, for any $\varepsilon > 0$ small and any $\mathrm{a}\in\mathbb{R}$, define

{eq:boperator}
$$\mathscr{B}_\varepsilon(\mathrm{a}) = \mathrm{a}\mathbf{1}[|\mathrm{a}|\leqslant\mathsf{N}^{\varepsilon}]. \tag{7.5}$$

We now let $\mathbf{W}^{\mathsf{N}}$ solve the same SDE as (7.4) (with the same initial data) but with two insertions of $\mathscr{B}_\varepsilon$-operators:

{eq:wsde}
$$\begin{aligned}
\mathrm{d}\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} &= \mathscr{T}_{\mathsf{N}}\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}[\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda\mathsf{N}^{\frac{1}{2}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\cdot}\mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{t}}\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{W}^{\mathsf{N}}]_{\mathrm{t},\mathrm{x}}\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}\mathscr{B}_\varepsilon(\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm})\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_1]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathbf{t}_{\mathbf{Av}}}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{X},\mathcal{Q}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm}\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}_\pm})\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{\mathrm{t},\mathrm{w}}\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}\mathscr{B}_\varepsilon(\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}})\mathrm{d}\mathrm{t} \\
&+ \sum_{\ell\in[\![1,\mathrm{L}_2]\!]}\sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{x},\mathrm{w}}\cdot\mathsf{N}\cdot\nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial}(\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant\mathrm{t}}\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{t}}]\mathbf{R}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}}\mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{w}})\mathrm{d}\mathrm{t}.
\end{aligned}\tag{7.6}$$

We now present the following analogue of Lemma 7.1 for $\mathbf{S}^{\mathsf{N},\mathrm{st}}$ and $\mathbf{W}^{\mathsf{N}}$.

{lemma:w}

Lemma 7.2. *Suppose that* $|\mathbf{S}^{\mathsf{N},\mathrm{st}}| \leqslant \mathsf{N}^{\varepsilon}$ *on* $[0,1]\times[\![0,\mathsf{N}]\!]$. *Then* $\mathbf{S}^{\mathsf{N},\mathrm{st}} = \mathbf{W}^{\mathsf{N}}$ *on* $[0,1]\times[\![0,\mathsf{N}]\!]$.

Proof. By assumption, we can replace $\mathbf{S}^{\mathsf{N},\mathrm{st}}$ by $\mathscr{B}_{\varepsilon}(\mathbf{S}^{\mathsf{N},\mathrm{st}})$ in (7.4), at which point we obtain that $\mathbf{S}^{\mathsf{N},\mathrm{st}}, \mathbf{W}^{\mathsf{N}}$ satisfy the same Itô SDE with the same initial data, so the claim follows. □

7.2. **Properties of $\mathbf{Y}^{\mathsf{N}}$.** We now gather a collection of facts about $\mathbf{Y}^{\mathsf{N}}$. In a nutshell, the goal of these properties is to reduce the proof of Proposition 3.9 to the proof of a comparison between $\mathbf{Y}^{\mathsf{N}}, \mathbf{W}^{\mathsf{N}}$. In particular, we compare $\mathbf{Y}^{\mathsf{N}}$ and $\mathbf{Q}^{\mathsf{N}}$, and additionally, we provide a priori estimates for $\mathbf{Y}^{\mathsf{N}}$ that will both facilitate a comparison between $\mathbf{Y}^{\mathsf{N}}$ and $\mathbf{W}^{\mathsf{N}}$, as well as provide a proof of the a priori estimates needed in Lemma 7.2.

{lemma:yfacts}

Lemma 7.3. *Fix any* $\varepsilon > 0$. *With high probability, we have*

$$\sup_{\mathrm{t}\in[0,1]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} |\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \leqslant \mathsf{N}^{\varepsilon}. \tag{7.7}$$ {eq:yfactsI}

Moreover, there exists $\gamma > 0$ *such that with high probability, we have*

$$\sup_{\mathrm{t}\in[0,1]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} |\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{Q}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\gamma}. \tag{7.8}$$ {eq:yfactsII}

Proof. This follows by the same argument as in the proof of (5.16) in Section 9.2 of [33]. Let us briefly illustrate the key points below. Take (7.3); by the Duhamel formula, we have

$$\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{0,\mathrm{t},\mathrm{x},\mathrm{w}} \mathbf{S}^{\mathsf{N}}_{0,\mathrm{w}} + \mathbf{\Lambda}^{(1)}_{\mathrm{t},\mathrm{x}} + \ldots + \mathbf{\Lambda}^{(4)}_{\mathrm{t},\mathrm{x}}, \tag{7.9}$$ {eq:yfactsI1}

where

$$\mathbf{\Lambda}^{(1)}_{\mathrm{t},\mathrm{x}} := \int_0^{\mathrm{t}} \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{s}} \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{w}} \cdot [\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}(\sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{R}^{\mathsf{N}}_{\mathrm{s},\cdot} \mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\cdot} \mathrm{d}\mathbf{b}_{\mathrm{s},\cdot})]_{\mathrm{w}},$$

$$\mathbf{\Lambda}^{(2)}_{\mathrm{t},\mathrm{x}} := \int_0^{\mathrm{t}} \mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{s}} \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{w}} \cdot \mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{Y}^{\mathsf{N}}]_{\mathrm{s},\mathrm{w}} \mathrm{d}\mathrm{s},$$

$$\mathbf{\Lambda}^{(3)}_{\mathrm{t},\mathrm{x}} := \sum_{\ell\in[\![1,\mathrm{L}_1]\!]} \int_0^{\mathrm{t}} \sum_{\mathrm{y}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} \cdot \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{y},\mathrm{w}} \cdot \mathsf{N} \cdot \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_{\mathbf{Av}}} (\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{s}} \cdot \mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{s},\mathrm{w}} \mathbf{R}^{\mathsf{N}}_{\mathrm{s},\mathrm{w}_\pm} \mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{w}_\pm}) \mathrm{d}\mathrm{s},$$

$$\mathbf{\Lambda}^{(4)}_{\mathrm{t},\mathrm{x}} := \sum_{\ell\in[\![1,\mathrm{L}_2]\!]} \int_0^{\mathrm{t}} \sum_{\mathrm{y}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} \cdot \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{y},\mathrm{w}} \cdot \mathsf{N} \cdot \nabla^{\mathbf{T},\mathrm{av}}_{\mathfrak{t}_\partial} (\mathbf{1}_{\mathfrak{t}_{\mathrm{st}}\geqslant \mathrm{s}} \mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{s}}] \mathbf{R}^{\mathsf{N}}_{\mathrm{s},\mathrm{w}} \mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{w}}) \mathrm{d}\mathrm{s}.$$

We claim that the following estimates hold, in which $\|\cdot\|^{2p}_{2p} := \mathbb{E}|\cdot|^{2p}$ (with explanations given afterwards):

$$\|\mathbf{\Lambda}^{(1)}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim_p \int_0^{\mathrm{t}} |\mathrm{t}-\mathrm{s}|^{-\frac{1}{2}} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{x}}\|^2_{2p} \mathrm{d}\mathrm{s}, \tag{7.10}$$ {eq:yfactsI2a}

$$\max_{\mathrm{k}=2,3,4} \|\mathbf{\Lambda}^{(\mathrm{k})}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim_p \mathsf{N}^{-\gamma} \sup_{\mathrm{s}\in[0,\mathrm{t}]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{x}}\|^2_{2p}. \tag{7.11}$$ {eq:yfactsI2b}

Above, $\gamma > 0$ is fixed. The first bound (7.10) is a standard bound for SHE-type equations; see, for instance, (32) in [25]. (The semigroup $\mathbf{e}^{\mathfrak{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}$, which does not appear in [25], does not change things since it is integration against a smoothing kernel.) As for (7.11) with $\mathrm{k}=2$, we note that for $\mathrm{s}\leqslant\mathfrak{t}_{\mathrm{st}}$, the operator $\mathrm{Err}[\mathbf{R}^{\mathsf{N}}\mathbf{Y}^{\mathsf{N}}]_{\mathrm{s},\mathrm{w}}$ is integration of $\mathbf{Y}^{\mathsf{N}}$ against an averaging kernel that is additionally multiplied by $\mathsf{N}^{-\gamma}$ for some $\gamma>0$ fixed; see (3.14)-(3.15). Thus, (7.11) with $\mathrm{k}=2$ follows essentially from Cauchy-Schwarz with respect to all space-time integrations and the triangle inequality for $\|\cdot\|_{2p}$. For $\mathrm{k}=3$, the same idea works with the following power-counting. We move the time-gradient onto the kernel $\sum_{\mathrm{y}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} \boldsymbol{\mathcal{K}}^{(\ell)}_{\mathrm{y},\mathrm{w}} = [\mathbf{e}^{(\mathrm{t}-\mathrm{s})\mathscr{T}_{\mathsf{N}}}(\boldsymbol{\mathcal{K}}^{(\ell)}_{\cdot,\mathrm{y}})]_{\mathrm{x}}$. By using time-regularity for this kernel (which follows by spatial regularity estimates in Proposition A.1 and the PDE (3.3) that turns time-regularity into spatial regularity), we get a factor of $\mathfrak{t}_{\mathbf{Av}} \leqslant \mathsf{N}^{-2/3}$ (see Definition 3.4). Since $\mathfrak{t}_{\mathrm{st}} \geqslant \mathrm{s}$, from $\mathbf{Av}^{\mathbf{X},\mathscr{Q}_\ell}\mathbf{R}^{\mathsf{N}}$, we get a factor $\lesssim \mathsf{N}^{-1/2+\delta_{\mathbf{S}}}$. So, the power of N in $\mathbf{\Lambda}^{(3)}$ is $\lesssim \mathsf{N}\cdot\mathsf{N}^{-2/3}\cdot\mathsf{N}^{-1/2+\delta_{\mathbf{S}}} \lesssim \mathsf{N}^{-\gamma}$. Lastly, for (7.11) with $\mathrm{k}=4$, we have a similar power-counting. Indeed, the time-gradient gives $\lesssim \mathfrak{t}_\partial \lesssim \mathsf{N}^{-3/2}$, and by the deterministic estimate for $\mathfrak{b}_\ell$ in (3.6), from $\mathfrak{b}_\ell[\boldsymbol{\phi}_{\mathrm{s}}]\mathbf{R}^{\mathsf{N}}$, we get a factor of $\mathsf{N}^{\mathrm{O}(\delta_{\mathbf{S}})}$. Lastly, by the localized support of $\boldsymbol{\mathcal{K}}^{\partial,\ell}$ in (3.17), we get a factor $\lesssim \mathsf{N}^{-1+\mathrm{O}(\delta_{\mathbf{S}})}$ from the sum over w of $\boldsymbol{\mathcal{K}}^{\partial,\ell}_{\mathrm{y},\mathrm{w}}$. So, the power of N in $\mathbf{\Lambda}^{(4)}$ is $\lesssim \mathsf{N}\cdot\mathsf{N}^{-1+\mathrm{O}(\delta_{\mathbf{S}})}\mathsf{N}^{-3/2} \lesssim \mathsf{N}^{-\gamma}$.

Observe that the LHS of (7.10)-(7.11), when added together, control $\|\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\|^2_{2p}$. Moreover, observe that we can replace the LHS of (7.10)-(7.11) by their respective maxima over $\mathrm{x} \in [\![0, \mathsf{N}]\!]$, since the RHS of these estimates are independent of the x-variable on the LHS. Thus, we obtain

$$\max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim 1 + \int_0^{\mathrm{t}} |\mathrm{t}-\mathrm{s}|^{-\frac{1}{2}} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{x}}\|^2_{2p} \mathrm{d}\mathrm{s} + \mathsf{N}^{-\gamma} \sup_{\mathrm{s}\in[0,\mathrm{t}]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{Y}^{\mathsf{N}}_{\mathrm{s},\mathrm{x}}\|^2_{2p}.$$

(The 1 on the RHS comes from estimating the first term on the RHS of (7.9) via the assumption (2.14).) It is now a standard consequence of the Gronwall inequality, more or less, that the LHS of the previous display is $\lesssim 1$. From this, the estimate (7.7) follows by a union bound over $[0,1] \times [\![0, \mathsf{N}]\!]$, where for the time-variable, we first union bound over a very fine discretization and then bootstrap to all of $[0,1]$ via standard short-time continuity.

We now discuss (7.8). Consider the SDE below with the same initial data as $\mathbf{Y}^{\mathsf{N}}, \mathbf{S}^{\mathsf{N}}$:

$$\mathrm{d}\mathbf{U}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \mathscr{T}_{\mathsf{N}} \mathbf{U}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} + \mathbf{1}_{\mathrm{t}_{\mathrm{st}} \geqslant \mathrm{t}} [\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}} (\sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot} \mathbf{U}^{\mathsf{N}}_{\mathrm{t},\cdot} \mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}}.$$

If we set $\mathbf{D}^{\mathsf{N}} := \mathbf{Y}^{\mathsf{N}} - \mathbf{S}^{\mathsf{N}}$, then we have

$$\mathbf{D}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} = \int_0^{\mathrm{t}} \mathbf{1}_{\mathrm{t}_{\mathrm{st}} \geqslant \mathrm{s}} \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \mathbf{H}^{\mathsf{N}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{w}} \cdot [\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}} (\sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{R}^{\mathsf{N}}_{\mathrm{s},\cdot} \mathbf{D}^{\mathsf{N}}_{\mathrm{s},\cdot} \mathrm{d}\mathbf{b}_{\mathrm{s},\cdot})]_{\mathrm{x}} + \mathbf{\Lambda}^{(2)}_{\mathrm{t},\mathrm{x}} + \mathbf{\Lambda}^{(3)}_{\mathrm{t},\mathrm{x}} + \mathbf{\Lambda}^{(4)}_{\mathrm{t},\mathrm{x}}.$$

Following the proof of (7.10) and using (7.11), we get the following (recall $\|\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim \mathsf{N}^{-\gamma}$ for some $\gamma > 0$):

$$\|\mathbf{D}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim \int_0^{\mathrm{t}} |\mathrm{t}-\mathrm{s}|^{-\frac{1}{2}} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{D}^{\mathsf{N}}_{\mathrm{s},\mathrm{x}}\|^2_{2p} \mathrm{d}\mathrm{s} + \mathsf{N}^{-\gamma}.$$

Another application of the Gronwall inequality argument from above yields $\|\mathbf{D}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}\|^2_{2p} \lesssim \mathsf{N}^{-\gamma}$ uniformly in x and locally uniformly in t. Finally, another union bound argument provides (7.8) with a possibly different $\gamma > 0$, and with $\mathbf{U}^{\mathsf{N}}$ in place of $\mathbf{Q}^{\mathsf{N}}$. Thus, it suffices to prove (7.8) after instead replacing $\mathbf{Y}^{\mathsf{N}}$ by $\mathbf{U}^{\mathsf{N}}$. To this end, observe first that we can drop $\mathbf{R}^{\mathsf{N}}$; before time t_{st}, this factor of $\mathbf{R}^{\mathsf{N}}$ is deterministically $1 + \mathrm{o}(1)$ (see (7.1)). Next, observe that with high probability, we can remove the indicator of $\mathrm{t}_{\mathrm{st}} \geqslant \mathrm{t}$ in the $\mathbf{U}^{\mathsf{N}}$ SDE because of (7.2); this follows by the same argument as in the proof of Lemma 7.1, for instance. Next, standard SHE estimates (see Proposition 5.4 in [25]) give spatial regularity for $\mathbf{U}^{\mathsf{N}}$ that let us remove the $\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}$-operator in the $\mathbf{U}^{\mathsf{N}}$ SDE. Indeed, $\mathrm{t}_{\mathsf{N}} \ll 1$ (see Definition 3.3), so $\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}$ is an approximate identity. (We clarify that all removals are at the cost of $\lesssim \mathsf{N}^{-\gamma}$-errors.) Thus, we arrive at the SDE (3.18) for $\mathbf{Q}^{\mathsf{N}}$, but with initial data that differs by another $\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}}$-operator that we can remove by the spatial regularity estimates in (2.14) for the initial data of $\mathbf{Z}^{\mathsf{N}}$. This completes the proof. □

7.3. **Final touches.** In view of Lemma 7.3, it now suffices to estimate $\mathbf{Y}^{\mathsf{N}} - \mathbf{W}^{\mathsf{N}}$ and prove the a priori estimate that is assumed in Lemma 7.2. We start with the former, then the latter.

{lemma:yw}

Lemma 7.4. *There exists $\gamma > 0$ such that with high probability, we have*

{eq:ywI}
$$\sup_{\mathrm{t}\in[0,1]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} |\mathbf{Y}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} - \mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}}| \lesssim \mathsf{N}^{-\gamma}. \tag{7.12}$$

Proof. We start with a few preliminaries. Consider the event $\mathscr{F}_{\mathrm{T}}$ on which

{eq:ywI0a}
$$\max_{\ell\in[\![1,\mathrm{L}_1]\!]} \int_0^{\mathrm{T}} \mathsf{N}^{-1} \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} |\mathbf{A}\mathbf{v}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}_{\mathrm{t},\mathrm{w}}| \mathrm{d}\mathrm{t} \lesssim \mathsf{N}^{-1-\beta_\star}, \tag{7.13}$$

{eq:ywI0b}
$$\max_{\ell\in[\![1,\mathrm{L}_2]\!]} \int_0^{\mathrm{T}} \sum_{\mathrm{dist}(\mathrm{w},\{0,\mathsf{N}\})\lesssim 1} |\mathbf{A}\mathbf{v}^{\mathbf{T},\mathbf{b}_\ell}_{\mathrm{t},\mathrm{w}}| \mathrm{d}\mathrm{t} \lesssim \mathsf{N}^{-\beta_\star}. \tag{7.14}$$

in which $\beta_\star > 0$ is fixed. We note that $\mathscr{F}_1 \subseteq \mathscr{F}_{\mathrm{T}}$ for any $\mathrm{T} \in [0,1]$. By Propositions 6.1 and 6.2, we know that $\mathscr{F}_1$ holds with high probability. Thus, if we define the stopping time

$$\mathfrak{t}_{\mathrm{BG}} := 1 \wedge \inf\{\mathrm{t} \in [0,1] : \mathbf{1}_{\mathscr{F}_\mathrm{t}} \neq 1\},$$

then $\mathfrak{t}_{\mathrm{BG}} = 1$ with high probability. (Use $\mathbf{1}_{\mathscr{F}_1} \leqslant \mathbf{1}_{\mathscr{F}_\mathrm{t}}$ for any $\mathrm{t} \in [0,1]$, so if $\mathbf{1}_{\mathscr{F}_\mathrm{t}} = 0$, then $\mathbf{1}_{\mathscr{F}_1} = 0$, and this occurs with low probability.) We now let $\mathbf{W}^{\mathsf{N},\mathrm{BG}}$ solve the following modification of (7.6) with the use of $\mathfrak{t}_{\mathrm{BG}}$:

{eq:wbgsde}
$$\mathrm{d}\mathbf{W}^{\mathsf{N},\mathrm{BG}}_{\mathrm{t},\mathrm{x}} = \mathscr{T}_{\mathsf{N}} \mathbf{W}^{\mathsf{N}}_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} + \mathbf{1}_{\mathrm{t}_{\mathrm{st}} \geqslant \mathrm{t}} [\mathbf{e}^{\mathrm{t}_{\mathsf{N}}\mathscr{T}_{\mathsf{N}}} (\sqrt{2}\lambda \mathsf{N}^{\frac{1}{2}} \mathbf{R}^{\mathsf{N}}_{\mathrm{t},\cdot} \mathbf{W}^{\mathsf{N},\mathrm{BG}}_{\mathrm{t},\cdot} \mathrm{d}\mathbf{b}_{\mathrm{t},\cdot})]_{\mathrm{x}} + \mathbf{1}_{\mathrm{t}_{\mathrm{st}} \geqslant \mathrm{t}} \mathrm{Err}[\mathbf{R}^{\mathsf{N}} \mathbf{W}^{\mathsf{N},\mathrm{BG}}]_{\mathrm{t},\mathrm{x}} \mathrm{d}\mathrm{t} \tag{7.15}$$

$$+\sum_{\ell\in[\![1,L_1]\!]}\sum_{w\in[\![0,N]\!]}\mathcal{K}^{(\ell)}_{x,w}\cdot N\cdot \mathbf{1}_{t_{BG}\geqslant t}\mathbf{1}_{t_{st}\geqslant t}\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}_{t,w}\mathbf{R}^N_{t,w_\pm}\mathscr{B}_\varepsilon(\mathbf{W}^{N,BG}_{t,w_\pm})dt$$
$$+\sum_{\ell\in[\![1,L_1]\!]}\sum_{w\in[\![0,N]\!]}\mathcal{K}^{(\ell)}_{x,w}\cdot N\cdot \nabla^{\mathbf{T},av}_{t_{\mathbf{Av}}}(\mathbf{1}_{t_{st}\geqslant t}\mathbf{Av}^{\mathbf{X},\mathcal{Q}_\ell}_{t,w}\mathbf{R}^N_{t,w_\pm}\mathbf{W}^{N,BG}_{t,w_\pm})dt$$
$$+\sum_{\ell\in[\![1,L_2]\!]}\sum_{w\in[\![0,N]\!]}\mathcal{K}^{\partial,\ell}_{x,w}\cdot N\cdot \mathbf{1}_{t_{BG}\geqslant t}\mathbf{1}_{t_{st}\geqslant t}\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{t,w}\mathbf{R}^N_{t,w}\mathscr{B}_\varepsilon(\mathbf{W}^{N,BG}_{t,w})dt$$
$$+\sum_{\ell\in[\![1,L_2]\!]}\sum_{w\in[\![0,N]\!]}\mathcal{K}^{\partial,\ell}_{x,w}\cdot N\cdot \nabla^{\mathbf{T},av}_{t_\partial}(\mathbf{1}_{t_{st}\geqslant t}\mathfrak{b}_\ell[\boldsymbol{\phi}_t]\mathbf{R}^N_{t,w}\mathbf{W}^{N,BG}_{t,w})dt.$$

Since $t_{BG}=1$ with high probability, the same proof of Lemma 7.1 implies that $\mathbf{W}^N=\mathbf{W}^{N,BG}$ on $[0,1]\times[\![0,N]\!]$ with high probability. Now, set $\mathbf{O}^N:=\mathbf{Y}^N-\mathbf{W}^{N,BG}$. By the SDEs (7.3) and (7.6), and the Duhamel formula,

$$\mathbf{O}^N_{t,x}:=\boldsymbol{\Xi}^{(1)}_{t,x}+\ldots+\boldsymbol{\Xi}^{(6)}_{t,x}. \tag{7.16}$$ {eq:ywI1}

where

$$\boldsymbol{\Xi}^{(1)}_{t,x}:=\int_0^t\mathbf{1}_{t_{st}\geqslant s}\sum_{w\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,w}\cdot[\mathbf{e}^{t_N\mathscr{T}_N}(\sqrt{2}\lambda N^{\frac12}\mathbf{R}^N_{s,\cdot}\mathbf{O}^N_{s,\cdot}d\mathbf{b}_{s,\cdot})]_w,$$
$$\boldsymbol{\Xi}^{(2)}_{t,x}:=\int_0^t\mathbf{1}_{t_{st}\geqslant s}\sum_{w\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,w}\cdot \mathrm{Err}[\mathbf{R}^N\mathbf{O}^N]_{s,w}ds,$$
$$\boldsymbol{\Xi}^{(3)}_{t,x}:=\sum_{\ell\in[\![1,L_1]\!]}\int_0^t\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}\cdot\sum_{w\in[\![0,N]\!]}\mathcal{K}^{(\ell)}_{y,w}\cdot N\cdot\nabla^{\mathbf{T},av}_{t_{\mathbf{Av}}}(\mathbf{1}_{t_{st}\geqslant s}\cdot\mathbf{Av}^{\mathbf{X},\mathcal{Q}_\ell}_{s,w}\mathbf{R}^N_{s,w_\pm}\mathbf{O}^N_{s,w_\pm})ds,$$
$$\boldsymbol{\Xi}^{(4)}_{t,x}:=\sum_{\ell\in[\![1,L_2]\!]}\int_0^t\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}\cdot\sum_{w\in[\![0,N]\!]}\mathcal{K}^{\partial,\ell}_{y,w}\cdot N\cdot\nabla^{\mathbf{T},av}_{t_\partial}(\mathbf{1}_{t_{st}\geqslant s}\mathfrak{b}_\ell[\boldsymbol{\phi}_s]\mathbf{R}^N_{s,w}\mathbf{O}^N_{s,w})ds,$$
$$\boldsymbol{\Xi}^{(5)}_{t,x}:=\sum_{\ell\in[\![1,L_1]\!]}\int_0^t\mathbf{1}_{t_{st}\geqslant s}\mathbf{1}_{t_{BG}\geqslant s}\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}\cdot\sum_{w\in[\![0,N]\!]}\mathcal{K}^{(\ell)}_{y,w}\cdot N\cdot\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}_{s,w}\mathbf{R}^N_{s,w_\pm}\mathscr{B}_\varepsilon(\mathbf{W}^{N,BG}_{s,w_\pm})ds,$$
$$\boldsymbol{\Xi}^{(6)}_{t,x}:=\sum_{\ell\in[\![1,L_2]\!]}\int_0^t\mathbf{1}_{t_{st}\geqslant s}\mathbf{1}_{t_{BG}\geqslant s}\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}\cdot\sum_{w\in[\![0,N]\!]}\mathcal{K}^{\partial,\ell}_{y,w}\cdot N\cdot\mathbf{Av}^{\mathbf{T},\mathfrak{b}_\ell}_{s,w}\mathbf{R}^N_{s,w}\mathscr{B}_\varepsilon(\mathbf{W}^{N,BG}_{s,w})ds.$$

The proofs of (7.10)-(7.11) are agnostic to $\mathbf{Y}^N$, and thus they yield

$$\|\boldsymbol{\Xi}^{(1)}_{t,x}\|^2_{2p}\lesssim_p\int_0^t|t-s|^{-\frac12}\max_{x\in[\![0,N]\!]}\|\mathbf{O}^N_{s,x}\|^2_{2p}ds, \tag{7.17}$$ {eq:ywI2a}
$$\max_{k=2,3,4}\|\boldsymbol{\Xi}^{(k)}_{t,x}\|^2_{2p}\lesssim_p N^{-\gamma}\sup_{s\in[0,t]}\max_{x\in[\![0,N]\!]}\|\mathbf{O}^N_{s,x}\|^2_{2p}. \tag{7.18}$$ {eq:ywI2b}

We now address $\boldsymbol{\Xi}^{(5)},\boldsymbol{\Xi}^{(6)}$. First, with the estimates (3.16)-(3.17), along with the fact that the $\mathscr{T}_N$-semigroup has, locally uniformly in time, bounded operator norm $\ell^\infty([\![0,N]\!])\to\ell^\infty([\![0,N]\!])$ (see Proposition A.1), we know

$$\sup_{s\in[0,t]}\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}|\mathcal{K}^{(\ell)}_{y,w}|\lesssim N^{-1+O(\delta_S)}, \tag{7.19}$$ {eq:ywI3a}
$$\sup_{s\in[0,t]}\sum_{y\in[\![0,N]\!]}\mathbf{H}^N_{s,t,x,y}|\mathcal{K}^{(\ell)}_{y,w}|\lesssim N^{-1+O(\delta_S)}\mathbf{1}_{\mathrm{dist}(w,\{0,N\})\lesssim1}. \tag{7.20}$$ {eq:ywI3b}

(For the latter estimate, the power of N comes from the explanation above; the indicator comes for free because the LHS does not change if we multiply by said indicator because of (3.17).) Now, take $\boldsymbol{\Xi}^{(5)}$. Because we restrict to $t_{st}\geqslant s$, we can use $\mathbf{R}^N_{s,w_\pm}\lesssim1$. Moreover, by construction of $\mathscr{B}_\varepsilon$, we can bound the $\mathscr{B}_\varepsilon(\cdot)$-term therein by $\lesssim N^\varepsilon$ for $\varepsilon>0$ small. Then, we use (7.19). This ultimately gives the deterministic bound

$$|\boldsymbol{\Xi}^{(5)}_{t,x}|\lesssim N^{O(\delta_S)+\varepsilon}\max_{\ell\in[\![1,L_1]\!]}\int_0^t\mathbf{1}_{t_{BG}\geqslant s}\cdot N^{-1}\sum_{w\in[\![0,N]\!]}N|\mathbf{Av}^{\mathbf{T},\mathbf{X},\mathcal{Q}_\ell}_{s,w}|ds$$

$$= \mathsf{N}^{\mathrm{O}(\delta_{\mathbf{S}})+\varepsilon} \max_{\ell\in[\![1,\mathrm{L}_1]\!]} \int_0^{\mathrm{t}_{\mathrm{BG}}} \mathsf{N}^{-1} \sum_{\mathrm{w}\in[\![0,\mathsf{N}]\!]} \mathsf{N} |\mathbf{Av}_{\mathrm{s,w}}^{\mathbf{T},\mathbf{X},\mathscr{Q}_\ell}| \mathrm{ds} \lesssim \mathsf{N}^{-\frac{1}{2}\beta_\star}. \tag{7.21}$$ {eq:ywI3c}

In the previous display, the last bound follows by definition of the stopping time t_{BG}, and because $\delta_{\mathbf{S}}, \varepsilon$ are small (though independent of N). Now, for $\boldsymbol{\Xi}^{(6)}$, we similarly control $\mathbf{R}^{\mathsf{N}}\mathscr{B}_\varepsilon(\cdot) \lesssim \mathsf{N}^{\varepsilon}$ for $\varepsilon > 0$ small, and we use (7.20). Thus, we have the following deterministic bound analogous to (7.21):

$$|\boldsymbol{\Xi}_{\mathrm{t,x}}^{(6)}| \lesssim \mathsf{N}^{\mathrm{O}(\delta_{\mathbf{S}})+\varepsilon} \max_{\ell\in[\![1,\mathrm{L}_2]\!]} \int_0^{\mathrm{t}} \mathbf{1}_{\mathrm{t}_{\mathrm{BG}}\geqslant \mathrm{s}} \cdot \sum_{\mathrm{dist}(\mathrm{w},\{0,\mathsf{N}\})\lesssim 1} |\mathbf{Av}_{\mathrm{s,w}}^{\mathbf{T},\mathfrak{b}_\ell}| \mathrm{ds} \lesssim \mathsf{N}^{-\frac{1}{2}\beta_\star}. \tag{7.22}$$ {eq:ywI3d}

We now combine (7.16), (7.17)-(7.18), and (7.21)-(7.22), so that

$$\|\mathbf{O}_{\mathrm{t,x}}^{\mathsf{N}}\|_{2p}^2 \lesssim \int_0^{\mathrm{t}} |\mathrm{t}-\mathrm{s}|^{-\frac{1}{2}} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{O}_{\mathrm{s,x}}^{\mathsf{N}}\|_{2p}^2 \mathrm{ds} + \mathsf{N}^{-\gamma} \sup_{\mathrm{s}\in[0,\mathrm{t}]} \max_{\mathrm{x}\in[\![0,\mathsf{N}]\!]} \|\mathbf{O}_{\mathrm{s,x}}^{\mathsf{N}}\|_{2p}^2 + \mathsf{N}^{-\frac{1}{2}\beta_\star}.$$

As in the proof of Lemma 7.3, we can insert a maximum over $\mathrm{x} \in [\![0,\mathsf{N}]\!]$ on the LHS. Then, we use the Gronwall inequality to get $\|\mathbf{O}_{\mathrm{t,x}}^{\mathsf{N}}\|_{2p}^2 \lesssim \mathsf{N}^{-\beta_0}$ for some $\beta_0 > 0$ fixed, and uniformly over $(\mathrm{t},\mathrm{x}) \in [0,1]\times[\![0,\mathsf{N}]\!]$. A union bound and short-time continuity argument, exactly like that in the proof of Lemma 7.3, now yields $|\mathbf{O}^{\mathsf{N}}| \lesssim \mathsf{N}^{-\gamma}$ (for some possibly different $\gamma > 0$) uniformly on $[0,1]\times[\![0,\mathsf{N}]\!]$ with high probability. Since $\mathbf{O}^{\mathsf{N}} = \mathbf{Y}^{\mathsf{N}} - \mathbf{W}^{\mathsf{N},\mathrm{BG}} = \mathbf{Y}^{\mathsf{N}} - \mathbf{W}^{\mathsf{N}}$ on $[0,1]\times[\![0,\mathsf{N}]\!]$ with high probability, we get the desired bound (7.12). □

Lemma 7.5. *For $\varepsilon > 0$ small but fixed, with high probability, we have $|\mathbf{S}^{\mathsf{N},\mathrm{st}}| \leqslant \mathsf{N}^{\varepsilon}$ on $[0,1]\times[\![0,\mathsf{N}]\!]$.* {lemma:sstap}

Proof. Let $\mathrm{t} \in [0,1]$ be the first (stopping) time such that $\|\mathbf{S}_{\mathrm{t},\cdot}^{\mathsf{N},\mathrm{st}}\|_{\ell^\infty([\![0,\mathsf{N}]\!])} \geqslant \mathsf{N}^{\varepsilon}$. The proof of Lemma 7.2 (based on uniqueness of Itô SDEs until stopping times) shows that $\mathbf{S}^{\mathsf{N},\mathrm{st}} = \mathbf{W}^{\mathsf{N}}$ up to (and at) time t. However, by (7.12) and (7.7) (the latter for, say, $\varepsilon/2$ instead of ε herein), we know $\|\mathbf{W}_{\mathrm{t},\cdot}^{\mathsf{N}}\|_{\ell^\infty([\![0,\mathsf{N}]\!])} \leqslant \mathsf{N}^{2\varepsilon/3}$. Because $\mathbf{W}^{\mathsf{N}} = \mathbf{S}^{\mathsf{N},\mathrm{st}}$ at time t, though, we get from this that $\|\mathbf{S}_{\mathrm{t},\cdot}^{\mathsf{N},\mathrm{st}}\|_{\ell^\infty([\![0,\mathsf{N}]\!])} \leqslant \mathsf{N}^{2\varepsilon/3}$, with high probability. Therefore, we cannot have that $\mathrm{t} < 1$ (with high probability), and thus $\mathrm{t} = 1$ on said high probability event. This finishes the proof. □

We are now in position to prove Proposition 3.9.

Proof of Proposition 3.9. As noted at the beginning of this section, the first estimate (3.19) follows from Lemma 5.11, so are left to show (3.20). This follows by the comparison of $\mathbf{Y}^{\mathsf{N}}$ to $\mathbf{Q}^{\mathsf{N}}$ in Lemma 7.3, the comparison of $\mathbf{Y}^{\mathsf{N}}$ to $\mathbf{W}^{\mathsf{N}}$ in Lemma 7.4, the high probability identification between $\mathbf{W}^{\mathsf{N}}$ and $\mathbf{S}^{\mathsf{N}}$ in Lemmas 7.1 and 7.2, and a comparison of $\mathbf{Z}^{\mathsf{N}}$ and $\mathbf{Y}^{\mathsf{N}}$ (which follows by (3.19) and (3.5)). This completes the proof. □

Appendix A. Technical discussions

A.1. **Heat kernel estimates.** Before we state the following, we recall the notation in Definition 3.3. {p:hkestimates}

Proposition A.1. *Throughout, we fix $\mathbf{a},\mathbf{b} \in \mathbb{R}$. For $0 \leqslant \mathrm{s} \leqslant \mathrm{t}$, set $\delta_{\mathrm{s,t}} := \max(|\mathrm{t}-\mathrm{s}|, \mathsf{N}^{-2})$. For any $0 \leqslant \mathrm{t} \lesssim 1$ and $\mathrm{x},\mathrm{y} \in [\![0,\mathsf{N}]\!]$, we have the following estimate:*

$$|\mathbf{H}_{\mathrm{s,t,x,y}}^{\mathsf{N},\mathbf{a},\mathbf{b}}| \lesssim_{\mathbf{a},\mathbf{b}} \mathsf{N}^{-1}\delta_{\mathrm{s,t}}^{-\frac{1}{2}}. \tag{A.1}$$ {:hkestimatesI}

Moreover, for any $0 \leqslant \mathrm{s} \leqslant \mathrm{t}$ and $\mathrm{x} \in [\![0,\infty]\!]$, we have

$$\sum_{\mathrm{y}\in[\![0,\infty]\!]} |\mathbf{H}_{\mathrm{s,t,x,y}}^{\mathsf{N},\mathbf{a},\mathbf{b}}| \lesssim_{\mathbf{a},\mathbf{b}} 1 \quad \text{and} \quad \sum_{\mathrm{y}\in[\![0,\infty]\!]} |\mathbf{H}_{\mathrm{s,t,x,y}}^{\mathsf{N},\mathbf{a},\mathbf{b}}|^2 \lesssim_{\mathbf{a},\mathbf{b}} \mathsf{N}^{-1}\delta_{\mathrm{s,t}}^{-\frac{1}{2}}. \tag{A.2}$$ hkestimatesII}

We now record regularity estimates. Fix $\vec{\mathrm{a}} = (\mathrm{a}_1,\dots,\mathrm{a}_{\mathrm{n}})$ and $\vec{\ell} = (\ell_1,\dots,\ell_{\mathrm{m}})$, where every entry in each vector is an integer. Fix any $0 \leqslant \mathrm{s} \leqslant \mathrm{t} \lesssim 1$ and $\mathrm{x},\mathrm{y} \in [\![0,\infty]\!]$ such that:

(1) We have $\mathrm{n},\mathrm{m} \lesssim 1$ and $\mathsf{N}|\mathrm{t}-\mathrm{s}|^{1/2} \gtrsim 1$.

(2) We have $|\mathrm{a}_1|,\dots,|\mathrm{a}_{\mathrm{n}}|,|\ell_1|,\dots,|\ell_{\mathrm{m}}| \lesssim \max(\mathsf{N}|\mathrm{t}-\mathrm{s}|^{1/2},1)$.

Then, we have the following regularity at length-scale $\mathsf{N}\delta_{\mathrm{s,t}}^{1/2}$:

$$|\nabla_{\vec{\mathrm{a}}}^{\mathbf{X}}\nabla_{\vec{\ell}}^{\mathbf{X}}\mathbf{H}_{\mathrm{s,t,x,y}}^{\mathsf{N},\mathbf{a},\mathbf{b}}| \lesssim_{\mathbf{a},\mathbf{b}} \mathsf{N}^{-1}\delta_{\mathrm{s,t}}^{-\frac{1}{2}} \cdot \prod_{\mathrm{i}\in[\![1,\mathrm{n}]\!]} \mathsf{N}^{-1}\delta_{\mathrm{s,t}}^{-\frac{1}{2}}|\mathrm{a}_{\mathrm{i}}| \cdot \prod_{\mathrm{j}\in[\![1,\mathrm{m}]\!]} \mathsf{N}^{-1}\delta_{\mathrm{s,t}}^{-\frac{1}{2}}|\ell_{\mathrm{j}}|. \tag{A.3}$$ kestimatesIII}

Above, the gradients $\nabla^{\mathbf{X}}_{\vec{\mathrm{a}}} := \nabla^{\mathbf{X}}_{\mathrm{a}_1} \dots \nabla^{\mathbf{X}}_{\mathrm{a}_n}$ *act on* x*, and the gradients* $\nabla^{\mathbf{X}}_{\vec{\ell}}$ *act on* y*. We also restrict to parameters so that the* LHS *above depends only on* $\mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\cdot,\cdot}$ *on* $[\![0,\infty]\!] \times [\![0,\infty]\!]$*.) Moreover, under the assumptions of (A.3),*

hkestimatesIV}

$$\sum_{\mathrm{y} \in [\![0,\infty]\!]} |\nabla^{\mathbf{X}}_{\vec{\mathrm{a}}} \nabla^{\mathbf{X}}_{\vec{\ell}} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}}| \lesssim_{\mathbf{a},\mathbf{b}} \prod_{\mathrm{i} \in [\![1,\mathrm{n}]\!]} \mathsf{N}^{-1} \delta_{\mathrm{s},\mathrm{t}}^{-\frac{1}{2}} |\mathrm{a}_{\mathrm{i}}| \cdot \prod_{\mathrm{j} \in [\![1,\mathrm{m}]\!]} \mathsf{N}^{-1} \delta_{\mathrm{s},\mathrm{t}}^{-\frac{1}{2}} |\ell_{\mathrm{j}}|. \tag{A.4}$$

Finally, we have the Chapman-Kolmogorov equation

{eq:hkestimates

$$\mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{t},\mathrm{x},\mathrm{y}} = \sum_{\mathrm{w} \in [\![0,\infty]\!]} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{r},\mathrm{t},\mathrm{x},\mathrm{w}} \mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}_{\mathrm{s},\mathrm{r},\mathrm{w},\mathrm{y}}. \tag{A.5}$$

Proof. These are the same estimates in Proposition A.1 of [33] (for the half-space heat kernel instead of the heat kernel on $[\![0,\mathsf{N}]\!]$). So, we now use the method-of-images representation for $\mathbf{H}^{\mathsf{N},\mathbf{a},\mathbf{b}}$ in terms of the half-space heat kernel to conclude. (See Section 3.2 of [25] for said method-of-images representation.) □

{lemma:kerneles

Lemma A.2. *Suppose that the kernel* $\boldsymbol{\mathcal{S}}_{\cdot,\cdot} : [\![0,\infty]\!]^2 \to \mathbb{R}$ *satisfies (4.27). Recall* $\mathfrak{l}_{\mathbf{Av}} = \mathsf{N}^{1-3\delta_{\mathbf{S}}/2}$ *from Definition 3.4. Recall the operator* $\mathscr{X} := \mathrm{Id} - \mathbf{Exp}(\tau_{\mathbf{Av}} \mathscr{T}_{\mathsf{N},0})$ *from the proof of Lemma 4.10. Given any* $\mathrm{D} > 0$*, there exists* $\mathrm{M} = \mathrm{O}_{\mathrm{D}}(1)$ *large enough such that*

{eq:kernelestir

$$|\mathscr{X}^{\mathrm{M}} \boldsymbol{\mathcal{S}}_{\mathrm{y},\mathrm{w}}| \lesssim_{\mathrm{D}} \mathsf{N}^{-\mathrm{D}}. \tag{A.6}$$

Proof. Roughly speaking, the operator $\mathscr{X}$ is an average of gradients with respect to length-scales of $\lesssim \mathsf{N}^2 \tau_{\mathbf{Av}}^{-2} = \mathfrak{l}_{\mathbf{Av}}$, and (4.27) shows that $\boldsymbol{\mathcal{S}}$ has regularity at scale $\gtrsim \mathsf{N}^{1-\delta_{\mathbf{S}}} \gg \mathfrak{l}_{\mathbf{Av}}$. Thus, every $\mathscr{X}$ gives a power-saving in N; accumulating enough of these gives (A.6). For a detailed argument, see the proof of Lemma A.3 in [33]. □

A.2. **Entropy estimates.** We record here the classical entropy inequalities (e.g. see Appendix 1.7 of [22]).

{lemma:entropy

Lemma A.3. *Consider any probability space* $(\boldsymbol{\Omega}, \mathscr{F}, \mathbb{P})$*, and let* $\mathbb{Q}$ *be another probability measure on* $(\boldsymbol{\Omega}, \mathscr{F})$ *so that* $\mathbb{Q} \ll \mathbb{P}$*. We let* $\mathbb{E}^{\mathbb{P}}$ *and* $\mathbb{E}^{\mathbb{Q}}$ *denote expectations with respect to* $\mathbb{P}, \mathbb{Q}$*, respectively. For any random variable* $\mathbf{X} : \boldsymbol{\Omega} \to \mathbb{R}$ *and any* $\kappa > 0$*, we have the inequality*

$$\mathbb{E}^{\mathbb{Q}} \mathbf{X} \leqslant \kappa^{-1} \mathbb{E}^{\mathbb{P}} \tfrac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} \log \tfrac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} + \kappa^{-1} \log \mathbb{E}^{\mathbb{P}} \mathrm{e}^{\kappa \mathbf{X}}.$$

Moreover, by choosing $\mathbf{X} = \mathbf{1}_{\mathscr{E}}$ *for some event* $\mathscr{E}$ *and optimizing* κ*, we have*

$$\mathbb{Q}(\mathscr{E}) \lesssim (1 + \mathbb{E}^{\mathbb{P}} \tfrac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} \log \tfrac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}) \cdot \Big\{ \log[\mathbb{P}(\mathscr{E})^{-1}] \Big\}^{-1}.$$

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